The Lorentz force: what a field does to a moving charge

Steady fields; a point charge of either sign moving at nonrelativistic speed unless stated.

The force law in words and symbols

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$F_B = |q| v B \sin\theta$$

The electric part pushes along E and does work; the magnetic part pushes perpendicular to the velocity and does none. Theta is the angle between v and B, so a charge moving parallel to B feels no magnetic force at all. The cross product fixes the direction: right hand along v, curl towards B, thumb gives v cross B, then apply the sign of q.

WHY / Split the motion into parts parallel and perpendicular to B before doing anything else; only the perpendicular part is turned.

WATCH / Never use the left hand for negative charges. Compute v cross B first and apply the sign of q afterwards; otherwise the next positive ion gets the wrong arrow.

Three properties worth memorising

$$\mathbf{F} \perp \mathbf{v} \;\Rightarrow\; K = \text{const}$$

$$\mathbf{F} \perp \mathbf{B} \;\Rightarrow\; \dot{\mathbf{v}} \perp \mathbf{B}$$

Because the force is perpendicular to the velocity, a magnetic field changes the direction of motion but never the speed. Because the force is also perpendicular to B, a charge cannot be pushed along the field; motion along B is unaffected. A static B field can therefore do no work and cannot by itself speed a particle up.

WHY / These two perpendicularities are the whole chapter compressed: circles, helices, selectors and mirrors all follow from them.

WATCH / Questions asking for the change in kinetic energy of a charge moving only in a magnetic field have answer zero. Electric fields do change energy; keep the two contributions separate.

Work, energy and a signed charge

$$W_B = 0$$

$$\Delta K = q(V_a - V_b) \text{ (electric part only)}$$

The magnetic force does no work at any instant, so the kinetic energy change of a charge in combined fields equals the electric work alone. For an electron moving against the electric field, the electric force is forward and it speeds up; the potential-energy bookkeeping carries the sign of q.

WHY / Work-energy answers must separate the two field contributions; otherwise a zero magnetic work term gets lost in an incorrect potential formula.

WATCH / The magnetic force is not always zero when a particle is deflected; it is the work that is zero. Deflection and energy change are different questions.

A feel for the tesla

$$1\ \mathrm{T} = 1\ \mathrm{N\,s\,C^{-1}m^{-1}}$$

$$B_{\oplus} \simeq 50\ \mu\mathrm{T}$$

One tesla exerts one newton on one coulomb moving at one metre per second. The Earth field is about 50 microtesla (half a gauss), a strong bar magnet about 1 T at its face, an MRI 1.5 to 3 T. Because the coulomb is enormous, the tesla is too; everyday laboratory fields are microtesla to tesla.

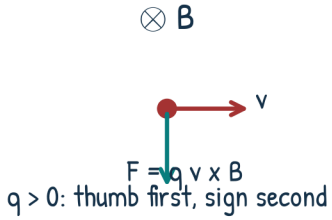
WHY / Large unit scales explain why magnetic forces on single static charges are hard to see yet forces on wire currents are ordinary.

WATCH / Gauss and tesla mix-ups are common: 1 G = 1e-4 T. Convert before comparing with the Earth field.

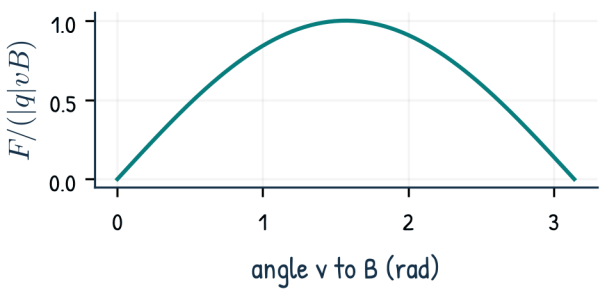
$$1\ \text{gauss} = 10^{-4}\ \mathrm{T}$$

CHECK / The Earth field of 50 microtesla is 0.5 gauss; MRI runs 1.5 to 3 T, fifteen to thirty thousand gauss.

Right hand along v, curl to B, thumb gives force; flip for negative q



Magnetic force peaks perpendicular and vanishes parallel to B



REMEMBER / Magnetic force is q v cross B: perpendicular to both v and B, so it turns motion and never changes speed or kinetic energy.

Circles and helices: the clock that ignores the speed

Uniform B; v not parallel to B; nonrelativistic mass m.

The circle from the force

$$r = \frac{mv_\perp}{|q|B}, \qquad \omega_c = \frac{|q|B}{m}, \qquad T = \frac{2\pi m}{|q|B}$$

The magnetic force supplies exactly the centripetal force: |q|vB = mv²/r gives the radius. The period equals circumference over speed, so the speed cancels: a faster particle runs a proportionally larger circle in the same time. The cyclotron frequency omega_c depends only on the charge-to-mass ratio and the field.

WHY / This speed-independent period is the single most useful fact of the chapter: cyclotrons, mass spectrometers and magnetic resonance all rest on it.

WATCH / Use the perpendicular component of velocity in r. Putting the full speed of a helix into the radius is the commonest error in trajectory problems.

$$t = \frac{\varphi}{\omega_c} = \frac{\varphi m}{|q|B}$$

Sense of rotation and the charge sign

ω_c from $q\,\mathbf{v} \times \mathbf{B}$

With B out of the page a positive charge moving right is pushed down and circles clockwise; a negative charge under the same motion circles anticlockwise. Either way the circulating charge is a tiny current loop whose own field at its centre opposes B: free-charge gyration is diamagnetic.

WHY / Get the direction from the cross product with a fixed convention; the sign of q flips the sense after the cross product is drawn.

WATCH / Curl sense for electrons is not a separate left-hand rule; it is the same cross product with a minus sign applied last.

CHECK / The circulating charge is a current loop whose own field opposes B: free-charge gyration is diamagnetic, the seed of diamagnetism in matter.

The helix

$$v_\parallel = v\cos\theta, \qquad v_\perp = v\sin\theta$$

$$p_{\text{helix}} = \frac{2\pi m v_\parallel}{|q|B}$$

An oblique entry separates into uniform motion along B and circular motion across it: a helix of radius r from v_perp and pitch = v_parallel times T. The circle lies in the plane perpendicular to B whatever the entry direction; the axis is along B. Theta is measured from B.

WHY / The two components never mix because each force component involves only the perpendicular velocity. Solving them separately avoids messy coupled equations.

WATCH / Pitch uses v_parallel and T together; do not use the full speed in either factor. A charge moving exactly along B travels in a straight line.

Numbers to anchor

$$e/m_e = 1.76 \times 10^{11}\ \text{C/kg}$$

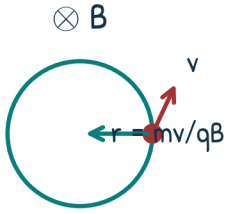
$$f_e \simeq 28.0\ \text{GHz/T}$$

An electron at 10^6 m/s in 1 mT has r = 5.7 mm and T = 36 ns, f = 28 MHz. A 10 MeV proton in 1 T: v about 4.4e7 m/s, r = 0.46 m, f = 15.2 MHz. Proton cyclotron frequency is 15.2 MHz per tesla, the number built into every cyclotron and MRI scanner.

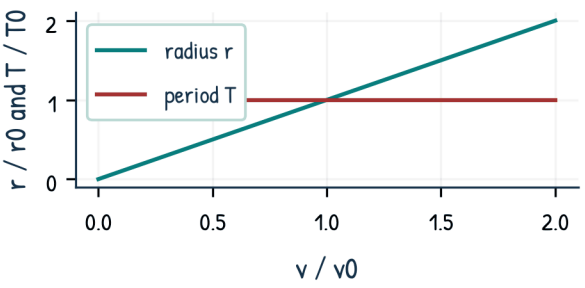
WHY / Order-of-magnitude anchors catch unit slips: a computed electron gyroradius of metres in a modest field is wrong.

WATCH / The proton spin precession frequency, 42.6 MHz per tesla, is a different constant from the orbital cyclotron frequency; resonance questions must say which one they mean.

Uniform B into page: force keeps turning v without changing speed



Radius grows with speed; the period stays fixed



REMEMBER / A charge perpendicular to uniform B circles with radius mv/qB and a period set only by q/m and B; speed changes the radius, never the time.

Parallel and crossed fields: the velocity selector

Uniform static E and B; crossed means perpendicular; v not relativistic.

The balance condition

$$qE = qvB \;\Rightarrow\; v = \frac{E}{B}$$

With E along +y and B along +z, a charge moving along +x feels the electric force qE along y and the magnetic force -qvB along y: the two are antiparallel whatever the sign of q because both flip together. They cancel at v = E/B. Slower particles bend towards the electric force side, faster ones the other way.

WHY / The condition loses both q and m, so the selector hands downstream apparatus a single speed without caring what it filters.

WATCH / The selector is for crossed fields only. If E has a component along B, the condition does not apply to that component.

Why sign and mass drop out

$$\frac{F_E}{F_B} = \frac{E}{vB}$$

$$\text{no } q, \text{ no } m$$

Both forces are proportional to q, so equality never depends on the sign or size of the charge. Neither force contains m, so heavy and light particles are selected identically. This is exactly why the selector pairs with a mass spectrometer: the spectrometer then measures m/q from the radius of one known speed.

WHY / A clean experimental chain separates speed selection from mass analysis; combining them corrupts the inference.

WATCH / For a beam of mixed species the selector transmits all with v = E/B; it does not purify charge or mass.

CHECK / Real selectors correct for fringing at the ends; the balance v = E/B applies deep inside the crossed region only.

Field along the motion

$$\mathbf{E} \parallel \mathbf{B}: \text{ helix of stretching pitch}$$

With E parallel to B the electric force accelerates along the field and the magnetic force only turns the perpendicular velocity. The result is a helix whose pitch grows or shrinks as v_parallel changes: a screw with a stretching thread. Radius is set by the (constant) perpendicular speed.

WHY / Parallel and crossed configurations must be recognised before computing; they ask different decompositions.

WATCH / Do not apply v = E/B to a parallel configuration; there the charge simply accelerates and the speed is not constant.

CHECK / With E parallel to B the perpendicular speed is constant and only the pitch changes; do not apply v = E/B to that geometry.

Numbers

$$E = 10^5 \text{ V/m},\; B = 0.10 \text{ T} \;\Rightarrow\; v = 10^6 \text{ m/s}$$

One hundred thousand volts per metre crossed with a tenth of a tesla selects one million metres per second. The region is usually a few centimetres long, so the transit time is tens of nanoseconds and stray fields at the entrance matter.

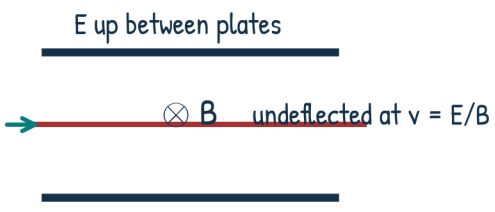
WHY / Checking the selected speed against the intended beam energy catches wiring or unit errors before the run.

WATCH / Real selectors have fringing fields at their ends; the ideal balance applies deep inside the crossed region.

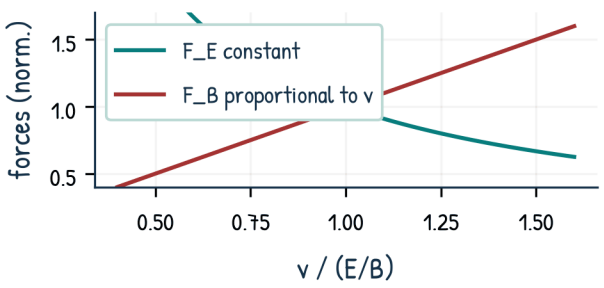
CHECK / Over a 3 cm plate region a 10^6 m/s beam spends 30 ns inside; entrance fringe fields, not the formula, set the real resolution.

CHECK / Transit times are tens of nanoseconds over centimetre plates.

Crossed E and B: one chosen speed passes straight through



Forces cross at v = E/B: slower bends one way, faster the other



REMEMBER / In crossed fields only particles with speed v = E/B pass undeflected, of either sign and any mass.

The three-region method

Field piecewise uniform in bounded regions; particle enters from outside.

The protocol

$$r = \frac{mv_\perp}{|q|B}, \qquad t = \frac{\varphi m}{|q|B}$$

In each uniform-field region the path is a circular arc centred on the perpendicular to v at the entry point, a distance r to the side the force points. Where the arc meets the boundary fixes the exit point; the tangent there is the exit direction; the angle varphi turned gives the time inside. Only a full circle takes one period.

WHY / Most exam trajectories are exactly this: geometry plus one radius formula. The perpendicular-at-entry construction fixes the centre without solving differential equations.

WATCH / Fringing at a magnet edge breaks uniformity; if the field varies strongly over the orbit use the mirror picture instead.

$$v_{\min} = \frac{|q|Bd}{m} \ \text{(strip)}$$

A strip of field

$$r > d: \ \sin\varphi = d/r$$

$$r \le d: \ t = \pi m/|q|B, \ \text{exit } 2r \text{ away}$$

A particle entering perpendicular to a strip of width d crosses if r exceeds d, turning through arcsin(d/r) and leaving displaced sideways by r(1-cos varphi). If r is smaller it never reaches the far edge: it turns through 180 degrees and comes back out the entry face after exactly half a period, displaced by 2r. The borderline r = d is the minimum-speed question v_min = qBd/m.

WHY / The crossing condition and the return condition cover both exam families; check which regime the numbers give before computing angles.

WATCH / The time in the strip is never the full period unless the particle closes a whole circle. For a half turn the time is exactly pi m/qB.

A circular region

$$\tan(\varphi/2) = R/r \ \text{for radial entry}$$

Entering a circular field region radially, symmetry makes the exit radial too: the arc centre, the entry point and the region centre form a right-angled figure with tan(varphi/2) = R/r. For r = R the deflection is a right angle. The result is geometry, not a new force law.

WHY / Radial entry is the special case that makes the deflection clean; off-radial entry needs the full three-step construction.

WATCH / Radial exit is guaranteed only for radial entry. Do not impose the symmetry when the beam enters off-axis.

CHECK / Radial entry into a circular region guarantees radial exit; off-radial entry breaks the symmetry and needs the full three-step construction.

Worked numbers

$$p \text{ at } 2\times10^6 \text{ m/s in } 0.30 \text{ T}: \ r = 6.96 \text{ cm}$$

Protons crossing a 5.0 cm strip: r exceeds d so they cross, turning 46 degrees in 28 ns (against a 219 ns full period) and displaced 2.1 cm sideways. Halve the speed and r halves to 3.48 cm: they turn back and exit 7.0 cm along the entry face after 109 ns, exactly half a period.

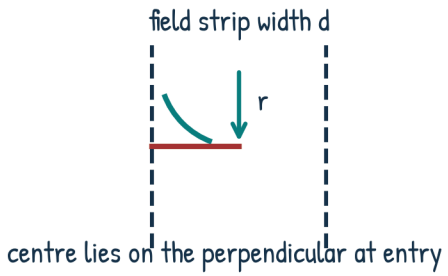
WHY / Same field, same particle, two regimes from one speed change: always compare r with d first.

WATCH / A 46 degree deflection is not 46 degrees of a full circle in time: use t = varphi m/qB with varphi in radians.

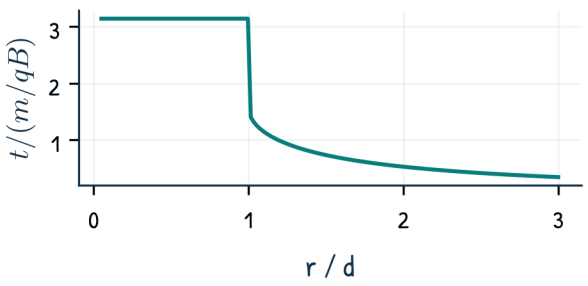
CHECK / Protons at 2 x 10^6 m/s in 0.30 T have r = 6.96 cm: crossing a 5 cm strip turns 46 degrees, halving the speed turns them back.

CHECK / Same field, two regimes from one speed change: always compare r with d first.

Three-region method: arc centred on the perpendicular to v at entry



Time in the strip: crossing branch then a jump to the half turn



REMEMBER / Mark the regions, draw an arc of radius r in each, and take the time from the angle turned, not from the full period.

The mass spectrometer

Ions accelerated through known V; uniform B; semicircular entry perpendicular to B.

From acceleration to radius

$$\tfrac{1}{2}mv^2 = qV, \quad r = \tfrac{mv}{qB} = \tfrac{1}{B}\sqrt{\tfrac{2mV}{q}}$$

Ions accelerated from rest through voltage V reach v = sqrt(2qV/m); the field then bends them into a circle whose radius carries both m and q. The detector sees the diameter 2r, so mass differences become distance differences: r is proportional to sqrt(m) at fixed q.

WHY / Eliminating the speed between the energy and radius equations leaves a purely measurable relation among V, B, r and m/q.

WATCH / This r uses the speed from the accelerating voltage only if the ion started from rest and V is the total drop. Double acceleration needs the summed voltage.

CHECK / The detector sees the diameter 2r, so the isotope spacing is twice the radius difference; never quote Delta r as the spot spacing.

Isotope separation

$$\frac{\Delta r}{r} = \frac{1}{2}\frac{\Delta m}{m}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{m_1}{m_2}}$$

Small mass differences halve in fractional radius, so resolution is limited by how narrowly the beam is defined, not by the formula. Singly charged uranium-235 and -238 at 10 kV in 0.50 T: radii 0.441 and 0.444 m, arriving 5.6 mm apart after their semicircles.

WHY / Half-relative-mass scaling is the working rule for estimating whether two isotopes resolve.

WATCH / The separation at the detector is twice the radius difference, because each beam completes a half circle. Do not quote Delta r as the spot spacing.

$$\frac{\Delta r}{r} = \frac{1}{2}\frac{\Delta m}{m}$$

With a velocity selector

$$v = E/B_1, \qquad r = \tfrac{mE}{qB_1B_2}$$

Bainbridge put a selector before the magnet: the ions then all have speed E/B1 regardless of how they were produced, and r is linear in m rather than its square root. The two instruments measure different things; combining them cleanly separates the variables.

WHY / Linearity in m simplifies calibration; the square-root form is for the pure acceleration design.

WATCH / A selector narrows speed but transmits every mass at that speed; the second field is still needed to separate them.

CHECK / Bainbridge linearises r in m by fixing the speed first; the pure-acceleration design gives the square root. Different instruments, different axes.

Numbers and validity

$$m = 235u: \; r = 0.441 \text{ m}$$

$$\Delta(2r) = 5.6 \text{ mm}$$

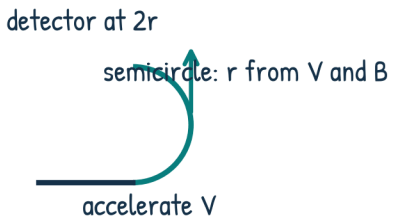
Atomic mass unit 1.66e-27 kg; charge one elementary charge. The ions travel metres in modest vacuum; collisions with residual gas scatter the beam and set the practical resolution. Relativistic mass increase is negligible at 10 keV per nucleon.

WHY / Checking that Delta m/m is small validates the half-relative approximation before using it.

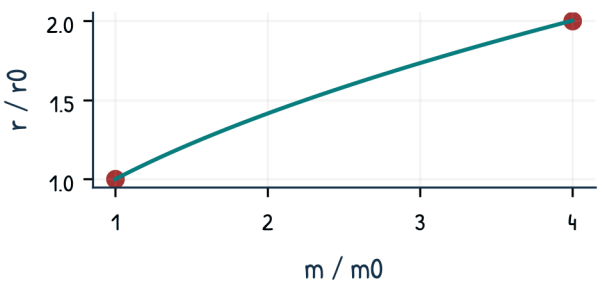
WATCH / For molecular ions with different charge states, q must come from the ion source settings; equal m does not mean equal r.

CHECK / For molecular ions read q from the ion source; equal m does not mean equal r when the charge states differ.

Spectrometer: acceleration then a semicircle onto a detector



Radius grows as the square root of mass at fixed q, V and B



REMEMBER / Heavier ions land farther out: radius measures the square root of mass for fixed q, V and B.

The cyclotron: a resonance machine

Uniform B between two hollow dees; alternating gap voltage; nonrelativistic ions.

Resonance condition

$$f = \frac{qB}{2\pi m}$$

$$K_{\max} = \frac{q^2 B^2 R^2}{2m}$$

The ion coasts on semicircles inside the dees where the field acts and is kicked by qV0 each gap crossing. If the voltage has reversed in exactly one half period, every crossing adds energy. The spiral grows in radius, not in period. Final energy is set by B and dee radius R; a smaller V0 only means more turns.

WHY / Fixed frequency works only because T is speed-independent; that is the whole machine in one sentence.

WATCH / Relativity breaks the fixed period by the factor gamma, which is why classical cyclotrons stop at tens of MeV for protons and never worked for electrons.

Turns and timing

$$N_{\text{turns}} = \frac{K_{\max}}{2qV_0}, \qquad t = N/f$$

Each full revolution gains 2qV0. Protons in 1.5 T with R = 0.50 m reach 27 MeV: resonance 22.9 MHz, about 270 turns at 50 kV gap amplitude, 12 microseconds total flight, a path of roughly 250 m.

WHY / Turn count and total time are separate outputs; a high-voltage machine is compact in time but not in energy.

WATCH / Energy per turn is 2qV0, not qV0: two gap crossings per revolution.

CHECK / Total flight time is turns divided by frequency: microseconds of acceleration over hundreds of metres of spiral path.

CHECK / Dee voltage fixes the turn count: 100 kV gaps give 135 turns for 27 MeV, so the spiral stays compact only while the frequency tracks the mass.

Focusing and size

$$K_{\max} \propto B^2 R^2$$

$$\gamma = 1 + K/mc^2$$

A perfectly uniform field lets ions drift out of the mid-plane; a field decreasing slightly with radius bends field lines and pushes strays back, the same physics as magnetic mirroring. Doubling the maximum energy needs four times the pole area at the same B. At 27 MeV the proton mass has already grown 3 per cent.

WHY / Scaling laws decide machine design before any detailed orbit calculation; pole area is the expensive part.

WATCH / The size scaling is in B^2R^2, so doubling B at fixed R quadruples the energy; quoting only V0 misses the physics entirely.

Numbers check

$$f_p = 15.2\ \text{MHz/T}$$

$$K = 27\ \text{MeV at } 1.5\ \text{T},\ R = 0.5\ \text{m}$$

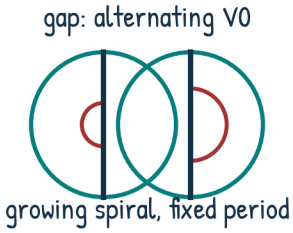
Proton cyclotron frequency 15.2 MHz per tesla; electron 28.0 GHz per tesla. The maximum kinetic energy 27 MeV equals (eBR)²/2m with eBR = 1.2e-19 in SI. Comparing with m_p c² = 938 MeV gives the relativistic correction 2.9 per cent at that energy.

WHY / Anchoring the frequency per tesla and the energy formula catches most cyclotron exam slips.

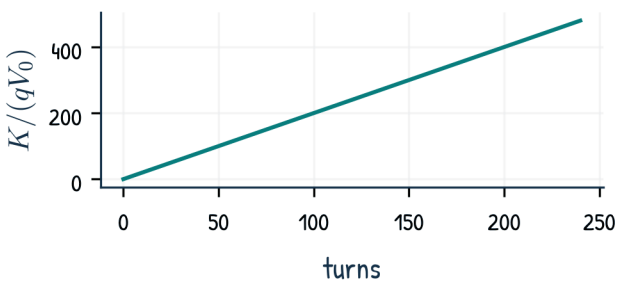
WATCH / Electrons are ultrarelativistic in any cyclotron-sized orbit; their machines are synchrotrons or linacs, not classical cyclotrons.

CHECK / Alphas in a proton machine reach the same energy at half the frequency (q/m halves); verify by scaling before recomputing orbits.

Cyclotron dees: each gap crossing adds energy at fixed frequency



Energy grows by 2qV0 per turn: linear until the exit radius



REMEMBER / Because the period ignores speed, a fixed-frequency gap voltage accelerates every turn until the ion leaves at the dee radius.

Magnetic mirrors, bottles and adiabatic invariance

Field varying slowly over one gyration; converging field lines; single particle.

Why the force points to weaker field

$$F_z = -\mu \frac{\partial B}{\partial z}, \qquad \mu = \frac{m v_\perp^2}{2B}$$

Where field lines converge, divergence-free B gives a small radial component B_r = -(r/2) dB/dz. The azimuthal velocity crossed with B_r pushes along the axis towards weaker field, for either sign of charge because sense of gyration and sign flip together. The particle behaves as a dipole of moment mu in a gradient.

WHY / The gradient force is the magnetic analogue of the electric dipole gradient force; the same number mu is the circulating charge of the gyromagnetic section.

WATCH / The push is towards weaker field for every charge sign. Reversing q reverses the gyration sense as well, not the net drift direction.

$$F_z = -\mu \frac{dB}{dz}, \qquad \mu = \frac{m v_\perp^2}{2B}$$

The adiabatic invariant

$$\mu = \frac{m v_\perp^2}{2B} = \text{const}$$

$$B_{\text{mirror}} = \frac{B_0}{\sin^2 \theta_0}$$

Magnetic force does no work so total kinetic energy is fixed; as v_perp grows with B the parallel energy shrinks. Writing the perpendicular energy as mu B and tracking the exchange shows mu itself is constant when B changes little per orbit. Reflection happens where all energy is perpendicular: at B = B0/sin²(theta0).

WHY / An invariant replaces solving the orbit: it converts a trajectory problem into two energy statements.

WATCH / Adiabatic invariance fails in sharply varying fields such as a pole edge; use the three-region construction there instead.

Bottles and the loss cone

$$\sin^2 \theta_0 > \frac{B_0}{B_{\max}} \ \text{trapped}$$

$$\theta_{\text{lc}} = \arcsin \sqrt{B_0 / B_{\max}}$$

Between two mirrors a particle is trapped when its pitch angle is large enough that reflection comes before the throat. Particles inside the loss cone of half-angle theta_lc escape; for mirror ratio 10 the cone is 18 degrees. The Earth field is such a bottle: trapped particles bounce pole to pole, and those in the loss cone make the aurora.

WHY / Loss-cone physics is the same for fusion mirror machines and the Van Allen belts: different machines, one geometry.

WATCH / Collisions scatter particles into the loss cone over time; a static bottle is never perfectly confining.

$$\sin^2 \theta_0 < \frac{B_0}{B_{\max}} \ \text{escapes}$$

Numbers

$$B_{\max}/B_0 = 10: \ \theta_{\text{lc}} = 18^\circ$$

$$\mu \text{ of a } 1 \text{ MeV } p$$

A 1 MeV proton mirroring in 31 microtesla has mu = K_perp/B of order 1e-13 A m² when perpendicular; the guiding-centre radius 37 km is tiny against planetary scales, which validates the averaging.

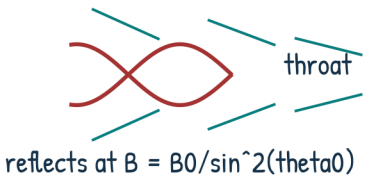
WHY / Separation of scales is what makes the invariant useful; check r_grad much less than B before applying it.

WATCH / The invariant is of the motion averaged over one gyration, not of the instantaneous gyro-phase orbit.

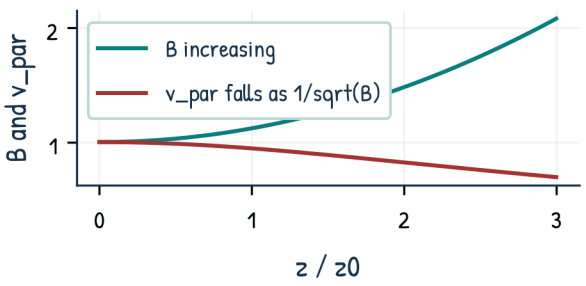
CHECK / A particle reflects where B = B0 / sin^2(theta0); at pitch 30 degrees it mirrors at 4 times the equatorial field.

CHECK / A particle at 40 degrees pitch mirrors at B = B0 / 0.41 = 2.4 times the equatorial field. Check the trapping inequality before quoting the loss cone.

Converging field lines squeeze the helix until it reflects



Invariant mu: parallel speed falls until reflection where B = B0/sin^2(theta0)



REMEMBER / In a slowly converging field the quantity mv_perp²/2B is conserved, so a particle reflects where the field is strong enough.

The Hall effect

Flat conductor; steady current; uniform transverse B; carrier density n of one sign.

The balance that makes the voltage

$$qE_H = qv_dB, \qquad V_H = \frac{IB}{nqt}$$

Carriers drifting along the current are pushed sideways by q v_d cross B and pile up on one edge until their transverse electric field balances the magnetic push. The steady Hall field is v_d B, and across the width w the voltage is IB/(nqt) with t the thickness along B.

WHY / Hall voltage is a transverse equilibrium, not an extra resistance; the field builds until the net transverse force vanishes.

WATCH / Thickness is along B, width is across the voltage. Swapping them inserts the wrong dimension and can give voltages a factor of ten large.

$$F_E = qE_H = qv_dB \;\Rightarrow\; E_H = v_dB$$

Carrier sign: one edge, two stories

$$V_H \text{ polarity} = \text{carrier sign}$$

$$R_H = \frac{1}{nq}$$

Positive carriers drifting with the current and negative carriers drifting against it are both pushed to the same edge, because q v_d is the same vector either way. But positive carriers make that edge positive and electrons make it negative: the voltmeter polarity reads the sign. Zinc, cadmium and p-type semiconductors conduct by holes — this is how it was learned.

WHY / One geometric fact (same edge) plus one bookkeeping fact (edge sign) covers every Hall question.

WATCH / Do not say electrons and holes go to opposite edges. Same edge, opposite voltage sign.

What the magnitude measures

$$V_H = \frac{IB}{nqt}$$

$$\mu_m = \frac{\sigma}{ne} = \sigma R_H$$

Combining Hall voltage with conductivity gives the mobility. Copper at 10 A in 1.0 T with n = 8.5e28 gives only 0.73 microvolts, while an n-type wafer with n = 1e21 at 10 mA in 0.50 T gives 0.31 V. The ratio is the carrier-density ratio: Hall sensors are semiconductors for exactly this reason.

WHY / Hall probes dominate magnetic sensing because low carrier density multiplies the signal.

WATCH / Hall coefficient sign is 1/nq with q signed; a positive R_H means dominant positive carriers, not a positive charge density in metals.

$$R_H = \frac{1}{nq}, \qquad \sigma = nq\mu$$

Geometry and numbers

$$V_H = \frac{IB}{nqt}$$

$$n = 8.5 \times 10^{28}\ \mathrm{m}^{-3}\ (\mathrm{Cu})$$

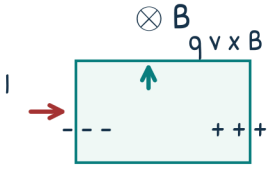
Microvolt copper signals need lock-in amplification; volt-level semiconductor signals do not. Multi-band metals can show odd behaviour because both carrier types contribute; the simple formula assumes one type. Ordinary laboratory fields and currents give Hall voltages from microvolts to volts.

WHY / Keeping one carrier model explicit states the assumption; two-carrier Hall is the extension.

WATCH / A reading of zero Hall voltage does not mean zero field: it can mean equal mixed carriers.

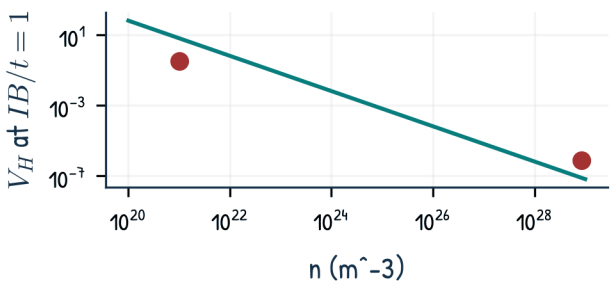
CHECK / Copper gives 0.7 microvolts for 10 A in 2 T across 0.2 mm; a 0.1 mm semiconductor strip gives 40 mV for 20 mA in 0.5 T.

Hall plate: carriers pile on one edge until q E_H balances q v B



same edge, opposite voltage

Hall voltage is inversely proportional to carrier density



REMEMBER / The polarity of the Hall voltage reveals the sign of the carriers; the size counts them.

Drifts and the cycloid

Crossed uniform E and B; or weak gradients with r grad B much less than B.

Drift by force balance

$$\mathbf{v}_d = \frac{\mathbf{E}\times\mathbf{B}}{B^2}, \qquad v_d = \frac{E}{B}$$

A velocity perpendicular to both fields makes the net force vanish: q(E + v_d cross B) = 0 solves to v_d = E/B. No q and no m appear, so electrons and ions drift together, which is why the drift carries no current in a neutral plasma.

WHY / The E cross B drift is a motion of the whole guiding centre, not a charge-selective force.

WATCH / Never call the drift a force. It is a velocity, and it is identical for opposite charges moving in opposite gyration senses.

CHECK / The drift is a velocity of the guiding centre: no q and no m, so a plasma drifts as one and carries no net current.

The cycloid from rest

$$x = \frac{E}{B}\left(t - \frac{\sin\omega_c t}{\omega_c}\right), \quad y = \frac{E}{B\omega_c}(1-\cos\omega_c t)$$

In the drifting frame the particle simply circles at speed E/B; back in the lab that circle is carried along: a rolling-wheel cycloid with cusps if the particle started from rest. The arch height is 2r_c = 2mE/qB², the peak speed 2E/B, and the energy budget qE times height matches the peak kinetic energy exactly.

WHY / The drifting frame solves crossed-field orbits in one line; integrating the lab equations of motion reproduces the same curve.

WATCH / The full speed at the arch top is 2E/B, twice the drift; energy checks must use the true speed, not v_d.

CHECK / The cycloid is a circle of radius mE/qB^2 carried at E/B: arch height 2 r_c, peak speed 2E/B, energy gained qE times the height.

Gradient and curvature drifts

$$\mathbf{v}_{\nabla B} = \pm\frac{v_\perp r_c}{2}\frac{\mathbf{B}\times\nabla B}{B^2}$$

Where B is stronger on one side, the orbit radius shrinks there and the guiding centre creeps sideways, perpendicular to both B and the gradient. The sign depends on the charge, so this drift separates charges and drives ring currents; the averaged force is -mu grad B, the same mirror force.

WHY / Gradient drift is charge-selective unlike E cross B drift; planetary ring currents and plasmas separate charge because of it.

WATCH / Only the perpendicular velocity drifts; particles with parallel motion can also pick up curvature drift, a second comparable term.

CHECK / Gradient drift reverses with the charge sign and drives ring currents; curvature drift needs the field line curvature, same order of magnitude.

Numbers

$$E = 10^4\ \mathrm{V/m},\ B = 0.1\ \mathrm{T}:\ v_d = 10^5\ \mathrm{m/s}$$

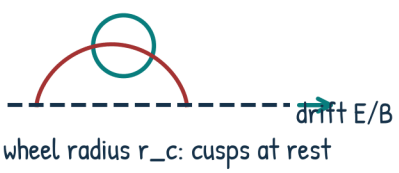
An electron released from rest has r_c = 5.7 micrometres and rises 11 micrometres per arch; a proton on the same fields drifts at the same 10^5 m/s with a 1.0 cm gyroradius, 1836 times larger. Drift speeds are comparable to common plasma flows; gyroradii span micrometres to kilometres.

WHY / Comparing electron and ion scales separates the fast gyration from the slow common drift.

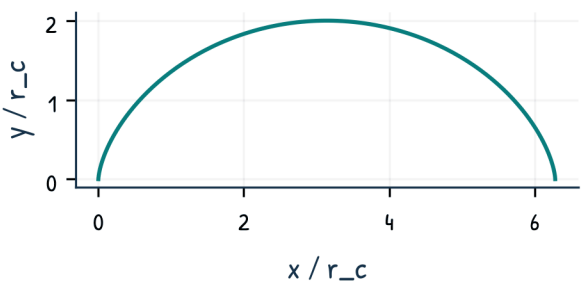
WATCH / Proton and electron arch heights differ by the mass ratio even though the drift matches; do not transfer one orbit size to the other.

CHECK / Electrons and protons in the same crossed fields drift together at 10^5 m/s but circle on radii differing by a factor of 1836.

Crossed fields from rest trace a cycloid carried at the drift speed



Cycloid from rest: cusps every period, arch height 2 r_c



REMEMBER / In crossed fields the guiding centre drifts at E cross B over B squared, the same velocity for every charge and mass.

Reading tracks: Thomson, momentum and detectors

Nonrelativistic where stated; known E and B; deflections small unless solved exactly.

Thomson crossed fields

$$\frac{q}{m} = \frac{2yE}{B^2L^2}$$

An electron beam first deflected by E alone then balanced by crossed B gives the classic e/m measurement: the electric deflection y over plates of length L at speed v = E/B eliminates v and leaves q/m = 2yE/(B²L²). Thomson measured it for cathode rays in 1897 and identified the electron.

WHY / This is the velocity selector used twice: once to know the speed, once to balance the deflection.

WATCH / The formula assumes small deflection and the selector speed throughout; large angles need circular-orbit geometry.

$$q/m = \frac{2yE}{B^2L^2}, \qquad v = E/B$$

Momentum from curvature

$$\frac{p}{q} = Br, \qquad p[\mathrm{MeV}/c] = 300\,B[\mathrm{T}]\,r[\mathrm{m}]$$

A charged particle of momentum p in field B curves with radius r = p/qB. This holds at any speed when p is the relativistic momentum: chambers measure momentum, not energy. The 300 Br rule with r in metres and B in tesla gives MeV/c directly.

WHY / Detector physics is momentum physics; energy must be inferred with a mass hypothesis or a calorimeter.

WATCH / Converting p to kinetic energy needs the right regime: p²/2m nonrelativistic, pc approximately ultrarelativistic, and something in between otherwise.

CHECK / Chambers measure momentum, not energy; converting needs the regime: K = p^2/2m below, K about pc above.

Bending magnets and spectrometers

$$\Delta x \simeq \frac{qB^2L^2}{2mv^2}\ \text{(slab)}$$

$$t = \varphi m/qB$$

In a bounded field region use the three-region method; in a thin lens approximation the impulse qBL over transit time gives the angular kick. Read a track by matching its curvature radius inside the field and the straight tangents outside.

WHY / Track reconstruction is the experimental form of the same circle geometry that exam strips ask for.

WATCH / Multiple scattering and field inhomogeneity set real resolution; ideal radius formulas overstate precision.

CHECK / A thin field region gives an angular kick qBL/p like a lens; inside the region use the arc, outside the straight tangents.

Scales of detectors

$$e/m_e = 1.76 \times 10^{11}\ \mathrm{C/kg}$$

$$B \sim 1\ \mathrm{T},\ r \sim \mathrm{m} \Rightarrow p \sim 300\ \mathrm{MeV}/c$$

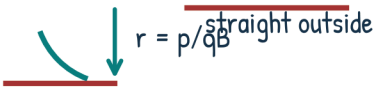
A 1 GeV/c track in a 1 T field has radius 3.3 m — the scale of modern detector bores. Electron and positron tracks bend oppositely, allowing pair identification; neutral particles leave no track at all.

WHY / Knowing detector scales turns abstract formulas into estimates an examiner recognises as physically literate.

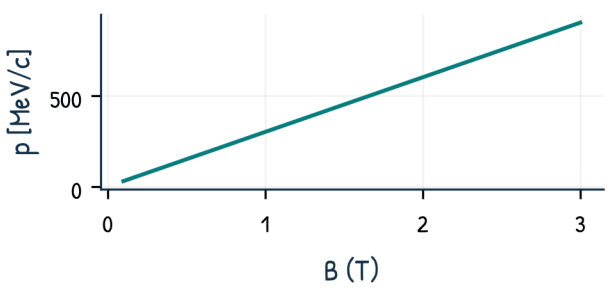
WATCH / Curvature fits near the end of a low-momentum track can fail as the particle loses energy; use the momentum at the point of measurement.

CHECK / 1 GeV/c in 1 T curves at 3.3 m; electron and positron tracks bend oppositely, and neutral particles leave no track at all.

Track curvature inside B measures momentum; tangents fix the entry



300 Br rule at r = 1 m: momentum is linear in the field



REMEMBER / Trajectories measure q/m or momentum: the 300 Br rule converts curvature to momentum at any speed.

Biot-Savart: the source law

Steady currents; vacuum; thin wires unless a density is specified.

The law

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\mathbf{l}\times\hat{\mathbf{r}}}{r^2}$$

$$\mathbf{B} = \frac{\mu_0}{4\pi}\int\frac{I\,d\mathbf{l}\times\hat{\mathbf{r}}}{r^2}$$

Each current element contributes a field perpendicular to both the element and the line to the field point, falling as 1/r². The constant mu0/4pi belongs to Biot-Savart; a 2pi in a final answer comes from the geometry, not the constant. Direction from the cross product; magnitude with the sine of the angle between element and r.

WHY / Superposition of vector elements is the universal method: any geometry can be cut into elements even when no symmetry shortcut exists.

WATCH / A straight segment whose line passes through the field point contributes nothing there — the cross product vanishes.

From moving charge to wire

$$\mathbf{B} = \frac{\mu_0}{4\pi}\frac{q\,\mathbf{v}\times\hat{\mathbf{r}}}{r^2} = \frac{1}{c^2}\,\mathbf{v}\times\mathbf{E}$$

A point charge q moving at v makes a field proportional to v cross r-hat: the magnetic field of a moving charge is its electric field crossed with v over c². A wire is a parade of such charges; the current element I dl replaces q v. The identity B = v cross E over c² previews the relativistic origin of magnetism.

WHY / The factor 1/c² is why magnetic forces between ordinary charges are tiny compared with electric ones; wires are neutral so the electric part cancels and magnetism survives.

WATCH / The moving-charge form assumes v much less than c; it is a quasi-static field, not radiation.

Units and bookkeeping

$$\mu_0 = 4\pi\times10^{-7}\ \mathrm{T\,m\,A^{-1}}$$

$$I\,d\mathbf{l} \leftrightarrow q\mathbf{v}$$

Current elements I dl, surface currents K dA and volume currents J dV all enter the same integral. A rotating charged ring of charge Q at angular speed omega is a loop with I = Q omega / 2pi. Choose the substitution matching the dimensionality of the source.

WHY / One integral form covers wires, sheets and clouds; the measure is the only thing that changes.

WATCH / Density factors are not optional corrections: a line current has no area density to multiply in later.

CHECK / A rotating charged ring is a steady loop: I = Q omega / 2 pi. Match the current model to the actual charge motion.

What the constant is not

$$\frac{\mu_0}{4\pi} = 10^{-7}\ \mathrm{T\,m\,A^{-1}}$$

$$\mu_0\varepsilon_0 = 1/c^2$$

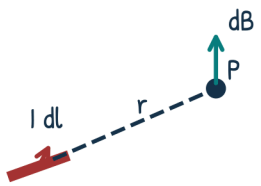
mu0 over 4pi is the Biot-Savart constant; mu0 over 2pi is the infinite-wire answer; mu0 over 4pi times 2 is the semi-infinite end field. The magnetic and electric constants are tied by mu0 eps0 = 1/c²: magnetism is relativity applied to moving charge.

WHY / Separating the three familiar factors prevents the commonest constant mix-up in wire problems.

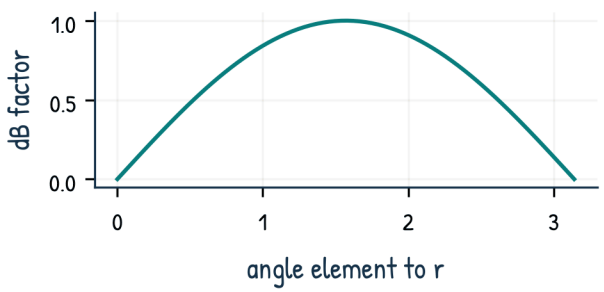
WATCH / A formula quoting mu0/2pi is a finished wire result, never the starting constant.

CHECK / mu0/4pi is the Biot-Savart constant, mu0/2pi the infinite-wire answer, mu0/4pi the semi-infinite end value: three factors, three situations.

Biot-Savart element: dB is perpendicular to both I dl and r



Biot-Savart weight: sin theta vanishes for elements along the line of sight



REMEMBER / Every magnetic field of steady currents comes from the Biot-Savart integral; symmetry may replace the integration but never the physics.

Straight wires: finite by angles

Thin straight conductor; field point perpendicular distance d from its line.

Finite wire by end angles

$$B = \frac{\mu_0 I}{4\pi d}(\sin\alpha + \sin\beta)$$

For a wire segment, the perpendicular distance d to the field point and the end angles alpha, beta measured at the point between the perpendicular and the ends give the field magnitude mu0 I (sin alpha + sin beta)/4 pi d. Both sines are positive when the point is beside the segment and the perpendicular foot lies inside it.

WHY / The angle method collapses the Biot-Savart integral in one step; extending angles beyond the foot uses signed sines.

WATCH / If the perpendicular foot falls outside the segment one sine enters with a minus sign. Blindly adding two positive sines doubles a contribution that should subtract.

$\alpha + \beta < \pi$ signed: subtract outside end

Limits

$$\alpha = \beta \to 90^\circ : \; B = \frac{\mu_0 I}{2\pi d}$$

$$\text{semi-infinite end} : \; \frac{\mu_0 I}{4\pi d}$$

Both angles 90 degrees give the infinite wire, the classic mu0 I / 2 pi d. A semi-infinite wire with the field point level with its end has one angle 90 and one zero: half the infinite field. At the end of a wire the perpendicular foot is exactly at the end, the marginal case.

WHY / The three formulas differ by factors of two that come from geometry; recognising which applies is the exam skill.

WATCH / mu0 I / 4 pi d is not the finite-wire formula evaluated carelessly; it is the specific end-on limit.

CHECK / Semi-infinite at its end level: exactly half the infinite field. The end value is the marginal case of the angle formula.

Compound wires add as vectors

$$\mathbf{B}_{\text{total}} = \sum_i \mathbf{B}_i$$

$$\text{straight through P: } B_i = 0$$

A bent wire is a set of straight pieces; each piece contributes by its own angles and direction, and the fields add as vectors. Any straight piece whose line passes through P contributes exactly zero there. A square loop at its centre is four such pieces; the leads of a circular coil often vanish.

WHY / Superposition with the zero-contribution rule removes most of the algebra from compound figures.

WATCH / A gap in a wire is not the same as the full wire minus nothing: model it as full plus a negative piece or as the two remaining segments.

Numbers

$$I = 10\ \text{A},\ d = 10\ \text{cm} : \; B = 20\ \mu\text{T}$$

$$\text{Earth} \simeq 50\ \mu\text{T}$$

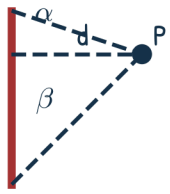
Ten amps ten centimetres away makes a field comparable to the Earth: the reason lab wiring deflects compasses. A square of side a carries B = 2 sqrt(2) mu0 I / pi a at its centre; a regular n-gon approaches the loop answer mu0 I / 2R from below as n grows.

WHY / Comparing with the Earth field immediately tells whether a geometry is a serious magnetic disturbance.

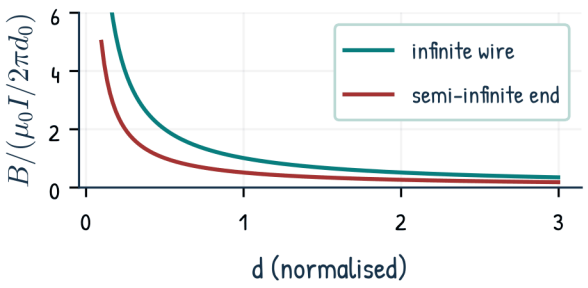
WATCH / Currents near compass measurements need magnetic hygiene: the wire field falls only as 1/d.

CHECK / Ten amps at ten centimetres give 20 microtesla, comparable to the Earth: lab wiring visibly deflects compasses.

Finite wire: field from the two end angles alpha and beta at P



Same 1/d falloff; the end-on wire is half the infinite field



REMEMBER / A finite straight wire is integrated once by angles; the infinite answer is a limit of the same formula.

Loops and arcs

Thin circular wire of radius R; field point at centre or on axis.

Centre field and arcs

$$B_{\text{centre}} = \frac{\mu_0 I}{2R}$$

$$B_{\text{arc}} = \frac{\mu_0 I \theta_0}{4\pi R}$$

At the centre every element is perpendicular to its radius, so the integral is just the circumference: mu0 I / 2R. An arc of angle theta0 (radians) is that fraction of a loop: mu0 I theta0 / 4 pi R. Straight pieces whose lines pass through the centre contribute nothing.

WHY / Arc results follow from the loop by proportion; recompute nothing.

WATCH / Theta0 must be in radians. A semicircle is pi, giving mu0 I / 4R, not 180 times anything.

$$B_{\text{centre}} = \frac{\mu_0 I}{2R}, \qquad B_{\text{arc}} = \frac{\mu_0 I \theta_0}{4\pi R}$$

The axial field

$$B_x = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

Each element on the ring is at distance s = sqrt(R²+x²) from the point on the axis; the perpendicular components cancel by symmetry and the axial projections add with factor R/s. At x = 0 this is mu0 I / 2R; far away it falls as 1/x³, the dipole law of the next section.

WHY / The axial formula is exact on the axis and the bridge to dipole fields; off axis there is no elementary closed form.

WATCH / Quoting mu0 I / 2R off the centre is wrong; the axis formula reduces to it only at x = 0.

CHECK / Far off the axis the loop field is the dipole field of its moment mu = I pi R^2: the axial formula reduces to 2 mu0 mu / 4 pi x^3.

The peak of the on-axis field

$$\frac{d}{dx}\left[\frac{R^2}{(R^2 + x^2)^{3/2}}\right] = 0 \text{ at } x = 0 \text{ (centre)}$$

For a single loop the axial field is maximum at the centre and decreases both ways. Two coaxial loops can make a flat region where the second derivative vanishes — the Helmholtz construction of the extension section. Multiple turns multiply the field by N without changing the shape.

WHY / Field shape questions are calculus questions: study the derivatives of the axial formula, not just its value.

WATCH / A maximum of B on axis is not automatically a point of zero gradient of the vector field; stability questions need the full components.

CHECK / The single-loop axial field peaks at the centre; the Helmholtz pair cancels the second derivative to build a flat region.

Turns, n-gons and compound loops

$$B_N = N\frac{\mu_0 I}{2R}$$

$$B_{n\text{-gon}} \to \frac{\mu_0 I}{2R}$$

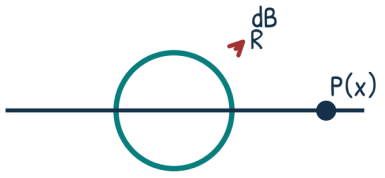
Multiple turns multiply every loop result by N: centre, axis and arc alike. A regular n-gon approaches the loop field as n grows; a square gives 2 sqrt(2) mu0 I / pi a and a hexagon sqrt(3) mu0 I / pi a at their centres.

WHY / The polygon family is a calibration ladder: each member checks the wire and loop formulas against each other.

WATCH / Circumscribed polygon radius and side length differ; write R in terms of a before comparing with the loop limit.

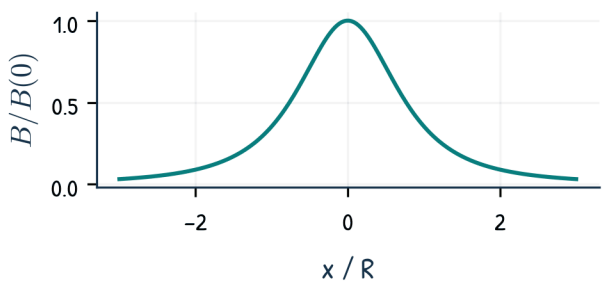
CHECK / Square: 2 sqrt(2) mu0 I / pi a; hexagon: sqrt(3) mu0 I / pi a; both climb toward mu0 I / 2R as the polygon grows.

Loop axis: every element is at distance sqrt(R^2 + x^2) from P



transverse parts cancel; net field is axial

Loop axis field: maximum at centre, dipole 1/x^3 tail far away



REMEMBER / A full loop gives mu0 I / 2R at its centre; an arc scales with its angle; the axis is one clean integral.

Solenoid and toroid

Long solenoid n turns per length; toroid with N total turns; steady current I.

Finite solenoid on axis

$$B = \frac{\mu_0 n I}{2}(\cos\theta_1 + \cos\theta_2)$$

Stacking loop fields along the axis gives a simple result: the angles are subtended at the ends between the axis and the line to the point, both taken positive for an interior point, so the bracket is 2 deep in the middle of a long coil and 1 at an end. Interior of an infinite solenoid: mu0 nI; at an end exactly half; outside far away it leaks of order R/L.

WHY / The end factor one-half is a good exam trap and a useful check: the end sees only half the turns ahead of it.

WATCH / The cosine convention must be stated: other texts measure both angles from one direction and subtract. Follow one convention through the whole problem.

$$B = \frac{\mu_0 n I}{2}(\cos\theta_1 + \cos\theta_2)$$

CHECK / The finite-solenoid cosine formula gives the interior plateau and the half value at the end from one expression.

The toroid

$$B(r) = \frac{\mu_0 N I}{2\pi r} \quad (a < r < b)$$

The Amperian circle inside the winding encloses all N turns: B = mu0 N I / 2 pi r, falling as 1/r across the core. Outside the winding the enclosed current is zero and the field vanishes exactly. A toroid is the ideal closed magnetic circuit.

WHY / No leakage outside makes the toroid the clean demonstration of the loop law and the model for transformer cores.

WATCH / Use total turns N for the toroid and turns per metre n for the solenoid; mixing them is a classic factor error.

CHECK / Inside the toroid B = mu0 N I / 2 pi r falls as 1/r across the winding; in the hole and outside it is exactly zero.

Ideal limits and reality

$$B_{\text{in}} = \mu_0 n I, \quad B_{\text{out}} \approx 0 \quad (L \gg R)$$

An infinite solenoid has perfectly uniform interior field and zero exterior field. A real one leaks at the ends and slightly outside the windings; the error is of order R/L. A solenoid of 1000 turns per metre at 1 A gives 1.26 mT, forty times the Earth field inside.

WHY / Knowing the size of leakage separates ideal formulas from apparatus statements.

WATCH / For a short coil the infinite formula misleads badly; use the finite cosine form with the actual geometry.

CHECK / Real solenoids leak of order R/L outside; the zero-outside statement is the infinite-length idealisation.

CHECK / End field falls to half of the interior plateau; a finite solenoid leaks of order R / L outside. Quote the cosine formula at the ends.

Field energy preview

$$U_B = \frac{B^2}{2\mu_0}\mathcal{V}$$

$$L = \mu_0 n^2 \ell \ \text{(long solenoid)}$$

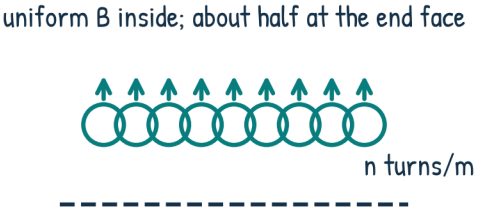
A long solenoid of volume V stores (B squared over 2 mu0) times V, consistent with half L I squared with L = mu0 n squared times length. The preview is the working check for the induction module and gives the end force when two halves are pulled apart.

WHY / Energy density times volume is the fastest sanity check on any solenoid formula in a derived result.

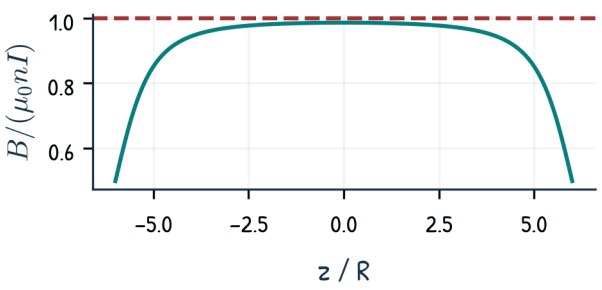
WATCH / Inductance per length assumes the infinite solenoid; a real coil has end corrections and stray field energy outside.

CHECK / Field energy of mu0 n I at 1 T is 0.4 MJ per cubic metre wherever the field lives: the preview of B^2 / 2 mu0.

Solenoid: stacked loops give a uniform interior field and zero outside ideal



End field falls to half of the interior value at the winding



REMEMBER / A long solenoid has uniform mu0 nI inside and zero outside; a toroid confines the field completely.

Magnetic moment and the dipole field

Loop small compared with distance r; moment mu = N I A; uniform magnetisation cases noted.

Defining the moment

$$\boldsymbol{\mu} = NI\mathbf{A}, \qquad \mu = NIA$$

The area vector follows the right-hand rule: fingers along the current, thumb along mu, which is also the direction of the loop field through its centre. For N turns multiply. A rotating charged ring of charge Q and angular speed omega has I = Q omega / 2pi and mu = Q omega R² / 2.

WHY / One vector fixes orientation, energy and torque; define it once and the dipole machinery runs.

WATCH / Area is a vector by convention tied to the current sense; a clockwise loop viewed from the chosen side has mu the other way.

$$\boldsymbol{\mu} = NIA\,\hat{\mathbf{n}}$$

CHECK / N turns multiply the moment; the right-hand rule around the current gives the normal, and the moment points along it.

The far field

$$B_{\text{axis}} = \frac{\mu_0}{4\pi}\frac{2\mu}{r^3}$$

$$B_{\text{eq}} = \frac{\mu_0}{4\pi}\frac{\mu}{r^3}$$

Far from a small loop the field is the electric dipole field pattern with p replaced by mu over eps0 c² bookkeeping: axial component twice the equatorial one at the same r, both falling as 1/r³. The full vector formula in components follows from the same expansion.

WHY / The factor two between axis and equator is the fingerprint of a dipole; every bar magnet far away obeys it.

WATCH / Dipole formulas need r much larger than the loop size; at the loop centre use mu0 I / 2R instead.

CHECK / On axis B = mu0 2 mu / 4 pi r^3 and at the equator half that: the same 1/r^3 law with two geometry factors.

Magnets are current loops

$$U = -\boldsymbol{\mu}\cdot\mathbf{B}$$

$$\boldsymbol{\tau} = \boldsymbol{\mu}\times\mathbf{B}$$

A bar magnet, a solenoid end and a current loop are the same object at long range. Superposing two coaxial antiparallel loops of opposite circulation cancels their interior field and leaves only exterior dipole fields: the picture of a physical dipole with poles.

WHY / Identifying every magnet with a moment lets dipole formulas apply to atoms, coils and planets alike.

WATCH / The pole model is a bookkeeping device; fields near the ends of real magnets are not inverse-square pole fields.

CHECK / A bar magnet and a solenoid of the same moment are indistinguishable from outside; the pole picture fails inside the material.

Numbers

$$\mu_{\text{loop}} = NIA$$

$$\mu_{\oplus} \simeq 8\times10^{22}\ \text{A m}^2$$

A 100-turn coil of radius 5 cm at 1 A has mu = 0.79 A m². The Earth dipole is 8e22 A m². The Bohr magneton is 9.27e-24 A m² — atomic moments are twenty orders below laboratory coils, yet they are all that matter in materials.

WHY / Spending a minute on moment scales shows why magnetism is atomic physics plus relativity.

WATCH / Moment of a magnet is independent of orientation but its torque is not; keep the vector and the scalar distinct.

CHECK / A 2 A, 50-turn coil of radius 5 cm carries mu = 0.785 A m^2; the Earth torque on it in 50 microtesla is 39 micronewton metres.

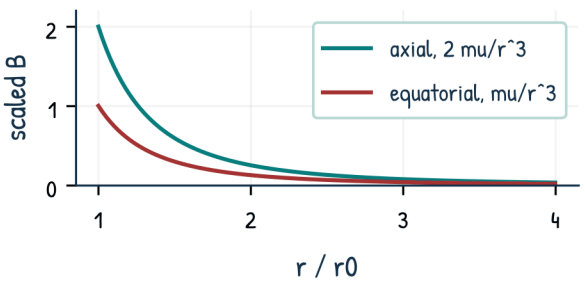
CHECK / The Earth torque on a small coil is mu B sin theta.

Loop moment mu = N I A with the right-hand area vector



fingers along I: thumb is mu

Dipole field: axial component twice equatorial at the same distance



REMEMBER / A current loop is a magnetic dipole: axial field twice equatorial, and the whole far field follows from mu = N I A.

Ampere's law: when it works

Steady currents; closed loop C; any surface bounded by C.

The law and its proof sketch

$$\oint_C \mathbf{B}\cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Walking once around the infinite wire at radius r adds B times 2 pi r = mu0 I, independent of r: the 1/r of the field cancels the r of the path. Any loop that threads the wire once gives the same circulation; superposition makes the total circulation count enclosed currents.

WHY / The same cancellation built Gauss law with $1/r^2$ and r^2; here it is 1/r and r.

WATCH / Zero circulation of a loop enclosing no current says nothing about the field along the loop: B can be large and tangential everywhere.

CHECK / The law is exact for steady currents; the shortcut is the symmetry: without it the loop law is true and useless.

Sign convention with a worked instance

$$I_{\text{enc}} = I_{\odot} - I_{\otimes}$$

$$\oint \mathbf{B}\cdot d\mathbf{l} = 2\ \text{A} \rightarrow 2.5\ \mu\text{T m}$$

Choose a loop direction; curl the right-hand fingers that way; the thumb is positive for threading currents. Wires of 5 A out and 3 A in threaded by an anticlockwise loop give enclosed 2 A and circulation 2.5 microtesla metres. The same loop reversed flips both signs.

WHY / A wrong direction choice flips both sides and preserves the physics; a wrong sign in copying does not.

WATCH / The enclosed current is the net through any surface bounded by the loop, counting sign; two opposite currents can cancel.

Honest usefulness

useful iff $\mathbf{B} \parallel d\mathbf{l}$ and $|B|$ constant on segments

The law is a constraint on an integral. It gives B only when symmetry guarantees the field is parallel to the path and constant along it, or perpendicular to some sides, or zero on them: infinite wire, long solenoid, toroid, thick cylinder, sheet. Finite solenoids and loops have no such symmetry; the law is true and silent.

WHY / Choosing between the loop law and Biot-Savart is choosing the method that the symmetry supports.

WATCH / A symmetric-looking but finite source defeats the shortcut; integrate element by element instead.

CHECK / Ampere on a finite solenoid or a loop gives nothing usable: Biot-Savart by angles and arcs is the honest route there.

Two methods agree

solenoid: $B = \mu_0 n I$ both ways

The long solenoid can be stacked from loop fields or obtained from one rectangle of Ampère loop; the answers match exactly. Agreement of independent methods is the exam check that a derived formula is right.

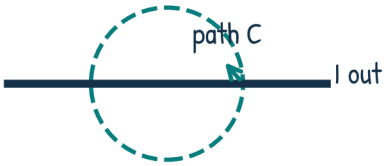
WHY / Whenever possible solve a symmetric case both ways once; it calibrates every later shortcut.

WATCH / Matching answers does not validate the method for the next geometry if its symmetry differs.

CHECK / The solenoid both ways: stacking loops gives the cosine profile; the loop law gives the interior value in one line. They agree.

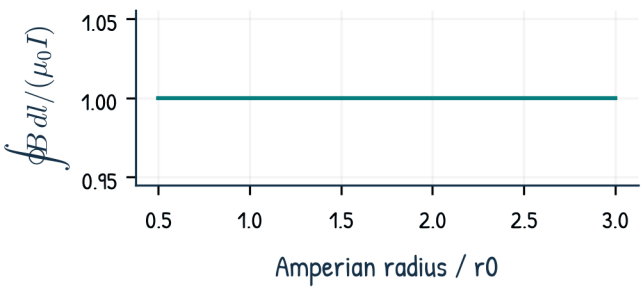
CHECK / The loop law gives nothing for a finite solenoid; Biot-Savart by angles is the honest route. Symmetry is the deciding criterion.

Amperian loop around a wire: circulation counts threaded current



loop direction sets the sign of enclosed current

Circulation is independent of radius: the loop law in one picture



REMEMBER / The circulation of B equals mu0 times the enclosed current: always true, useful only where symmetry reduces the integral to B times length.

Ampere applications: the eight fields

High symmetry: wire, thick wire, cylinder, sheet, solenoid, toroid, coax, plate.

Wire family

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B_{\text{thick}} = \frac{\mu_0 I r}{2\pi a^2} \ (r < a)$$

Outside a long straight wire the answer is mu0 I / 2 pi r. Inside a thick wire of radius a with uniform current density, only the enclosed fraction counts: B grows linearly to mu0 I / 2 pi a at the surface. Two parallel wires with opposite currents between them add fields in the gap.

WHY / The enclosed fraction is the entire method: write I_enc(r) and divide by the path length.

WATCH / Inside the wire the current is a fraction of I; multiplying the exterior formula by r/a twice gives the wrong power.

CHECK / Inside a thick wire B grows linearly to mu0 I / 2 pi a at the surface; a shell carries all its current at the skin.

Sheets and solenoids

$$B_{\text{sheet}} = \frac{\mu_0 K}{2}$$

$$B_{\text{solenoid}} = \mu_0 n I, \ B_{\text{end}} = \frac{\mu_0 n I}{2}$$

An infinite sheet of surface current K has uniform field mu0 K / 2 on each side, opposite in direction, independent of distance. A solenoid wall is such a sheet wrapped up: interior mu0 nI, exterior zero. Two sheets of opposite K give mu0 K in the gap.

WHY / Sheet fields are the building blocks of the boundary-condition section and of overlap tricks.

WATCH / The sheet formula has no distance because the sheet is infinite; a finite plate has edge fringing.

CHECK / Between two opposed sheets the fields add to mu0 K; outside they cancel. The solenoid is this construction rolled up.

Coax and hollow cylinders

$$J \text{ inner}, \ I_{\text{enc}} = I\frac{r^2 - b^2}{c^2 - b^2}$$

$$B_{\text{gap}} = \frac{\mu_0 I}{2\pi r}$$

For current distributed between radii b and c the enclosed current is the area fraction. A neutral coaxial cable with return current in the outer shell confines its field to the gap: inside the inner conductor and outside the cable the fields cancel exactly.

WHY / Coax is the exam case where the return current matters; write both contributions before canceling.

WATCH / Ideal thin shells and uniform current densities are assumptions; surface currents change the face values but not the enclosed ledger.

CHECK / Coax: zero inside the inner conductor and outside the cable; all the field lives in the gap annulus.

The toroid and the decision

$$B_{\text{toroid}} = \frac{\mu_0 N I}{2\pi r}$$

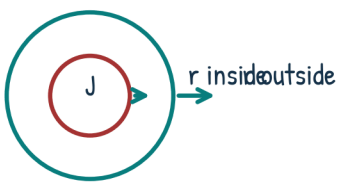
$$\text{no symmetry: Biot-Savart}$$

Use the loop law for infinite wires, long solenoids, toroids, thick cylinders and sheets. Use Biot-Savart for loops, arcs, finite wires, off-axis points and bent conductors. The decision table in the source chapter maps each geometry to one method.

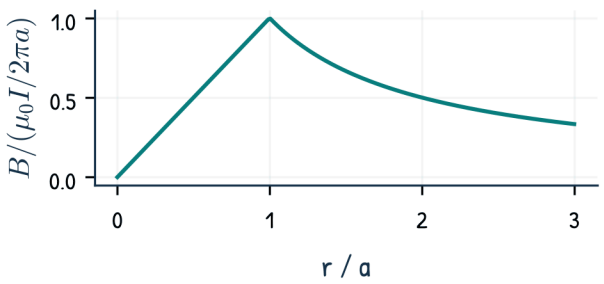
WHY / A method decision made before algebra saves the exam; mixing methods mid-problem is where sign errors breed.

WATCH / Compounding two symmetries (moving from wire to sheet) requires the new enclosed current, not a renamed old one.

Thick wire: enclosed fraction grows as r squared inside, full current outside



Inside linear growth, outside 1/r: one continuous B(r)



REMEMBER / Eight standard geometries cover most exam fields; learn the enclosed-current ledger for each.

Superposition tricks and no monopoles

Linear media and vacuum; uniform current densities; steady fields.

Uniform fields from cylinders

$$\mathbf{B} = \tfrac{\mu_0}{2}\mathbf{J} \times \mathbf{r}_{\perp}$$

$$\mathbf{B}_{\text{gap}} = \tfrac{\mu_0}{2}\mathbf{J} \times \mathbf{d}$$

A long cylinder of uniform current density J produces at internal offset r a field mu0 J cross r / 2. Two overlapping cylinders of opposite J leave a hole whose field is the difference: uniform, mu0 J cross d / 2, where d points from the removed centre to the full-body centre.

WHY / The overlap trick turns a hard geometry into two symmetric ones; the same method covers offset holes and missing arcs.

WATCH / The uniform result is only in the overlap region where both bodies exist; outside it fields vary again.

$$\mathbf{B}_{\text{gap}} = \tfrac{\mu_0}{2}\mathbf{J} \times \mathbf{d}$$

The gap in a loop

$$\mathbf{B}_{\text{gap}} = \mathbf{B}_{\text{full}} - \mathbf{B}_{\text{missing arc}}$$

A circular loop missing an arc equals the complete loop minus that arc taken with opposite current. The full loop gives mu0 I / 2R at the centre; the missing arc by proportion. Compound and subtract with vector signs, never scalar magnitudes.

WHY / Negative-current bookkeeping is the magnetic twin of missing-charge superposition in electrostatics.

WATCH / Field lines of pieces are not field lines of the sum; draw each contribution before combining.

CHECK / A gapped loop is the full loop minus the missing arc taken with opposite current; compound with vector signs, never scalar magnitudes.

CHECK / The overlap field is uniform and transverse to J. Moving the hole changes only d, never the field magnitude.

No magnetic monopoles

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0 \quad \forall S$$

$$\nabla \cdot \mathbf{B} = 0$$

Magnetic flux through any closed surface is zero: field lines that enter also leave. This is why there is no magnetic charge, no inverse-square magnetic Coulomb law, and why field lines close through magnets. It is exact, not an approximation of the current chapter.

WHY / Divergence-free B is the structural counterpart of Gauss law and the reason magnetic circuits are loops.

WATCH / A point where B = 0 does not imply a magnetic charge absence test; only the closed-surface integral is identically zero.

$$\oint \mathbf{B} \cdot d\mathbf{A} \equiv 0$$

Field lines close

outside N to S, inside S to N

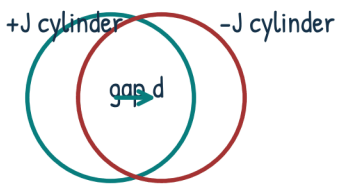
$$\oint \mathbf{B} \cdot d\mathbf{A} \equiv 0$$

Bar magnet field lines run through the magnet itself, forming closed loops; cutting a magnet makes two smaller closed systems, never an isolated pole. Superconducting flux conservation and the absence of monopoles are the same statement in different language.

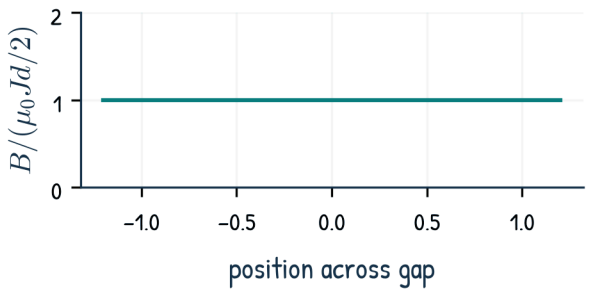
WHY / Closed field lines are why magnetic circuits are analogised to electric circuits with reluctance but no magnetic voltage source inside the gap.

WATCH / Field-line maps drawn only outside a magnet invite pole thinking; draw the interior loop at least once.

Overlap trick: two opposite uniform cylinders leave a uniform-field gap



Field in the overlap gap is uniform and transverse to J



REMEMBER / A hole is a full body plus a negative body; a gap is a full loop minus an arc; and no closed surface ever nets magnetic flux.

Force on a current

Uniform B on a straight segment, or local B along a curved path; steady I.

From carriers to the wire

$$\mathbf{F} = I\mathbf{L} \times \mathbf{B}$$

$$d\mathbf{F} = I\, d\mathbf{l} \times \mathbf{B}$$

Each carrier feels q v_d cross B; a length L holds nAL carriers, so the total is nAL q v_d cross B = I L cross B since I = n q v_d A. The sign of the carriers is already inside the current direction: no extra sign remains to apply.

WHY / Deriving the wire force once from the carrier force shows where the Hall field enters and why the lattice feels the push.

WATCH / For a negative carrier drifting opposite to conventional I the force is the same; do not flip the cross product.

$$F = nALqv_dB = ILB\sin\theta$$

The chord theorem

$$\mathbf{F} = I\left(\int d\mathbf{l}\right) \times \mathbf{B}$$

$$\oint d\mathbf{l} = 0$$

In a uniform field the integral of dl along any path is the straight vector from start to end: a curved wire feels the force of its chord. A semicircle of radius R perpendicular to B feels 2IRB; a closed loop feels no net force at all in uniform B.

WHY / Chord reduction replaces path integration with two points; closed loops always cancel in uniform fields.

WATCH / A closed loop can still turn; zero net force does not mean zero effect. Torque comes from non-collinear side forces.

Numbers

$$I = 10\ \text{A},\ L = 0.5\ \text{m},\ B = 0.2\ \text{T}:\ F = 1\ \text{N}$$

One newton is the weight of 100 grams: motors stack many such lengths in strong fields. Force per metre of a 1 kA conductor in 1 T is 1 kN, which is why busbars are braced.

WHY / Newton-per-metre anchors keep design estimates honest in both directions of scale.

WATCH / Sin theta in ILB sin theta is the angle between wire and field, not to the field normal; parallel wire to B feels nothing.

CHECK / 1 kA in 1 T feels 1 kN per metre: the structural design number for busbars and pulse-magnet windings.

CHECK / Opposite currents 5 cm apart at 500 A press at 1 N per metre: substation busbars need bracing at every joint. Sin theta uses the angle between wire and field: perpendicular gives the full force.

Non-uniform fields: the net force

$$\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$$

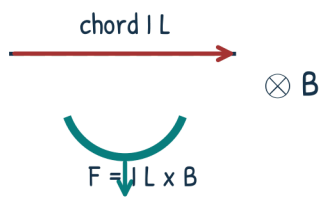
$$F \simeq \mu\, dB/dx \ \text{(aligned)}$$

A closed loop in a gradient feels a net force because the two sides see different fields. For an aligned moment the force pulls toward stronger field with magnitude mu times the gradient; anti-aligned reverses it. This is the electromagnet lifting a nail and the trap behind equal-and-opposite thinking.

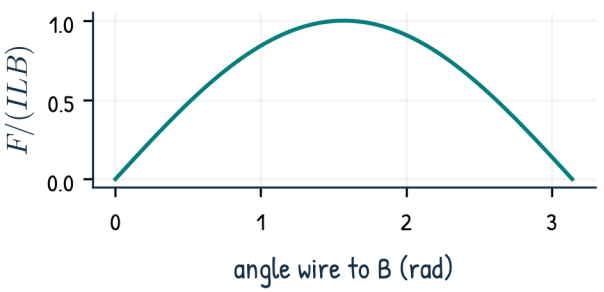
WHY / The gradient force is the working principle of magnetic lifting and of mirror confinement; uniform-field results silently drop it.

WATCH / Only the gradient along the moment orientation gives the simple mu dB/dx; transverse gradients produce sideways forces and torques.

Curved wire in uniform B feels the force of its straight chord



Force peaks when wire is perpendicular to B and vanishes when parallel



REMEMBER / The force on a wire is I L cross B; a curved wire in uniform field feels the force of its chord.

Torque on a loop: energy and the galvanometer

Small planar loop of moment mu in uniform B; torsion constant kappa for instruments.

Couple and energy

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}, \qquad U = -\boldsymbol{\mu} \cdot \mathbf{B}$$

Forces on opposite sides of a loop in uniform field are equal, opposite and displaced: a pure couple of magnitude mu B sin theta. The potential energy is lowest when mu aligns with B and highest anti-aligned; the force on the moment in a gradient is toward stronger field when aligned.

WHY / Torque and energy answer different questions: torque is the tendency to turn at fixed position; energy compares orientations and gives forces via gradients.

WATCH / A loop at anti-alignment is in unstable equilibrium; torque vanishes there too, so zero torque does not imply the minimum.

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos\theta$$

The oscillating magnet

$$T = 2\pi\sqrt{\frac{I_m}{\mu B}}, \qquad \mu B = \frac{4\pi^2 I_m}{T^2}$$

A magnet swinging in the horizontal component B_H of the Earth field is a torsional pendulum for small angles: period set by moment of inertia I_m and the product mu B_H. Oscillation measures the product; deflection of a compass by the same magnet measures the ratio. Two measurements, two unknowns — the basis of absolute magnetic measurement.

WHY / Gauss method separates mu and B_H without any external calibration; this is why the old unit of B bore his name.

WATCH / The period uses the horizontal component only; a magnet swinging near the dip poles has nothing to restore it.

The galvanometer

$$\varphi = \frac{NAB}{\kappa} I$$

current balance

A coil of N turns, area A in field B carrying current I twists a suspension of torsion constant kappa by angle proportional to I: the moving-coil galvanometer. Calibrated once, it reads current; paired with a known resistor it reads voltage.

WHY / The instrument is the torque law in hardware; its sensitivity is NAB/kappa.

WATCH / Angle scales are linear only within the spring law; stop pins exist for a reason.

CHECK / Current sensitivity of a moving-coil meter is NAB/kappa; adding a series resistor turns it into a voltmeter of range IR.

Fixed current versus fixed flux

$$F = +\partial U_B / \partial x \ (I \text{ fixed})$$

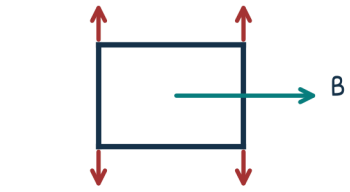
$$F = -\partial U / \partial x \ (\Phi \text{ fixed})$$

Virtual work on magnetic systems depends on the constraint. A current-driven circuit feeds energy from the source and the mechanical force follows the positive gradient of field energy; an isolated superconducting loop conserves flux and the sign flips. State the constraint before differentiating.

WHY / The two conventions are the commonest source of sign errors in force questions; the physics is identical, the bookkeeping is not.

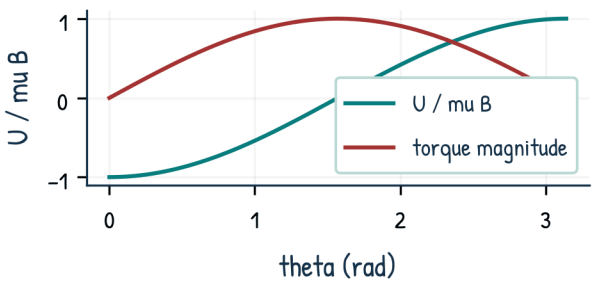
WATCH / Never mix the two rules in one problem; a battery-held coil and a persistent superconducting loop are different thermodynamic systems.

Loop in uniform B: net force zero, torque turns the plane toward alignment



displaced forces: couple mu x B

Aligned loop sits at the energy minimum; torque vanishes there too



REMEMBER / A loop in uniform B feels no net force but a torque mu cross B; its energy is -mu dot B and its small oscillations are the magnetometer.

Forces between currents

Long straight parallel wires; steady currents; vacuum between.

The force law

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

Wire 2 sits in wire 1 field mu0 I1 / 2 pi d and feels I2 L cross B, giving the force per length. Same-direction currents attract; opposite directions repel. Each wire feels the other field only: a wire never pushes on itself.

WHY / Mutual force depends on the product of currents and falls as 1/d; the geometry is the same inverse-distance law as the field.

WATCH / Adding the two fields at the midpoint before computing the force double-counts; each wire responds to the other field alone.

$$F/L = \frac{\mu_0 I_1 I_2}{2\pi d}$$

CHECK / Wire 2 feels wire 1 field only; a wire never pushes on itself, whatever the current distribution inside it.

The ampere, defined

$$1\ \text{A}: \ \frac{F}{L} = 2 \times 10^{-7}\ \text{N/m at } d = 1\ \text{m}$$

The classical definition: one ampere is the current in each of two long parallel wires one metre apart giving two tenths of a micronewton per metre. Since 2019 the ampere is fixed by the elementary charge, but the force formula is unchanged.

WHY / The tiny force shows again that wire magnetism is weak per amp, yet measurable because currents can be large.

WATCH / Post-2019 definitions change the unit bookkeeping, not Maxwell equations; never rewrite mu0 in the force formula.

CHECK / Since 2019 the ampere is fixed by the elementary charge; the force formula is unchanged and still defines the practical scale.

Newton third law

$$\mathbf{F}_{1\to 2} = -\mathbf{F}_{2\to 1}$$

momentum to fields

The pair forces are equal and opposite even though each wire feels only the other field. When currents change, the mechanical forces still balance but field momentum changes; the missing momentum is stored in B, not lost.

WHY / Third law holds for steady circuits as a statement about wires; the field carries momentum in time-dependent cases.

WATCH / Do not add self-force: an element of wire is not pushed by its own field; the divergence of the wire model is an exam classic.

CHECK / Steady circuits satisfy the third law as wire forces; during switching the field momentum changes and balances the books.

Scaling

$$F \propto I_1 I_2 / d$$

parallel attract, anti repel

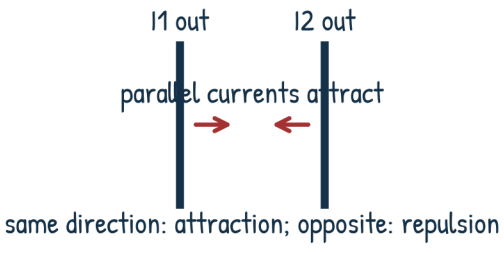
Doubling one current doubles the force; halving the distance doubles it. Two 1 kA busbars 10 cm apart press at 2 N per metre — structural bracing is standard. Pulse magnets see forces proportional to I² and can burst their windings.

WHY / I² scaling underlies both motor design and the pinch effect; one mental square catches many factor-of-two slips.

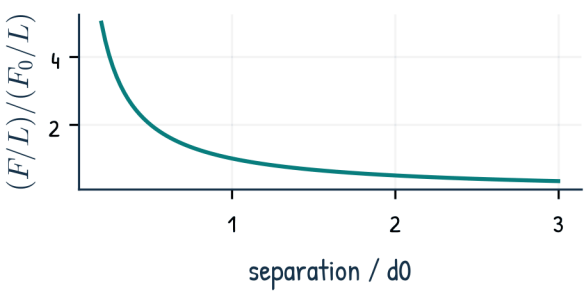
WATCH / Currents induced during switching can repel or attract contrary to the steady-state expectation; watch the direction of each current at the instant considered.

CHECK / Two 1 kA busbars 10 cm apart press at 2 N per metre; pulse forces scale as I squared and can burst windings.

Two parallel wires: each feels the field of the other only



Force per length falls as 1/d: same currents, half distance, double force



REMEMBER / Parallel currents attract and antiparallel repel: mu0 I1 I2 / 2 pi d per unit length, and the ampere is defined from it.

Magnetic pressure and field energy

Surface currents and field discontinuities; ideal sheets; uniform fields on each side.

Force on a current sheet

$$\frac{F}{A} = \frac{B_1^2 - B_2^2}{2\mu_0}$$

from strong side to weak

A sheet with surface current K separates fields B1 and B2. The sheet feels only the field of the rest, the average of the two sides; with B1 - B2 = mu0 K the force becomes the difference of squares over 2 mu0. At 1 T this is four atmospheres on the windings.

WHY / The factor one-half is the conductor-stress factor again: a current sheet cannot push on its own field.

WATCH / The direction is from strong field to weak regardless of current direction; the squares erase the sign.

$$\frac{F}{A} = \frac{B_1^2 - B_2^2}{2\mu_0}$$

Solenoid end force and energy density

$$u_B = \frac{B^2}{2\mu_0}, \qquad F_{\text{halves}} = \frac{B^2 A}{2\mu_0}$$

Cut a long solenoid and the halves attract with pressure times area: field lines behave as if under tension along themselves and pressure across. A 1 T solenoid of 10 cm² bore pulls together with 400 N. The same B²/2 mu0 is the energy density: 0.4 MJ per cubic metre at 1 T.

WHY / Energy methods give the end force without integration: separating halves by dx creates field volume A dx at energy density B²/2 mu0.

WATCH / The energy argument assumes constant current maintained by a source; at fixed flux the sign of the virtual-work rule flips.

CHECK / B^2 / 2 mu0 is both the pressure on the windings and the energy density: 4 atmospheres and 0.4 MJ per cubic metre at 1 T.

Boundary conditions

$$B_{1t} - B_{2t} = \mu_0 K$$

$$B_{1n} = B_{2n}$$

Across any surface current the tangential B jumps by mu0 K and the normal component is continuous. The solenoid field steps from mu0 nI to zero exactly at the winding sheet K = nI. A perfect diamagnet with B = 0 inside feels the full B²/2 mu0 from outside.

WHY / Boundary conditions are the local form of the loop and pillbox arguments; use them to fix field values at surfaces.

WATCH / Outside an ideal long solenoid the field is zero, so the hoop stress is inside-to-out; a finite solenoid leaks and reduces the ideal stress a little.

CHECK / Across a surface current the tangential B jumps by mu0 K and the normal component stays continuous: the local form of the loop argument.

Numbers and the motor puzzle

$$1\ \text{T} \to 3.98 \times 10^5\ \text{Pa} \approx 4\ \text{atm}$$

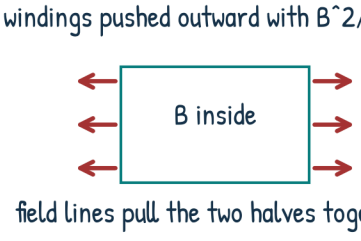
$$u_B = 0.4\ \text{MJ/m}^3$$

At 10 T the pressure is 400 atmospheres: high-field magnets are built like pressure vessels. A running motor still has back-EMF accounting; the magnetic forces transmit mechanical power while B itself does no work on charges — the energy comes from the source through the electric field the source maintains.

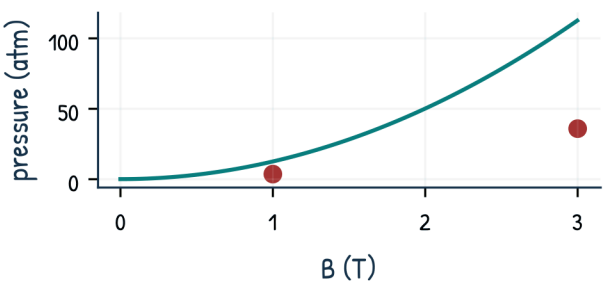
WHY / Pressure scales as B², so doubling field quadruples stress: the practical limit of pulsed and superconducting magnets.

WATCH / A quenching MRI magnet dumps megajoules of field energy into heating; B²/2 mu0 is real stored energy, not a fiction of the formula.

Solenoid hoop stress: the field presses on the current that confines it



Magnetic pressure grows as B squared: 4 atm at 1 T, 36 atm at 3 T



REMEMBER / A field pushes on the currents that confine it with pressure B squared over 2 mu0; a solenoid wears its own field like a pressure vessel.

Rotating charges: the gyromagnetic ratio

Rigid bodies whose charge is distributed like their mass; nonrelativistic rotation.

Ring to body

$$\frac{\mu}{L} = \frac{Q}{2m}, \qquad I = \frac{Q\omega}{2\pi}$$

A spinning charged ring is a current Q omega / 2pi with moment mu = Q omega R squared over 2 and angular momentum L = m R squared omega; the radius cancels in the ratio. A stack of rings with the same charge-to-mass ratio keeps the same mu/L. The gyromagnetic ratio is a property of Q/m alone.

WHY / Universality of Q/2m lets one check any rotating-charge geometry against one ratio.

WATCH / The ratio is classical; electron spin has g = 2 and is not a spinning ball. Quote the anomaly, do not hide it.

$$\frac{\mu}{L} = \frac{q}{2m} \ \text{(uniform)}$$

CHECK / For charge distributed like mass the gyromagnetic ratio is q/2m; the electron spin value is twice it, a quantum fact.

The Bohr magneton

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24}\ \mathrm{A\,m^2}$$

For an electron on an orbit of angular momentum hbar the moment is one Bohr magneton, the natural unit of atomic moments. Paramagnetic salts are measured in mu_B per atom; the Curie law uses the same unit.

WHY / Atomic magnetism is measured in moments that are tiny per atom but enormous per mole.

WATCH / Spin moment is close to mu_B while spin angular momentum is hbar/2: the factor two is the famous g = 2.

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24}\ \mathrm{A\,m^2}$$

CHECK / The Bohr magneton is the natural unit of atomic moments; paramagnetism is measured in multiples of it.

CHECK / Spin g-factor is twice the orbital value. The nuclear magneton is smaller by the proton-to-electron mass ratio.

Rotating disc and shell

$$B_{\text{disc}} = \frac{\mu_0 \sigma \omega R}{2}$$

$$B_{\text{shell}} = \frac{2}{3}\mu_0 \sigma \omega R$$

A rotating charged disc is stacked rings: each ring of width dr contributes mu0 dI / 2r = mu0 sigma omega dr / 2 at the centre, the same for every ring, giving mu0 sigma omega R / 2. A rotating uniformly charged shell gives a uniform interior field two-thirds mu0 sigma omega R, the same surface current as the magnetised sphere.

WHY / Ring stacking is the first real use of the solenoid logic; the shell case previews magnetisation in matter.

WATCH / The disc centre field grows with R at fixed sigma and omega; that is not an approximation error but more current at larger radii.

Numbers

$$\mu = 3.1 \times 10^{-6}\ \mathrm{A\,m^2} \text{ for } 1\ \mu\mathrm{C},\ R = 10\ \mathrm{cm},\ 100\ \mathrm{rev/s}$$

A macroscopic spinning charge is a feeble magnet: its centre field 0.6 nT is a hundred-thousandth of the Earth. At atomic scales the same physics with hbar-sized angular momentum gives mu_B, perfectly measurable in a lab balance of moments.

WHY / Contrasting human and atomic scales explains why magnetism is an atomic phenomenon despite being discovered in lodestones.

WATCH / Charged macroscopic rotors radiate and lose energy; the static formulas assume steady maintained currents.

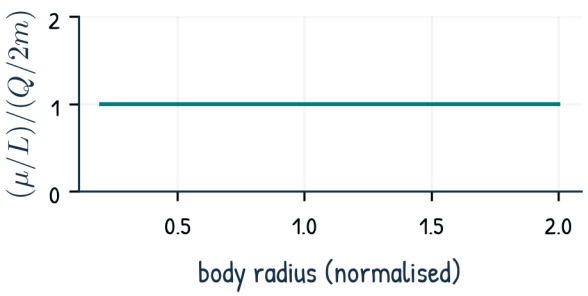
CHECK / A ring of 2 microcoulombs at 50 rev/s on a 5 cm radius: I = 100 microamp, mu = 7.85e-7 A m^2, and mu/L = Q/2m exactly.

Rotating charge: moment and angular momentum share the ratio Q/2m



same mu/L = Q/2m for ring, disc and sphere

The gyromagnetic ratio Q/2m is independent of size and speed



REMEMBER / Any such rotating charged body has mu/L = Q/2m: the same ratio for ring, disc or sphere.

The magnet as a current distribution

Uniform magnetisation M; linear response only where stated; ellipsoids for demagnetising factors.

Bound currents from M

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}, \qquad \mathbf{J}_b = \nabla \times \mathbf{M}$$

A slab magnetised along its normal is a stack of current loops: interiors cancel, the rim does not. Matching the slab moment MAt to a rim loop gives current M per unit height, K_b = M cross n. Where M varies, a volume current curl M remains.

WHY / Every magnetostatics problem in matter becomes a free-current problem once bound currents replace the material.

WATCH / Bound surface current direction follows the right-hand rule with M and the outward normal; flipping the normal flips K.

$$K_b = M, \qquad J_b = \nabla \times \mathbf{M}$$

The magnetised sphere

$$B_{\text{inside}} = \tfrac{2}{3}\mu_0 M$$

$$\mu = \tfrac{4}{3}\pi R^3 M$$

A uniformly magnetised sphere has K_b = M sin theta, the same as a rotating charged shell: uniform interior field two-thirds mu0 M and a pure dipole outside with moment M times volume. A long cylinder instead has B = mu0 M inside - larger because its demagnetising field is smaller.

WHY / Demagnetising factors depend on shape: sphere one third, long cylinder zero, thin plate one. Shape decides the internal field for the same M.

WATCH / The interior sphere field is uniform despite the varying surface current; do not expect a solenoid-like axial falloff inside a sphere.

$$B_{\text{in}} = \tfrac{2}{3}\mu_0 M \ \text{(sphere)}$$

Cutting and poles

cut a magnet: two smaller dipoles

no monopoles

Cutting a bar magnet creates two complete magnets, each with its own current distribution and closed field lines. The pole picture is bookkeeping: north and south are never separated because field lines close through the material.

WHY / Bound-current language never invites monopoles; pole language always does. Prefer currents for derivations.

WATCH / Demagnetising fields of short bar magnets are strong: a short fat magnet magnetises itself less than a long thin one of the same M.

CHECK / Cutting a magnet makes two complete dipole systems; the poles are bookkeeping, not particles, and no bound pole charge exists.

The slab moment ledger

$$M = \mu / \mathcal{V}$$

$$I_b = Mt \ \text{(slab rim current)}$$

A slab of thickness t magnetised along its normal has total moment M A t equal to a single rim loop of current I_b around the edge: I_b = M t, a current M per unit height. Two half-thick slabs stacked give the same total rim current as one full slab.

WHY / The ledger is the cleanest derivation of bound surface current and rehearses the solenoid identity n I = M.

WATCH / The rim current is bound current: no free charge moves and no wires are attached. Confusing it with conduction current breaks the H definition.

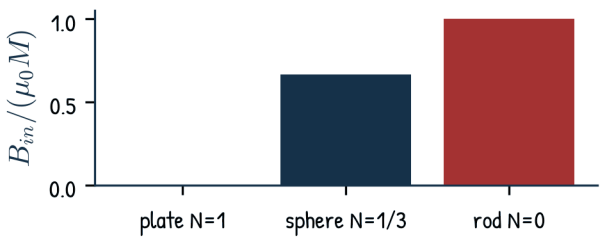
CHECK / The moment ledger: mu = M V with V the volume; a 1 cm sphere at M = 10^6 A/m carries 4.19 A m^2 and matches its exterior dipole exactly.

Uniformly magnetised slab: surface bound current K_b = M cross n



bulk cancels; rim current survives

Demagnetising factor sets the interior field of uniform magnetisation



REMEMBER / Uniform M makes only surface bound current K_b = M cross n; a magnetised cylinder is literally a solenoid with nI = M.

B, H and M inside matter

Linear media M = chi H where stated; ferromagnets use loops, not constants.

The auxiliary field

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}, \qquad \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{free}}$$

Ampères law with all currents is awkward because bound currents depend on the material response. Solving for the free-current circulation defines H: its loop integral counts free current only. B remains the force-giving field; H is a bookkeeping convenience.

WHY / Using H makes magnetic circuits and solenoid problems independent of the material magnetisation details.

WATCH / H is measured in A/m and B in tesla. Numerically similar-looking values in gauss and oersteds come from mixed unit systems; keep SI.

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}), \qquad \mathbf{M} = \chi\mathbf{H}$$

CHECK / H is the field the free currents make; M is the material response; B is what drives charges.

Linear media

$$\mathbf{M} = \chi\mathbf{H}, \qquad \mathbf{B} = \mu_r\mu_0\mathbf{H}, \qquad \mu_r = 1 + \chi$$

For linear isotropic media M follows H with susceptibility chi, and B = (1+chi) mu0 H = mu_r mu0 H. In a long solenoid filled with such material H = nI stays fixed and B multiplies by mu_r. Diamagnets have tiny negative chi, paramagnets tiny positive; iron is neither linear nor constant.

WHY / The pair (H fixed by free current, B amplified by material) is the whole of linear magnetic circuits.

WATCH / mu_r is not a constant for iron: it is a slope on the hysteresis loop and collapses at saturation. Quoting one mu_r for iron is an approximation at a stated operating point.

$$\mu_r = 1 + \chi$$

Units and magnitudes

$$H = nI \ [\text{A/m}]$$

$$\chi_{\text{water}} \approx -9 \times 10^{-6}$$

A 1000 turns per metre solenoid at 1 A has H = 1000 A/m and B = 1.26 mT in vacuum. Water chi about -9e-6, aluminium about 2e-5, iron can reach mu_r of thousands below saturation. Beware oersted-era confusion: 1 A/m corresponds to 4 pi times 1e-7 T in vacuum.

WHY / Practical susceptibility values span ten orders; knowing which side of unity a material sits on prevents order-of-magnitude disasters.

WATCH / Do not insert mu_r into mu0 itself; write B = mu_r mu0 H as a product at the end.

CHECK / H is measured in A/m: a 1000 turn/m solenoid at 0.5 A gives H = 500 A/m and B = 0.63 mT in air.

The gapped circuit

$$B \approx \frac{\mu_0 N I}{g + \ell/\mu_r}$$

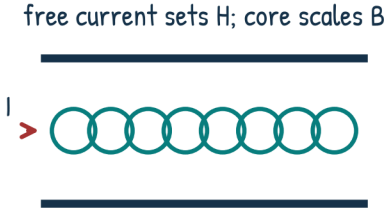
A magnetic circuit of iron length ell with gap g under N ampere-turns has flux set by total reluctance: gap g dominates when mu_r is large. This is how transformer cores and electromagnets are designed: reluctance adds like resistance and the gap controls B.

WHY / Reluctance thinking turns field design into a circuit problem; mu0 NI plays the role of EMF.

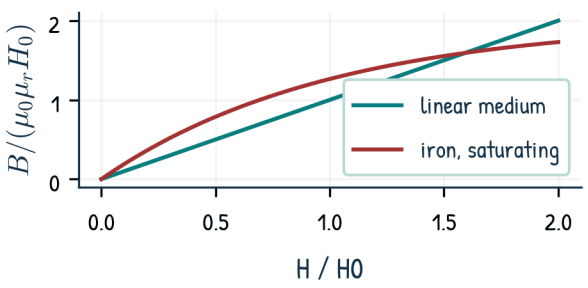
WATCH / If the gap is wide the flux fringes out of the pole area and the simple formula overestimates B; if iron saturates the formula fails differently.

CHECK / Gapped cores: B = mu0 N I / (g + l/mu_r); the air gap dominates once g exceeds l/mu_r, which is how real chokes are designed.

Filled solenoid: free current fixes H while the material scales B



Linear media keep constant mu_r; iron bends towards saturation



REMEMBER / H counts free currents only; B is the real field. In a long solenoid H = nI and B = mu_r mu0 nI.

Dia- and paramagnetism

Closed shells for diamagnetism; independent moments for paramagnetism; weak fields where linearised.

Larmor diamagnetism

$$\omega = \omega_0 \pm \frac{eB}{2m}, \qquad \Delta\mu = -\frac{e^2 r^2}{4m} B$$

An applied field changes orbital speeds so as to oppose itself: one sense of orbit slows, the other speeds up, both reducing the net moment along B. Averaging over orientations gives chi = -mu0 n Z e squared <r squared> / 6 m_e, tiny and negative, independent of temperature.

WHY / Diamagnetism is universal: every material has it; it is visible only where permanent moments are absent.

WATCH / The classical electron orbit model gives the right magnitude but quantum mechanics is the real story; the sign and scaling are what the exam needs.

$$\chi_{\text{dia}} = -\frac{\mu_0 n Z e^2 \langle r^2 \rangle}{6 m_e}$$

Curie paramagnetism

$$M = n\mu \tanh \frac{\mu B}{k_B T}, \qquad \chi = \frac{\mu_0 n \mu^2}{k_B T}$$

Independent atomic moments of size mu align against thermal agitation: magnetisation is the Brillouin-tanh of x = mu B / kT, and for weak fields linear: Curie law chi = C/T. At room temperature a Bohr magneton in 1 T gives x of order 1/300 - weak alignment.

WHY / Curie law is a thermometer as much as a magnetisation law: measuring chi versus T extracts the moment per atom.

WATCH / Curie law holds for independent moments only. Interacting moments make Curie-Weiss, and locked moments make ferromagnets.

$$M = n\mu \tanh \frac{\mu B}{k_B T}, \qquad \chi = \frac{\mu_0 n \mu^2}{k_B T}$$

CHECK / Paramagnetism is alignment against disorder: Curie law at low field, saturation when mu B exceeds k_B T.

Numbers

$$\chi_{\text{dia}} \sim -10^{-5}$$

$$\chi_{\text{para}} = \mu_0 n \mu^2 / k_B T$$

Water -9e-6, copper -1e-5, bismuth -1.7e-4, pyrolytic graphite up to -4e-4 across planes. A paramagnetic salt at 1 K in 1 T can reach substantial M because x = mu B / kT grows as T falls.

WHY / Order-of-magnitude tables are the fastest exam check on a computed susceptibility.

WATCH / Superconductors are the exception with chi = -1, a perfect diamagnet; their levitation is not explicable by the Larmor formula.

CHECK / Water chi about -9e-6, aluminium +2e-5, iron huge; at 1 K and 1 T the paramagnetic argument x = mu B/k_B T reaches 0.67.

Adiabatic demagnetisation

$$T_f = T_i \frac{B_f}{B_i} \quad (\text{fixed } M)$$

Since M depends on B/T only, an isentropic field reduction from Bi to Bf scales the temperature the same way: a salt at 1 K in 1 T demagnetised to 0.01 T reaches 10 mK. The same magnetocaloric effect drives magnetic refrigerators near gadolinium Curie point.

WHY / The one-line scaling is an exam favourite: ideal paramagnet plus adiabatic gives direct temperature ratio.

WATCH / Internal field and interactions set a floor; the ideal ratio fails at millikelvin and near ordering transitions.

CHECK / Adiabatic demagnetisation: order paramagnets at high field and low T, remove the field adiabatically, and the spin temperature falls with the entropy.

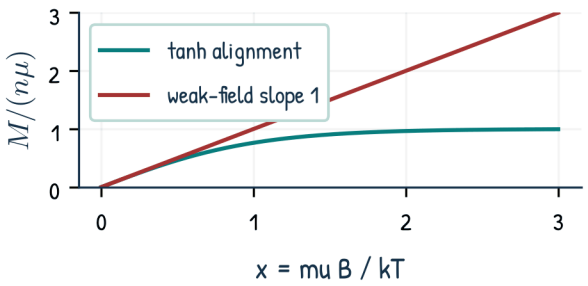
CHECK / Adiabatic demagnetisation reaches millikelvin from a 1 K start.

Atomic orbit in applied B: frequency shifts by the Larmor amount



orbit shift opposes B: Larmor

Paramagnetic alignment: linear at small x, saturated at large x



REMEMBER / Diamagnetism opposes any applied field and is temperature-independent; paramagnetism aligns existing moments and follows Curie law.

Ferromagnetism and hysteresis

Exchange-coupled moments; domain structure; loop area is loss, not theory noise.

Exchange, not dipole coupling

$$k_B T_c \approx \frac{zJ}{3}$$

$$U_{dd} = \frac{\mu_0 \mu_B^2}{4\pi a^3}$$

Two Bohr magnetons a quarter nanometre apart interact magnetically with only 5.5e-25 J, equivalent to 0.04 K: dipole forces cannot order iron at room temperature. Exchange energy J from Coulomb plus Pauli exclusion with z neighbours gives Tc of the right thousand-kelvin order.

WHY / The estimate that rules out the wrong mechanism is worth more than a memorised Curie temperature.

WATCH / Exchange is quantum mechanical and has no classical dipole derivation; quote it as the mechanism and estimate its size.

$$T_c \sim \frac{zJ}{3k_B}$$

Domains and hysteresis

$$w_{\text{cycle}} = \oint H\, dB$$

$$\text{area} = \text{loss per volume per cycle}$$

Domains are regions of aligned moments; walls move and moments rotate under H. Trained material follows a loop: B lags H, remanence is what stays when H returns to zero, coercivity is what must be reversed to erase it. The loop area in the (H,B) plane equals energy lost per unit volume per cycle.

WHY / Hysteresis is thermodynamics with a picture: source work minus stored energy is heat in the walls.

WATCH / Soft and hard materials are not weak and strong magnets; they are narrow and wide loops - transformers u

Numbers

$$B_s(\text{Fe}) = 2.1\ \text{T}, \quad T_c = 1043\ \text{K}$$

$$\text{loss} = \oint H\, dB$$

Silicon steel loses about 1 J per kilogram per cycle; a 20 kg core at 50 Hz with 150 J/m³ loop area dissipates 20 W. Iron saturates near 2.1 T: no circuit formula can deliver more. Curie-Weiss chi = C/(T-Tc) describes the approach from the paramagnetic side.

WHY / Loop area times frequency times volume is the engineering loss formula every transformer question uses.

WATCH / Above Tc the domains vanish; applying Curie law below Tc inverts the actual physics.

CHECK / Silicon steel at 1.2 T and 50 Hz loses about 1.5 W/kg; a 20 kg core at 60 Hz dissipates roughly 20 W of hysteresis heat.

Soft versus hard

$$\text{soft: narrow loop, low } H_c$$

$$\text{hard: wide loop, high } H_c$$

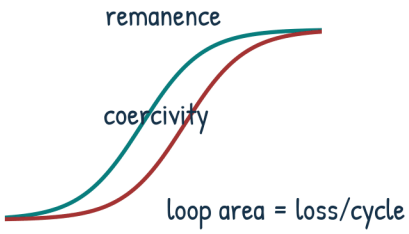
Transformer cores want narrow loops and laminations to cut eddy losses; permanent magnets want wide loops so demagnetising fields do not erase them. A hard material in a core wastes its loop area as heat every cycle.

WHY / Choosing a magnetic material is choosing a loop shape for a duty, not choosing a bigger mu_r.

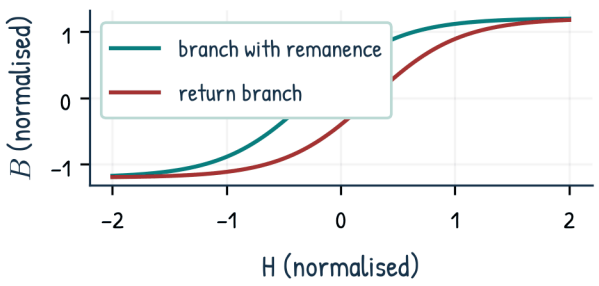
WATCH / Laminations interrupt eddy currents whose loss scales as thickness squared; solid cores run hot at power frequency.

CHECK / Soft for transformer cores and shields, hard for permanent magnets; above the Curie temperature a ferromagnet becomes paramagnetic abruptly.

Domains wall motion leaves B lagging H: the hysteresis loop



Hysteresis loop: remanence at H=0 and coercivity at B=0



REMEMBER / Iron locks moments with exchange, not dipole coupling; its hysteresis loop is both memory and energy loss.

Materials in circuits: cores, shields and forces

Linear approximations below saturation; gap-dominated reluctance; soft iron for channelling.

Cores and shielding

$$B = \mu_r \mu_0 n I$$

$$\text{shield: } \mu_r \gg 1 \text{ around region}$$

An iron core multiplies the solenoid field at the same free current; a closed iron shell shields its interior by guiding flux around it. The same construction builds electromagnets, transformer cores and magnetic screens. Saturation at 2.1 T caps the multiplication.

WHY / Magnetic circuits obey a reluctance version of Ohm law: ampere-turns drive flux through total reluctance.

WATCH / Shielding works only for the field components the flux path intercepts; uniform static fields are hard to cancel with partial shells.

$$B = \mu_r \mu_0 n I \text{ (below } B_s \approx 2.1 \text{ T)}$$

Forces on materials

$$F = \frac{B^2 A}{2\mu_0}$$

$$\text{toward stronger field when aligned}$$

A magnetisable body in a gradient is pulled toward stronger field, roughly with stress B²/2 mu0 over the pole area: the paperclip jump. A superconducting plate supports the same stress in repulsion; a 2 kg magnet over 4 cm² needs 0.35 T at the plate.

WHY / Force from energy density in the gap is one line and agrees with virtual work at fixed current.

WATCH / Saturation caps iron force: once M cannot grow, B in the gap stops scaling and levitation saturates.

CHECK / Induced dipoles are attracted to field maxima either way; diamagnets and superconductors are pushed toward weaker field.

The gapped electromagnet

$$B \approx \frac{\mu_0 N I}{g + \ell/\mu_r}, \qquad \text{reluctance} = \frac{\ell}{\mu A}$$

The gap dominates when mu_r is large: B is set almost entirely by ampere-turns over gap width. Fringing adds effective gap; saturation subtracts effective mu_r. Design means deciding which limit the magnet sits in.

WHY / Reluctance in series is the design equation of motors, speakers and MRI yokes.

WATCH / Never leave the gap out of a magnetic circuit calculation; iron reluctance is usually negligible, gap reluctance is not.

CHECK / The gapped electromagnet: mu0 N I over gap plus reluctance of the iron path; fringe fields at wide gaps steal flux from the pole face.

Saturation caps every circuit

$$B_s(\text{Fe}) \approx 2.1 \text{ T}$$

$$\mu_r \to 1 \text{ as } B \to B_s$$

Once iron saturates, mu_r collapses toward unity and extra ampere-turns buy almost no flux: the gapped formula fails upward. Electromagnet lifting force stops growing with current and transformer cores distort waveforms. Design means staying below the knee of the B(H) curve with margin.

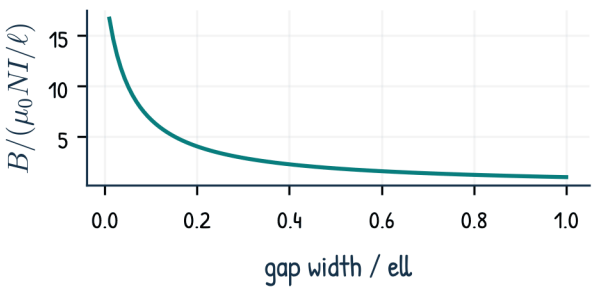
WHY / Saturation is the engineering limit behind every magnetic circuit number; the linear formulas are local, not global.

WATCH / Heating raises losses and shifts the knee; pulsed magnets may exploit deep saturation briefly but cannot hold it.

Magnetic circuit: ampere-turns drive flux through iron and gap in series



Gap-dominated circuit: B falls roughly as the inverse gap width



REMEMBER / Iron channels flux like a wire channels current; the gap sets B; and field gradients pull iron towards stronger field.

The Earth as a dipole

Central tilted dipole plus stated local anomalies; slow secular variation.

The three angles

$$B_H = B\cos\theta_{\text{dip}}, \quad B_V = B\sin\theta_{\text{dip}}$$

$$\tan\theta_{\text{dip}} = \frac{B_V}{B_H}$$

Declination is compass deviation from true north, east or west; dip is the tilt of a freely pivoted needle below the horizontal. The horizontal component is what compasses and tangent galvanometers use; total field across India runs 25 to 65 microtesla. The magnetic pole near geographic north is a south pole of the dipole.

WHY / Angles and components answer different instruments; read the instrument, then take the matching component.

WATCH / Dip is not declination, and the magnetic pole near the Arctic is not a north pole of the dipole.

$$B_H = B\cos\theta_{\text{dip}}, \quad B_V = B\sin\theta_{\text{dip}}$$

Dipole geometry

$$\tan\theta_{\text{dip}} = 2\tan\lambda$$

$$B = B_{\text{eq}}\sqrt{1+3\sin^2\lambda}$$

At magnetic latitude lambda the radial component is 2 cos-like and the tangential one sine-like in the dipole field, so the dip angle satisfies tan(dip) = 2 tan(lambda). Field strength grows from equator to pole as sqrt(1+3 sin^2 lambda) times the equatorial value.

WHY / One identity links a station dip to its latitude and checks internal consistency of field tables.

WATCH / Local crustal anomalies shift both angles; the dipole relation is a model, and departures are real data.

$$\tan\theta_{\text{dip}} = 2\tan\lambda$$

Measuring B_H

$$B_{\text{coil}} = B_H\tan\theta$$

$$T = 2\pi\sqrt{I_m/\mu B_H}$$

A tangent galvanometer in the magnetic meridian deflects a compass by theta with B_coil = B_H tan theta. A swinging magnet gives the product mu B_H; a deflected compass gives the ratio mu / B_H. Together they give both unknowns - Gauss absolute method.

WHY / Absolute magnetic measurement needs no external magnet reference: geometry, mass and time suffice.

WATCH / Coil must lie in the magnetic meridian; misalignment measures a component and biases theta.

CHECK / Tangent galvanometer: B_coil = B_H tan theta; oscillation gives mu B_H. Both measure the horizontal component alone.

Earth and core numbers

$$\mu_\oplus \simeq 8\times10^{22}\ \text{A m}^2$$

$$I_{\text{core}} \sim 2\times10^{9}\ \text{A}$$

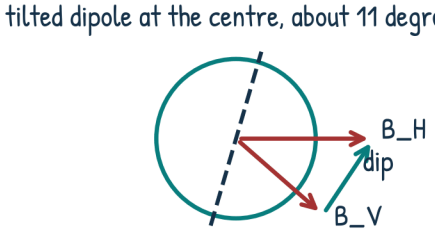
Equatorial surface field about 30 microtesla gives the dipole moment from mu = B_eq R cubed over 1e-7 in SI. A core-sized loop of radius 3.5e6 m would need two billion amperes: liquid-iron convection in a dynamo, decaying in a few 10^4 years if not maintained.

WHY / A one-line estimate shows why the field needs a dynamo and has persisted for aeons.

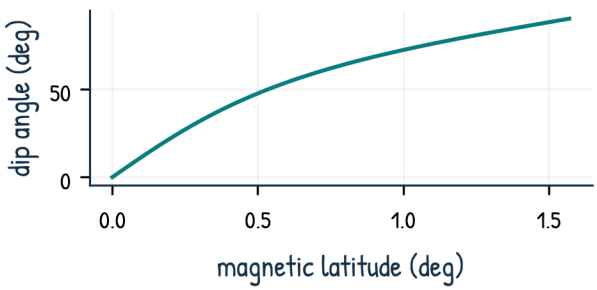
WATCH / The dipole tilts about 11 degrees from the rotation axis and wanders; the geomagnetic poles are not fixed points.

CHECK / B about 25 to 65 microtesla over the globe; the dipole moment 8e22 A m^2 needs a core current of about 2 GA to sustain it.

Station field components: B_H horizontal, B_V vertical, dip their angle



Dip angle grows from 0 at the equator to 90 at the pole



REMEMBER / Declination, dip and the two components describe a station; tan(dip) = 2 tan(latitude) is the dipole identity.

Flux: definition, sign and the changing bookkeeping

Flat loop or coil; uniform B or a known profile; normal chosen once and kept.

The flux integral

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{A} = BA\cos\theta$$

For a uniform field through a flat loop of area A the flux is BA cos theta with theta between B and the chosen normal: full BA face-on, zero edge-on, negative when the field crosses against the normal. A coil of N turns links N times the flux of one turn; the weber is one tesla square metre, and one weber per second is one volt.

WHY / The surface is any surface bounded by the loop; no-monopoles guarantees they all give the same answer.

WATCH / Reversing the normal reverses Phi and every EMF computed from it. Fix the normal with the right-hand rule on the first line of the solution and never revisit it.

Non-uniform fields

$$\Phi = \frac{\mu_0 I a}{2\pi}\ln\frac{d+b}{d}$$

A rectangle a by b with its near side at distance d from a long straight wire: integrate mu0 I a dr / 2 pi r across the strip. The logarithm is the flux of the wire through the loop and simultaneously the mutual inductance of a wire and a loop per unit current.

WHY / One integral, two readings: flux now, mutual inductance later. Keep the geometry exact and the later sections inherit it.

WATCH / The integral runs across the loop width b, not along it; swapping a and b silently changes the answer.

CHECK / At d = 1 cm, b = 5 cm, a = 10 cm and 10 A the flux is 3.2 micro-weber: millimetre geometry makes microweber flux.

Faraday and the charge that flows

$$\mathcal{E} = -N\frac{d\Phi}{dt}, \qquad q = \frac{\Delta\Phi}{R}$$

An induced EMF appears whenever the flux through the circuit changes and never while it is steady, however large. The EMF is work per unit charge around the loop; in a resistance R it drives i = E/R, and the total charge passed during a flux change is Delta Phi over R, independent of how fast the change happened.

WHY / Charge depends only on the flux change; rate affects current and power but not the integrated charge: the flip-coil and the ballistic galvanometer.

WATCH / Steady flux means no EMF even at tesla levels; only the derivative counts. And q = Delta Phi / R needs the circuit resistance R, not the load alone.

$$q = \frac{N\Delta\Phi}{R}$$

The sign protocol

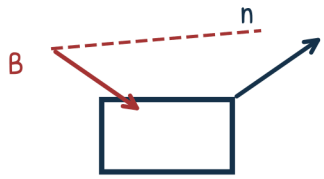
$$\mathcal{E} = -\frac{d\Phi}{dt} \text{ (four steps)}$$

Choose the positive direction round the loop; let the right hand give the normal; compute Phi with that normal and its sign; compute dPhi/dt with its sign; then E = -dPhi/dt decides the current direction. The final Lenz check: the induced current opposes the change of flux. Worked drill: field into the page grows 0.2 to 0.6 T in 2 s with clockwise positive gives E negative, so the current runs anticlockwise, making outward flux against the growth.

WHY / Every sign error in the chapter is a skipped step; the protocol is Lenz law made mechanical and it exits with an independent check.

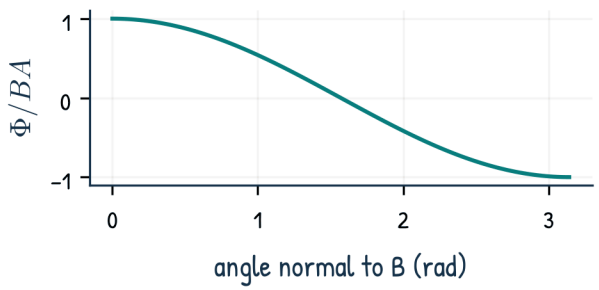
WATCH / Do not flip the normal halfway through to make a sign come out; recompute instead. If the Lenz check fails, the error is in step three or four.

Loop flux: the angle between the field and the normal decides everything



$$\Phi = BA\cos\theta$$

Flux through the loop: BA face-on, zero edge-on, negative when reversed



REMEMBER / Flux is a scalar with a sign; choose the normal once, keep it, and the sign of every EMF follows.

Faraday experiments and the loop rule

Circuits of known resistance; flux changes by motion, rotation, deformation or switching.

The four experiments

$$\mathcal{E} = -N\,d\Phi/dt \text{ in all four}$$

A magnet moved into a coil deflects a galvanometer one way; moved out, the other way; a second coil switched on gives a single flick; a loop rotated in a steady field gives a sinusoid. Same law, four mechanisms: motion of the source, motion of the circuit, switching, rotation. None produces EMF while the flux is constant.

WHY / Recognising which of the four a problem is decides what to differentiate: a position, a current, or an angle.

WATCH / The flick on switching lasts only while the current builds; after the transient the flux is steady and the meter returns to zero.

CHECK / The induced current direction always opposes the change: growth is opposed, decay is supported. Say opposes the change, not opposes the field.

The charge protocol

$$i = \frac{\mathcal{E}}{R} = -\frac{N}{R}\frac{d\Phi}{dt}$$

Integrate the current over the change: q = N Delta Phi / R whatever the history. A 100-turn coil of 50 cm^2 flipped in 0.1 s from face-on to face-on reversed in 0.5 T passes q = 2 Delta Phi N / R weber-turns over R; measure the kick on a ballistic meter and you have B without timing the flip.

WHY / Ballistic measurements exploit that charge forgets the rate; current measurements do not.

WATCH / If the resistance changes during the flip the simple formula fails; integrate the actual i dt instead.

$$\bar{\mathcal{E}} = N\frac{\Delta\Phi}{\Delta t} \text{ (average over the flip)}$$

CHECK / A 100-turn coil of 50 cm2 flipped in 0.5 T passes q = 2 N Delta Phi / R. The ballistic galvanometer reads the charge, not the current.

EMF around the loop

$$\oint \mathbf{E}\cdot d\mathbf{l} = \mathcal{E} \text{ (total EMF)}$$

The EMF is the line integral of the total electric field around the circuit, magnetic and electrostatic parts together. In a uniform-resistance loop the current makes the same fraction of the EMF appear across each segment; in a circuit with a battery the two EMFs add or oppose by their directions around the loop.

WHY / Kirchhoff with induction: sum of EMFs equals sum of IR drops; the induced EMF enters as a distributed source around the loop.

WATCH / An induced EMF distributed around a loop cannot be assigned to one point; only the potential drops of the ohmic parts are local.

What the law asserts and assumes

$$\mathcal{E} = -\frac{\partial}{\partial t}\int \mathbf{B}\cdot d\mathbf{A} \;+\; \oint (\mathbf{v}\times\mathbf{B})\cdot d\mathbf{l}$$

Faraday law in general has two pieces: a changing field through a stationary circuit, and a moving circuit sweeping flux. The compact flux rule packages both whenever the circuit moves with its material; when the boundary slides through the conductor the package fails and only the charges obey a universal law.

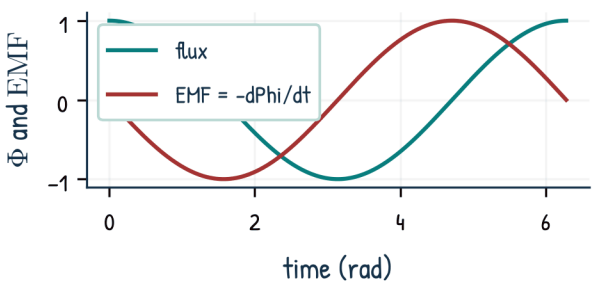
WHY / Separating the two mechanisms before substituting is what keeps the sign and the physics straight in mixed problems.

WATCH / The flux rule is a theorem with hypotheses, not a definition; a sliding contact or a spinning disc needs the charges treatment.

The magnet drives the flux; the meter sees only the rate of change



Flux and EMF a quarter-cycle apart: peak EMF where flux vanishes



REMEMBER / A current appears only while the flux changes; the four experiments differ in mechanism and agree in law.

Motional EMF: the rod as a battery

Conductor moving through uniform B; v, B and the rod mutually perpendicular unless decomposed.

From the Lorentz force

$\mathcal{E} = Blv$

Free electrons in the moving rod feel q v cross B along the rod and pile up at one end until their electrostatic field balances the magnetic push: qE = qvB gives E = vB inside the rod and a potential difference Blv between the ends. The rod is a battery whose positive terminal is the end toward which v cross B pushes positive charge.

WHY / No circuit and no flux were used: the physics is the force on the carriers, valid for a lone rod in space.

WATCH / The polarity comes from v cross B for positive carriers; never from the current direction elsewhere in the circuit.

CHECK / A 50 cm rod at 5 m/s in 0.4 T gives 1.0 V; open-circuit, it stands ready with one volt across its ends.

The same result from flux

$\left|\frac{d\Phi}{dt}\right| = Blv = |\mathcal{E}|$

With the rod sliding on rails the enclosed area grows at rate lv, so the flux rule gives Blv as well. The two derivations agree because the loop moves with its material; the Lorentz derivation is the physics, the flux rule is the shortcut that packages it.

WHY / Two derivations of one number is the check; a disagreement means the flux rule was applied outside its hypotheses.

WATCH / The flux derivation needs a closed moving loop; the rod alone has EMF between its ends whether or not anything closes the circuit.

$d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$

CHECK / A 30 cm rod at 5 m/s in 0.4 T gives 0.6 V.

General form and decomposition

$$\mathcal{E} = \int_{\text{rod}} (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

For a rod at an angle to v or B only the component of v cross B along the rod counts. A rod rotating about one end sweeps elements at different speeds; a rod moving parallel to B generates nothing; a rod moving perpendicular to its own length but at 45 degrees to B takes the sine factor into the integral.

WHY / The general integral replaces every special formula; the special formulas are its evaluations.

WATCH / Blv sin theta uses the angle between v and B; the angle between rod and v cross B is a separate factor. Name both.

CHECK / Blv at an angle: only the component of v cross B along the rod contributes; decompose before integrating.

Where the flux rule needs care

follow the charges: $d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$

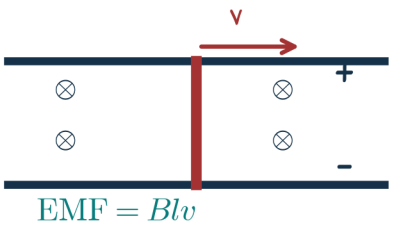
When the circuit boundary slides through the conductor the flux through the circuit is ambiguous: a Faraday disc, a rolling wheel with a rim contact, a contact jumping wires. Compute the force on the charges that actually move; choosing a circuit that rotates with the material restores the flux rule exactly.

WHY / The flux rule is exact for loops that carry their material; the Lorentz force is exact always.

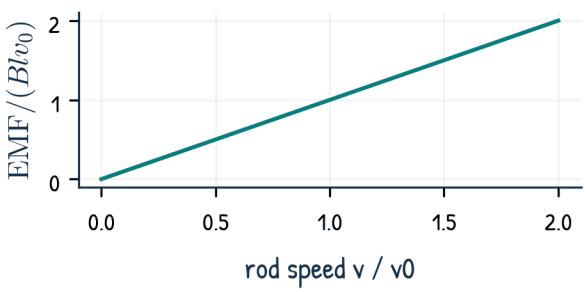
WATCH / A spinning axisymmetric magnet with a still disc gives zero EMF; spin the disc and the magnet may stay or turn with it: only conductor motion counts.

CHECK / Spinning only the axisymmetric magnet gives zero. The Faraday disc is the flux-rule embarrassment and the Lorentz-force easy victory.

A moving rod in a B field: charge is driven along it like a battery



Motional EMF grows linearly with speed: the rod is a battery of Blv



REMEMBER / A moving rod is a battery of EMF Blv with its positive terminal at the v cross B end; no flux argument is required.

Lenz law as energy conservation

Any induction problem with a movable part or a changing flux; steady fields.

Statement and algorithm

$$\text{induced B opposes } \Delta\Phi$$

Find the change of flux and its sense; the induced current makes a field opposing the change, adding flux when flux is lost and subtracting when it grows; the right-hand rule gives the current; and the mechanical consequence, force or torque, opposes the motion causing the change.

WHY / Three steps: change, opposing field, mechanical reaction. The reaction is what makes every generator need a torque to turn.

WATCH / Lenz opposes the change, not the field: a decreasing inward flux is supported by an induced inward current. Say opposes the change every time.

CHECK / Closing a switch on a nearby coil: the induced current in the second coil opposes the growth; opening it, the current tries to sustain what is being lost.

The energy audit

$$W_{\text{mech}} = \int i^2 R\,dt + \Delta U_{\text{field}}$$

If the induced current attracted the approaching magnet, the magnet would accelerate, the flux would change faster, the current would grow, and energy would appear from nothing. With the correct sign the mechanical work done against the force equals the heat produced: the audit balances and the perpetual machine dies.

WHY / Write the ledger before solving: mechanical work in, heat in R, field energy change. It catches every wrong sign.

WATCH / The audit includes the field energy change when currents are still building; ignoring it is the standard way to lose a minus sign.

The magnet in the copper tube

$$v_t \propto \frac{m}{\mu^2}, \qquad F = Kv$$

A magnet dropped through a conducting tube induces currents in every ring of the wall whose fields oppose the fall; the drag is proportional to speed and the magnet reaches a terminal velocity of tenths of a metre per second, falling like a feather through the tube.

WHY / Lenz law you can hold: the demonstration survives every scaling argument because the drag law is linear at these speeds.

WATCH / The magnet does not stop dead; it approaches a terminal speed. Saying the currents halt the magnet confuses drag with a wall.

CHECK / Halving the tube radius slows the fall sixteen times: the drag scales as one over a to the fourth for a thin wall.

Action and reaction

$$\mathbf{F}_{\text{on rod}} = -\mathbf{F}_{\text{on source}}$$

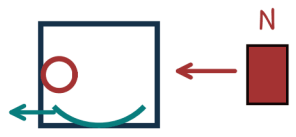
The force on the moving part always opposes the motion relative to the field; the equal and opposite force acts on whatever produces the field. In a generator the crank feels this reaction; in a motor the field feels the reaction of the drive force.

WHY / Lenz law is Newton third law written for induction; the pair is always mechanical force and electromagnetic reaction.

WATCH / The reaction is on the source of B, not on the field itself; track where the momentum goes before writing the balance.

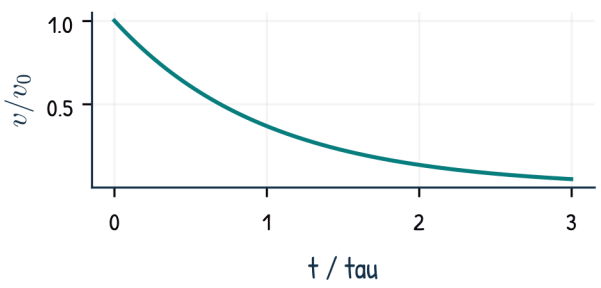
CHECK / The reaction pair is always mechanical force on the moving part and torque on the source of B. Track the momentum before writing the balance.

Lenz law: the induced current always pushes back against what produced it



induced I opposes the change

Lenz drag decays the speed exponentially: tau = mR over B squared l squared



REMEMBER / The induced current opposes the change of flux; that sign is the only one consistent with energy conservation.

The rod on rails: coasting and driven

Parallel rails distance l apart, uniform B perpendicular, resistance R; frictionless unless stated.

Coasting: the exponential

$$v = v_0 e^{-t/\tau}, \qquad \tau = \frac{mR}{B^2 l^2}$$

A rod given v0 generates Blv, current Blv/R and retarding force B squared l squared v over R; Newton gives exponential decay with tau = mR over B squared l squared. The rod travels a finite distance v0 tau and dissipates exactly its initial kinetic energy as heat.

WHY / One time constant governs the whole family: larger mass or resistance slows the decay, larger B squares it away.

WATCH / The rod coasts forever in theory but stops within a few tau in practice; the stopping distance is v0 tau, not v0 times some stopping time.

$$x_\infty = v_0\tau, \qquad \tfrac{1}{2}mv_0^2 = \int i^2 R\,dt$$

Driven: terminal speed

$$v = v_t(1 - e^{-t/\tau}), \qquad v_t = \frac{FR}{B^2 l^2}$$

A constant force F drives the rod against the same magnetic drag; the speed approaches the terminal value FR over B squared l squared where the applied power Fv is entirely converted to heat. Same tau, same curve shape, inverted: the driven problem is the coasting problem plus a constant.

WHY / Both exponentials are the same first-order system seen from two ends; after one tau each has covered 63 per cent of the gap.

WATCH / The terminal current is F over Bl, not F over R; the mechanical force balance sets the current, and R then sets the speed.

The energy ledger

$$Fv_t = i^2 R, \qquad \tfrac{1}{2}(m+M)v_t^2 \text{ approached}$$

At terminal speed every watt of mechanical input is ohmic heat; during the transient the missing part goes into kinetic energy. For the coasting rod the ledger is one-shot: initial kinetic energy equals total Joule heating, provable by integrating B squared l squared v squared over R.

WHY / The ledger is the physics; the exponential is just the solution. Check both against each other.

WATCH / With friction the split between rail heat and resistor heat depends on the whole history; only the total is fixed by energy.

$$P_{\text{mech}} = Fv, \qquad P_{\text{elec}} = \mathcal{E}i = i^2 R$$

CHECK / At terminal speed the applied power F v equals the Joule heat exactly. The transient stores the missing part as kinetic energy.

Time constants compared

$$\tau_{\text{rail}} = \frac{mR}{B^2 l^2}, \qquad \tau_{\text{RL}} = \frac{L}{R}$$

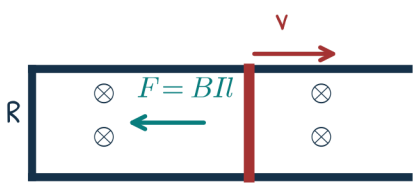
The rod time constant is mechanical mass over a magnetic drag coefficient; the RL constant is inductance over resistance. They are the same mathematics: both are first-order approaches to a steady state, and both scale linearly with the inertia and the damping inverse.

WHY / Recognising one equation in two disguises is the fastest route to the answer in mixed rod-circuit problems.

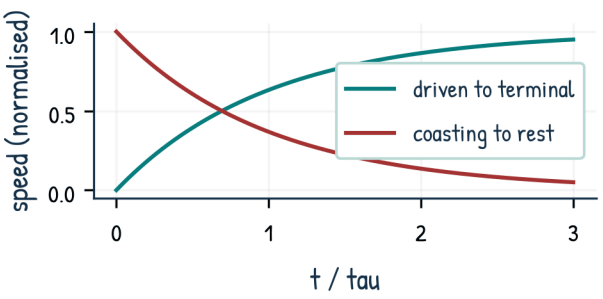
WATCH / The rail tau contains no inductance and the RL tau contains no mass; conflating them gives seconds where milliseconds belong.

CHECK / The rail time constant has no inductance and the RL constant has no mass. Conflating them gives seconds where milliseconds belong.

Rails close the circuit: current flows, and the rod feels the drag force



The rod family: one time constant, decay or approach seen from two ends



REMEMBER / The rod family is one first-order equation: exponential decay to rest or exponential approach to a terminal speed.

Rotating rods, discs, coils and loops

Uniform B perpendicular to the rotation axis; constant omega; thin conductors.

The rotating rod

$$\mathcal{E} = \tfrac{1}{2} B\omega L^2$$

A rod of length L turning about one end sweeps elements at v = omega r; each contributes B omega r dr and the integral is half B omega L squared, the same as a rod moving at the average speed. A rod pivoted at its centre has zero EMF between its ends but one eighth B omega L squared from centre to either end.

WHY / Average-speed thinking reproduces the integral: the midpoint speed times length is the whole answer.

WATCH / The pivot choice matters: end-pivot gives half B omega L squared, centre-pivot zero end to end. Read the geometry before substituting.

CHECK / A 0.5 m rod at 100 rad/s in 0.5 T gives 6.25 V; the flux swept per second is the same number in webers.

The Faraday disc

$$\mathcal{E} = \tfrac{1}{2} B\omega a^2$$

A spinning disc with brushes at axle and rim is a stack of rotating rods in parallel: a unipolar generator delivering enormous current at a fraction of a volt. Its circuit has no fixed loop; follow the charges or choose a radial line rotating with the disc to make the flux rule work.

WHY / The disc is the classic flux-rule embarrassment and the Lorentz-force easy victory; examiners love it precisely for that.

WATCH / Spinning only the magnet gives zero: an axisymmetric field is unchanged by its own rotation and no charges move through it.

CHECK / 10 cm disc at 300 rad/s in 0.5 T: 0.75 V and, into a milliohm, 750 A with a 1.9 N m braking torque.

The rotating coil

$$\mathcal{E} = NBA\omega \sin \omega t, \qquad \mathcal{E}_0 = NBA\omega$$

A coil rotating about its axis in uniform B has Phi = NBA cos omega t and the EMF is the sine, largest when the flux is zero and the sides cut field fastest, zero when the flux peaks. This is the AC generator and the sinusoid every AC section assumes.

WHY / The quarter-cycle offset between flux and EMF is the single most-missed fact: maximum flux means zero EMF.

WATCH / Do not argue EMF is maximum because flux is maximum; differentiate the cosine and watch the sine vanish there.

CHECK / 100 turns, 100 cm^2, 0.2 T at 50 Hz: E0 = 62.8 V and RMS 44 V: the household scale in one line.

CHECK / A 50 cm disc at 300 rad/s in 0.5 T delivers 0.75 V. Into a milliohm this is 750 A with 1.9 N m braking torque.

Loops entering and leaving fields

$$\mathcal{E} = Blv \text{ while straddling}$$

$$\mathcal{E} = 0 \text{ wholly in or out}$$

A rectangular loop pulled out of a field region has motional EMF only in the side still inside; wholly inside or outside the two long sides cancel and the flux is constant. A loop sitting still in a changing field has transformer EMF A dB/dt instead: no motion, no motional EMF.

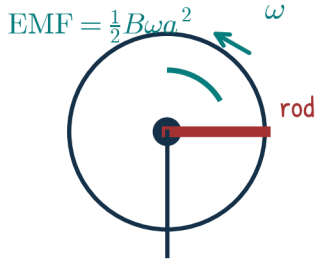
WHY / Two EMF species in one picture: motional from motion through B, transformer from changing B through a still loop.

WATCH / While the loop is wholly inside a uniform field there is no current, even though the field is large; only the boundary crossing matters.

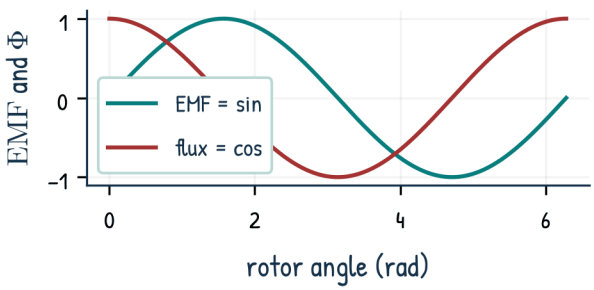
$$\mathcal{E} = -A\tfrac{dB}{dt} \text{ (still loop, changing field)}$$

CHECK / The meter spikes only while an edge crosses the boundary: square pulses separated by quiet gaps.

A rotating rod sweeps area at rate half B omega a squared: the disc generator



Rotating coil: EMF largest at zero flux, zero at maximum flux



REMEMBER / Rotation converts a steady field into a sinusoid or a DC voltage: half B omega L squared for a rod, NBA omega for a coil.

The rod with every attachment

Rails joined through R, C, L, a spring or a hanging mass; constant B; one-dimensional motion.

Hanging mass

$$(m+M)\dot{v} = Mg - \frac{B^2l^2}{R}v$$

A hanging mass M pulls the rod through the pulley; the pair accelerates under Mg minus magnetic drag, reaching terminal speed MgR over B squared l squared with time constant (m+M)R over B squared l squared. At terminal speed the falling weight supplies exactly the resistor heat.

WHY / The combined mass sets the approach time; only the hanging weight sets the terminal speed.

WATCH / The tension is not Mg during acceleration; write Newton for the pair or subtract the two equations, never both inconsistently.

CHECK / M = 50 g on the 0.5 T circuit gives terminal 4.9 m/s with tau = 1.5 s: half again the coasting time constant.

CHECK / Tension is not Mg during acceleration.

Capacitor instead of resistor

$$a = \frac{F}{m + CB^2l^2}, \qquad i = CBla \text{ constant}$$

With a capacitor there is no steady current; charge q = CBlv grows with the speed and the current is proportional to the acceleration. The magnetic force CB squared l squared dv/dt acts as an added inertia, so the rod accelerates constantly at F over m plus CB squared l squared. No terminal speed exists.

WHY / The capacitor rod is a mass problem with extra mass; the current is fixed by the acceleration, not by the speed.

WATCH / Writing i = Blv/R for a capacitor circuit is the classic error; there is no resistor and the current is set by CBl dv/dt.

$$q = C\mathcal{E} = CBlv, \qquad i = CBl\frac{dv}{dt}$$

CHECK / B = 2 T, l = 0.5 m, C = 0.1 F, m = 0.1 kg under 1 N: added mass 0.1 kg, a = 5 m/s^2, constant current 0.5 A.

Friction and spring

$$m\ddot{x} + \frac{B^2l^2}{R}\dot{x} + kx = 0$$

Kinetic friction adds a constant opposing force: the coasting rod then stops in finite time, unlike pure magnetic drag. A spring makes a damped oscillator with damping constant B squared l squared over R; replace R by C and the damping disappears into extra mass, leaving slower undamped SHM.

WHY / A velocity-proportional drag never stops a body in finite time; a constant one does. That distinction is the whole friction problem.

WATCH / With a spring the natural frequency is sqrt(k/m) only when the magnetic drag is absent; the damping shifts both amplitude and, lightly, the frequency.

$$\omega = \sqrt{\frac{k}{m + CB^2l^2}} \quad \text{(capacitor + spring)}$$

Inductor on the rails

$$\omega = \frac{Bl}{\sqrt{mL_{\text{ind}}}}$$

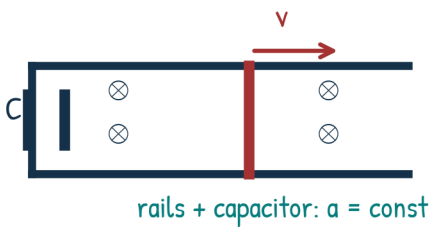
Rails joined through a pure inductor make the rod oscillate for ever: Blv = L di/dt and m dv/dt = -Bi l combine into SHM with omega = Bl over root(mL). The kinetic energy of the rod exchanges with the inductor field energy: a mass-spring oscillator with the roles renamed.

WHY / The rod plays the capacitor in this LC circuit; the analogy table of the inductance sheets reads the answer directly.

WATCH / No resistance means no decay; adding even a small R turns the oscillation into a ring-down with time constant 2m/b.

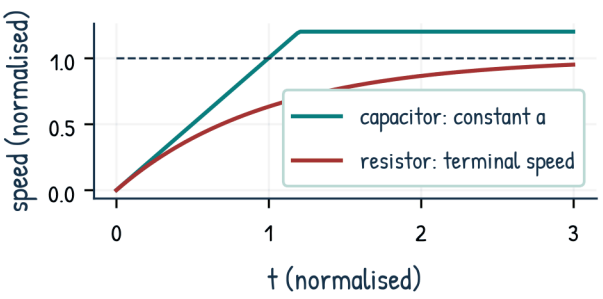
CHECK / The loop equation and Newton combine by differentiation; the first integral is the energy ledger, not another differential equation.

Swap the terminal element and the whole motion changes: R, C or L



rails + capacitor: a = const

Resistor gives a terminal speed; capacitor gives constant acceleration; inductor gives SHM



REMEMBER / Each attachment changes the equation of motion: drag, added mass, damping, or oscillation.

Induced electric fields

Axisymmetric changing flux, especially a long solenoid of radius R with uniform dB/dt inside.

The field with no conductor

$$\oint \mathbf{E}\cdot d\mathbf{l} = -\frac{d\Phi}{dt}$$

Faraday law without a circuit: a changing B creates an electric field whose circulation around any closed path equals minus the rate of change of flux through it. The field lines close on themselves around the changing region; it is non-conservative and has no potential difference between points.

WHY / This is the field the electrostatics rules forbade: closed field lines, non-zero circulation, no potential.

WATCH / The electrostatic conductor argument fails here; a conductor in a changing flux has circulating induced E and an EMF, not an equipotential body.

$$\oint \mathbf{E}\cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{A}$$

The solenoid geometry

$$E_{\text{in}} = \frac{r}{2}\dot{B}, \qquad E_{\text{out}} = \frac{R^2}{2r}\dot{B}$$

By symmetry the induced field is azimuthal; a circle of radius r gives E times 2 pi r equal to the enclosed flux rate. Inside, E grows linearly to half R dB/dt at the winding; outside it falls as 1/r where the flux rate is the full solenoid value even though B itself is zero there.

WHY / Outside the solenoid there is E with no B: the flux through the loop, not the field on it, is what counts.

WATCH / The outside field is not zero just because B is zero there; the enclosed flux still changes. Use the full solenoid flux rate for r above R.

CHECK / R = 5 cm at 100 T/s: E is 1.0 V/m at 2 cm, 2.5 V/m at the winding, 1.25 V/m at 10 cm.

CHECK / The profile rises linearly inside and falls as 1/r outside. Maximum E sits at the winding itself.

A charge accelerated by the induced field

$$v = \frac{q\Phi_f}{2\pi r m} \ \text{(ring impulse)}$$

A bead on a frictionless ring around the changing flux feels tangential force qE; for r beyond R the impulse per unit charge round the ring is the flux change divided by the circumference, whatever the rate. Free to move on a straight track it accelerates along the track tangent at qE.

WHY / The impulse depends on the total flux change, not its rate: the betatron principle in embryo.

WATCH / The energy gained per revolution is q times the EMF around the ring; the rate of gain depends on dPhi/dt but the per-turn gain does not.

CHECK / A proton on a 5 cm ring outside a 3 cm solenoid gains 1.3 m/s from a 1e-4 Wb flux rise: rate cancels, impulse does not.

Why no potential exists

$$\oint \mathbf{E}\cdot d\mathbf{l} \neq 0 \ \Rightarrow \ \text{no } V$$

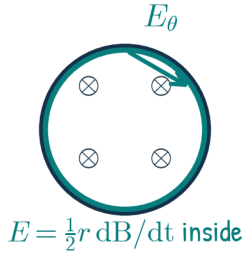
A potential exists only when the circulation vanishes for every loop; here loops enclosing changing flux have circulation minus dPhi/dt. The line integral of E between two points depends on the path by exactly the enclosed flux rate, which is why the word EMF exists and why voltmeters disagree near changing flux.

WHY / The voltmeter paradox of the later sheet is this one equation read twice.

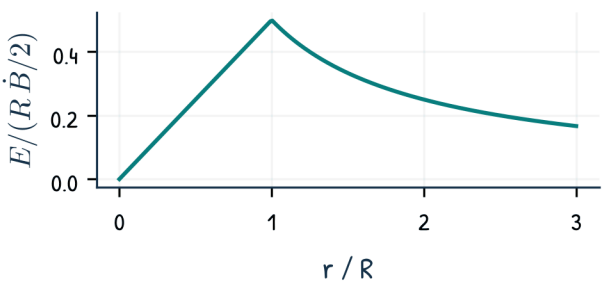
WATCH / Speaking of the potential difference between two points of an inductive loop is undefined until the path is named; use EMF along a stated path.

CHECK / Two points on the loop have no unique potential difference.

A changing B field makes electric loops even with no wire present



Induced E grows linearly inside the solenoid and falls as 1/r outside: peak at the winding



REMEMBER / A changing magnetic field makes an electric field that circles the flux; it has no potential and exists where B is zero.

The betatron: accelerate with a changing flux

Axisymmetric field profile; electrons at fixed orbit radius R; nonrelativistic derivation, relativistic-safe result.

Guiding and accelerating, two rates

$$p = eB_{\text{orb}}R, \qquad \dot{p} = eE = \frac{eR}{2}\frac{d\langle B\rangle}{dt}$$

To keep an electron of momentum p on a fixed radius R the orbit field must satisfy p = e B_orb R, so its rate of change must match the momentum gain. The induced field on the orbit is R/2 times the average-field rate; equating the two rates gives the field at the orbit rising at half the average rate.

WHY / Two demands on one magnet: bend and accelerate. Equating their rates is the design condition.

WATCH / Do not confuse the field at the orbit with the average over the disc; the factor of two is exactly that distinction.

$$B_{\text{orb}} = \tfrac{1}{2}\langle B\rangle \ \text{(2:1 condition)}$$

The 2:1 condition

$$\langle B\rangle = \frac{\Phi}{\pi R^2} = 2B_{\text{orb}}$$

Pole pieces are shaped so the field is stronger inside the orbit than on it. A uniform field fails: momentum then grows only half as fast as the orbit condition demands and the electrons spiral inward. The condition holds at every instant of the ramp, not just on average over the cycle.

WHY / A uniform-field betatron is a contradiction in terms; the shaping is the machine.

WATCH / The classical cyclotron fails for electrons because of relativity; the betatron works because it tracks p = eBR, valid at any speed.

CHECK / R = 0.5 m with B_orb ramped to 0.5 T in 5 ms: final pc = 75 MeV after 4.8e5 turns of about 157 eV each.

The stability window

$$B \propto r^{-n}, \qquad 0 < n < 1$$

Radial stability needs the available focusing force to fall off more slowly than the 1/r demand: n below 1. Vertical stability needs the field lines to bow outward: n above 0. Weak focusing lives in the window between; the same n keeps the 2:1 profile satisfied across the orbit.

WHY / One exponent decides whether the beam stays put; machine design is this inequality made of steel.

WATCH / A perfectly uniform field (n = 0) fails vertical stability too: the beam is free to drift out of the mid-plane.

CHECK / Displace an electron outward at fixed p: centripetal demand falls as 1/r, supply as r^-n; restoring needs n below 1.

CHECK / Displace the orbit outward at fixed p: the demand falls as 1/r while the supply falls as r to the minus n.

Energy accounting

$$e\frac{d\Phi}{dt}\ \text{per turn}, \qquad K_{\max} = eB_{\text{orb}}cR\ \text{(relativistic)}$$

Each turn the electron gains e times the flux rate, about 157 eV in the worked machine; the total after half a million turns is the 75 MeV set by e B_orb R c. The argument used p throughout, which is why relativity costs nothing here.

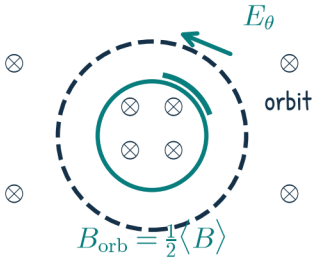
WHY / Accounting by turns checks the ramp: gain per turn times turns must equal the final p c.

WATCH / The betatron does not need a fixed frequency; the orbit period changes as the electron speeds up and nothing resonates.

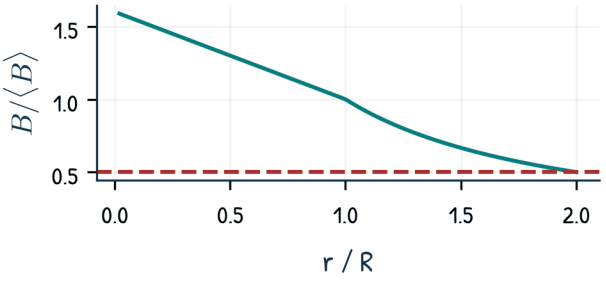
$$\Delta\Phi = 2\pi R\frac{p}{e} = \pi R^2\langle B\rangle$$

CHECK / The argument used p throughout, which is why the betatron works for electrons where the cyclotron fails. Gains of 157 eV per turn over 4.8e5 turns give 75 MeV.

Betatron: induced E accelerates the beam while the 2:1 rule holds the orbit



Betatron poles: average field inside is twice the field on the orbit (dashed 2:1)



REMEMBER / The average field inside the orbit must be twice the field on it: the 2:1 condition.

Eddy currents: braking, heating and laminating

Conductors in non-uniform or changing fields; ohmic metal of conductivity sigma and density rho_m.

Magnetic braking

$$F = k\sigma B^2 V_{\text{eff}}\, v, \qquad \tau = \frac{m}{k\sigma B^2 V_{\text{eff}}}$$

A plate moving through a field edge drives current density sigma v B through the metal; dissipated power fixes a retarding force proportional to speed. The velocity decays exponentially with a time constant depending on material and field, not on the plate size: tau about rho_m over k sigma B squared.

WHY / The brake is a velocity-proportional drag: it never stops a body in finite time by itself, only exponentially.

WATCH / Constant friction stops in finite time; eddy drag does not. Keep the two regimes distinct in every stopping problem.

CHECK / Copper in 0.5 T with k = 0.1 gives tau about 6 ms; aluminium, with half the rho/sigma, stops twice as fast.

The slotted pendulum

$$P \sim \sigma B^2 v^2 V_{\text{eff}}$$

A solid copper bob swinging between magnet poles stops within one swing; slot the bob so the eddy paths are cut, k collapses, and it swings freely. The demonstration is the lamination principle in one motion: interrupt the current loops and the loss dies.

WHY / Every eddy-current remedy is the same move: break the loop paths with slots, laminations or gaps.

WATCH / The slot must cut across the current direction; a slot parallel to the currents changes nothing.

CHECK / A 50 g bob at 0.5 T is far past critical damping: it returns to rest without overshooting.

CHECK / The slotted bob keeps its mass and loses its eddy paths. At 0.02 T the bob keeps about 8 visible swings; at 0.5 T it stops in one.

Induction heating and skin depth

$$P/V \propto \sigma f^2 B^2, \qquad \delta = \sqrt{\frac{2}{\mu_0 \mu_r \sigma \omega}}$$

An alternating field drives eddy currents whose power grows with frequency squared until the skin effect confines them to a depth delta. At 25 kHz iron has delta about 0.1 mm: the current concentrates in a thin high-resistance skin and the pan heats. Aluminium at the same frequency has delta about 0.5 mm and mostly reflects the field.

WHY / Induction hobs need ferromagnetic pans not for coupling but for resistance: the skin must be thin.

WATCH / The power law holds below the skin-effect regime only; quoting f squared at megahertz without delta is a trap.

$$\delta_{\text{Fe}}(25\ \text{kHz}) \approx 0.1\ \text{mm}$$

CHECK / The power law holds below the skin-effect regime only. Iron at 25 kHz has 0.1 mm skin: thin and resistive, which is why the pan heats.

Laminations: the d squared law

$$\frac{P}{V} = \frac{\sigma \omega^2 B_0^2 d^2}{24}$$

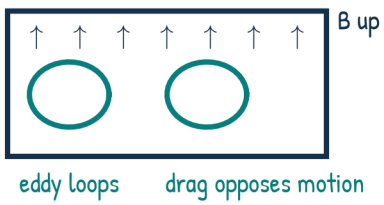
A core slab of thickness d carrying B0 cos omega t parallel to its faces develops E = x dB/dt at depth x; averaging sigma E squared gives power per volume proportional to d squared. Cutting the core into sheets ten times thinner reduces eddy loss a hundredfold: 0.35 mm silicon steel at 1.2 T and 50 Hz loses 0.19 W/kg where the solid core would glow.

WHY / The square law is why transformer and motor cores are laminated, painted and packed.

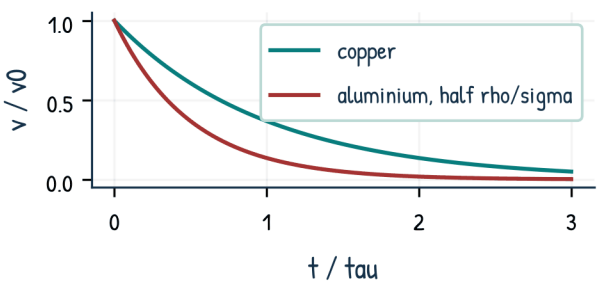
WATCH / Hysteresis loss is the other half of core heating and scales with frequency, not with d squared; the two losses have different remedies.

CHECK / Same core solid at 2 cm thick: 4.7 MW/m^3 against 1.45 kW/m^3 laminated — a factor of 3300.

Eddy currents circulate in bulk metal; slots break the loops and brake less



Eddy braking is exponential; the material ratio rho_m over sigma sets the time



REMEMBER / Eddy currents oppose the change that makes them; their drag is proportional to speed and their heat scales with the square of thickness.

Generators, motors and back-EMF

Coils in uniform B driven at constant omega; armature resistance R; steady load.

The AC generator

$$\mathcal{E} = NBA\omega \sin \omega t$$

A rotating coil delivers a sinusoid of peak NBA omega; the EMF is largest when the sides cut field fastest, which is when the flux through the coil is zero. Large power stations turn a magnet inside fixed coils at 50 or 60 Hz; the frequency is set by the turbine speed and the pole count.

WHY / Faraday plus rotation is the whole generator; the phase offset between flux and EMF is the exam favourite.

WATCH / Maximum EMF at zero flux, not maximum flux; this inversion is checked by differentiating the flux.

$$V_{\text{rms}} = \frac{NBA\omega}{\sqrt{2}}$$

CHECK / Peak EMF occurs at zero flux: the sides cut field fastest when the coil is edge-on. Differentiate the cosine to see the sine vanish at maximum flux.

Commutation and motors

$$V = \mathcal{E} + IR \ (\text{motor}), \qquad V = \mathcal{E}_{\text{gen}} - IR$$

A DC motor uses a commutator to flip the coil connections twice per turn; a DC generator the same device in reverse. In a running motor the back-EMF opposes the supply: V minus E equals IR, so the current is set by how fast the motor turns, not by R alone.

WHY / Back-EMF is Faraday fighting the supply; the armature current is the difference divided by R.

WATCH / A stalled motor has zero back-EMF and draws V/R: 24 A for a 12 V motor at 0.5 ohm, enough to burn it in seconds.

CHECK / 12 V motor with 0.5 ohm: 24 A at start, 4 A running at back-EMF 10 V. The stall current is the design limit.

CHECK / A stalled motor draws V / R: 24 A for a 12 V motor at 0.5 ohm.

Why a generator gets harder to turn

$$\tau_{\text{mech}} = \frac{\mathcal{E}i}{\omega}, \qquad P_{\text{mech}} = \mathcal{E}i = i^2(R + R_{\text{int}})$$

The armature current in the field feels a torque opposing the rotation; the crank supplies exactly the electrical power delivered plus the coil heat. Open the circuit and the crank turns freely; short it and it is hardest of all. A brighter bulb means a smaller load resistance, more current and more torque.

WHY / The hand on the crank pays for every watt: mechanical input equals electrical output plus losses, identically.

WATCH / The opposition torque scales with the current; confusing it with friction loses the whole load dependence.

$$\mathcal{E}i = i^2 R_L + i^2 R_{\text{int}} \ (\text{energy audit})$$

CHECK / A generator feeding a battery pushes against its voltage: the shaft supplies exactly E times I.

Regenerative braking and pickups

$$\mathcal{E} = Blv \ \Rightarrow \ i = \frac{\mathcal{E}}{R}$$

Run the traction motor as a generator and the car recharges its battery while the current torque slows it: Lenz law as fuel saving. A guitar pickup is a magnet and coil around a steel string whose vibration changes the flux at the string frequency; a dynamic microphone is the same coil run backwards.

WHY / Devices are chapters in hardware: motor, generator and pickup are one Lorentz-force family.

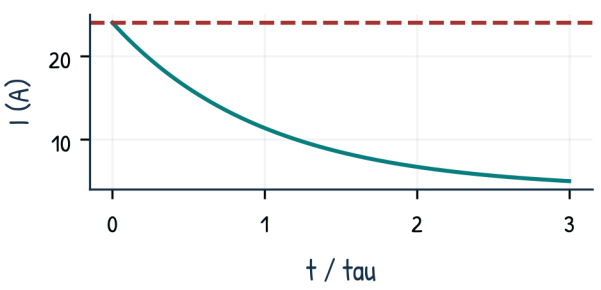
WATCH / A pickup needs a ferromagnetic string to concentrate the flux; nylon strings give almost no signal.

CHECK / The loudspeaker is the motor principle on a coil in a radial field; reverse the current and it becomes the microphone.

A motor is a generator running backwards: back-EMF limits the current



Start-up current falls from V/R = 24 A to the running 4 A as back-EMF builds



REMEMBER / A generator under load is hard to turn because its own current reacts on the crank; a motor under load slows because back-EMF falls.

The voltmeter paradox and everyday induction

Loops containing changing flux; real voltmeter leads forming their own loops.

Two voltmeters, two readings

$$\oint \mathbf{E} \cdot d\mathbf{l} = 1.0 \text{ V (lead loop)}$$

A loop of two resistors around a changing flux carries one current; a voltmeter whose leads bow left encloses part of the flux and reads the IR drop minus that EMF; leads bowing right enclose the rest and read the opposite residual. Both readings are correct line integrals of E along their own paths.

WHY / The voltmeter measures the line integral along its leads; with induction the answer depends on the path.

WATCH / There is no error in the meters and no paradox in the physics: the question the meters answer is path-dependent by construction.

EMF is not a potential difference

$$V_{AB} = \int_A^B \mathbf{E} \cdot d\mathbf{l} \text{ (path-dependent)}$$

Between two points of an inductive circuit the integral of E depends on which way the path goes around, by exactly the enclosed flux rate. Potential difference exists only where the circulation vanishes; with changing flux it does not, and the word EMF names what does exist.

WHY / The word potential should not be used for an induced EMF; the correct object is an EMF along a named path.

WATCH / Keep voltmeter leads twisted and away from changing flux, or accept that you are measuring the flux.

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} \neq V_+ - V_-$$

Induction around us

$$\mathcal{E} = -N\frac{d\Phi}{dt} \text{ in every device}$$

Metal detectors read eddy-current loading of a search coil; induction hobs heat the pan as a shorted secondary; wireless charging is a transformer with an air gap; maglev lifts via induced track currents; regenerative braking runs the motor backwards. One law, a dozen devices.

WHY / Every device is one of the four Faraday experiments in production form.

WATCH / Device vocabulary hides the physics; translate each demo into flux change before computing.

CHECK / The coin detector signal falls as one over distance to the sixth: dipole coupling, not the inverse square law.

CHECK / Large search coils see deep and coarse, small coils fine and shallow: the detectorist trades depth for resolution.

The energy path

$$S = EH \text{ (Poynting: energy through the gap)}$$

No charge crosses a transformer; the energy travels in the fields around the core. The wires guide the Poynting flux from source to load; in a DC circuit too the energy flows through the air around the wire and enters the resistor through its surface.

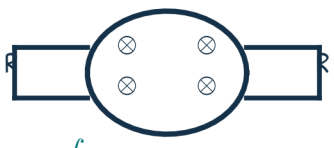
WHY / Field energy transport is the honest picture; the wire is a guide, not a conveyor belt.

WATCH / Asking where the energy is in a circuit: in the fields, with the sources and sinks at the ends.

CHECK / The same E cross H flux carries power along a coaxial cable at the speed of the wave front.

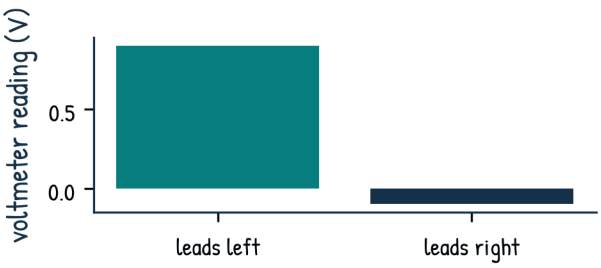
CHECK / The Poynting vector near a coax points along the gap and integrates to exactly EI at every cross-section.

Two paths around a changing flux: Kirchhoff is not enough, Faraday is



$V = \int E \cdot dl$ is path dependent

Two voltmeters on the same points read different path integrals when flux is changing



REMEMBER / With changing flux inside a circuit, two voltmeters on the same two points can legitimately read different numbers.

Flux linkage and self-inductance

Coils of N turns in vacuum or linear media; steady geometry; current I.

The definition

$$L = \frac{N\Phi}{I}, \qquad \mathcal{E}_{\text{self}} = -L\frac{dI}{dt}$$

Close a switch and the current does not jump: the growing flux induces an EMF opposing the growth. Open it and the collapsing flux drives the current on, the spark. The henry is one weber per ampere: a 1 H coil makes 1 V when its current changes at 1 A/s.

WHY / Inductance is electrical inertia; the definition mirrors the capacitor but resists current change instead of voltage.

WATCH / L = N Phi / I with Phi the flux through ONE turn: N enters twice, hence L grows as N squared.

$$1\ \text{H} = 1\ \text{Wb A}^{-1} = 1\ \text{V s A}^{-1}$$

CHECK / The 10 mH coil on 20 V through 10 ohm settles in about a millisecond; resistance alone sets the final 2 A.

Geometry alone

$$L \propto N^2 \ \text{(same shape)}$$

Because B is proportional to I for steady currents in vacuum, L is pure geometry: turns, size, shape. A 200-turn coil of the same shape has four times the inductance of a 100-turn coil. The exception is a ferromagnetic core whose mu depends on B; iron-cored L falls as the core saturates.

WHY / Scaling by N squared answers half the exam questions without any integration.

WATCH / Saturating iron makes L current-dependent; quote the small-signal value and state the assumption.

CHECK / Doubling the turns doubles the field and doubles the linkage: the square law is two factors of N.

CHECK / Iron-cored L falls as the core saturates: quote the small-signal value and state the assumption. A 200-turn coil has four times the inductance of a 100-turn coil.

Where the energy lives

$$U = \tfrac{1}{2}LI^2 = \int \frac{B^2}{2\mu_0}\,dV$$

The energy is in the field, not the wire: half L I squared equals the B squared over 2 mu0 integrated over all space. A gapped core stores most of its energy in the gap even though the winding is on the iron: field energy follows B squared times volume.

WHY / Saying the energy is in the coil is the bookkeeping error behind most wrong gap estimates.

WATCH / A small inductance can store large energy at large current; energy scales as I squared, inductance linearly.

CHECK / A 10 H magnet at 500 A stores 1.25 MJ: the kinetic energy of a 1.5-tonne car at 150 km/h.

CHECK / Doubling the gap roughly doubles the stored energy at the same flux before saturation: the gap buys headroom.

Real inductors

$$L_{\text{real}} = L_{\text{ideal}}(1 + \text{parasitics})$$

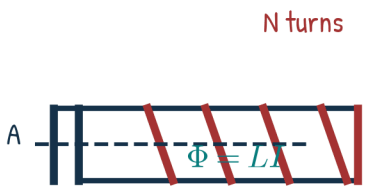
Every real coil has winding resistance in series and turn-to-turn capacitance in parallel; the two resonate at a self-resonant frequency above which the inductor behaves as a capacitor. Cores add saturation and hysteresis; high-frequency coils are air-cored or ferrite, small and wound to minimise capacitance.

WHY / The ideal formulas are exact for idealised geometry and a good first estimate; the parasitics decide the usable band.

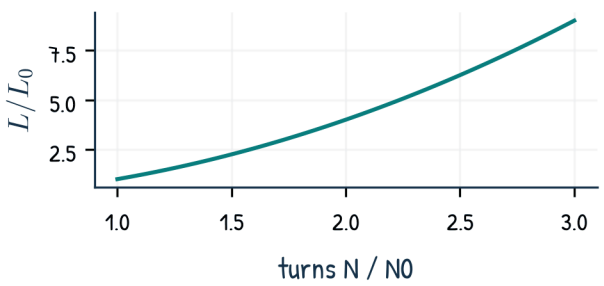
WATCH / Above self-resonance the impedance rises again: the component has stopped being an inductor.

$$f_{\text{self}} = \frac{1}{2\pi\sqrt{LC_{\text{parasitic}}}}$$

One turn owns flux Phi = LI; N turns own N times it: the N squared law



Inductance grows as N squared: N makes the field and N links it



REMEMBER / A coil resists changes in its own current; its measure is the flux linkage per ampere: L = N Phi / I.

Computing L: four standard geometries

Long solenoid, toroid, coaxial cable, two-wire line; vacuum or uniform core.

The long solenoid

$$L = \mu_0 n^2 A\ell = \frac{\mu_0 N^2 A}{\ell}$$

B = mu0 n I links each of N turns through area A: L = mu0 n squared A l. The per-litre view L/V = mu0 n squared is the useful thinking: a given winding density gives a given inductance per unit volume. With a mu_r core multiply, below saturation.

WHY / Compute the flux through one turn, multiply by N, divide by I; the same recipe works for every geometry.

WATCH / Using n instead of N in the numerator is the common slip; check dimensions: mu0 n squared A l has henrys.

CHECK / 1000 turns on 20 cm of 2 cm radius: L = 7.9 mH in air.

CHECK / The per-litre view L / V = mu0 n squared gives inductance per winding volume.

Toroid and coaxial cable

$$L' = \frac{\mu_0}{2\pi}\ln\frac{b}{a}$$

The toroid is the solenoid bent shut: L = mu0 N squared A / 2 pi r, with no external field. The coax cable has flux per length mu0 I ln(b/a) / 2 pi in its gap annulus: L prime = mu0 ln(b/a) / 2 pi, about 0.32 microhenry per metre for b/a = 5.

WHY / Both are one flux integral; the toroid hides its ends and the coax hides its return path.

WATCH / The cable inductance omits the flux inside the conductors, a small low-frequency correction of mu0 / 8 pi per conductor.

$$L_{\text{toroid}} = \frac{\mu_0 N^2 A}{2\pi r}$$

CHECK / The toroid confines every field line inside the winding: zero in the hole and outside. The coax puts all its flux in the gap annulus.

Two-wire line and single loop

$$L'_{\text{2wire}} = \frac{\mu_0}{\pi}\ln\frac{d}{a}, \qquad L_{\text{loop}} \approx \mu_0 R\left[\ln\frac{8R}{a} - 2\right]$$

A two-wire line of spacing d and wire radius a has mu0 ln(d/a) / pi per length; a single loop of radius R and wire radius a has L about mu0 R times the logarithm of 8R/a minus 2: 0.59 microhenry for R = 10 cm and a = 1 mm. Every loop is about a microhenry per metre of wire.

WHY / The logarithm is the signature of flux inside a geometry with one small dimension.

WATCH / The single-loop formula diverges logarithmically as the wire thins: real answers always quote the wire radius.

CHECK / Doubling the coax outer radius adds only 0.14 microhenry per metre: cables of very different sizes have similar impedances.

CHECK / A single loop diverges logarithmically as the wire thins; real answers always quote the wire radius.

L prime C prime and the speed

$$L'C' = \mu_0\varepsilon_0 = \frac{1}{c^2}, \qquad Z_0 = \sqrt{\frac{L'}{C'}}$$

With the capacitors note C prime = 2 pi eps0 / ln(b/a) for the same cable: the product L prime C prime is mu0 eps0, so signals travel at the speed of light in the dielectric, independent of geometry. The ratio gives the characteristic impedance 60 ln(b/a) ohms divided by root kappa.

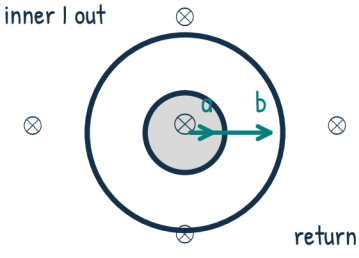
WHY / The cable pair L prime, C prime carries the whole transmission-line story: speed from the product, impedance from the ratio.

WATCH / Only the impedance depends on b/a; the speed depends on the dielectric alone.

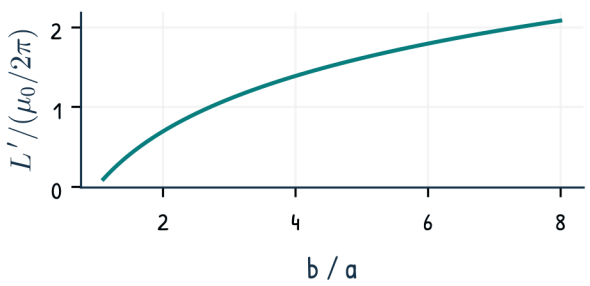
$$Z_0 = 60\ln\frac{b}{a}\ \Omega\ \text{(vacuum)}$$

CHECK / Going from b over a = 2 to b over a = 4 adds just 60 ln 2 = 42 ohm: the geometry factor is logarithmic.

Coax: flux between the conductors stores the inductance per metre



Cable inductance per length grows only logarithmically with the outer radius



REMEMBER / Inductance is flux linkage per ampere computed from the geometry: mu0 n squared A l for a solenoid.

Mutual inductance and coupling

Two coils; linear media; current in one linking flux through the other.

Definition and reciprocity

$$M = \frac{N_2 \Phi_{21}}{I_1}, \qquad \mathcal{E}_2 = -M\frac{dI_1}{dt}$$

A current in coil 1 links flux through coil 2; linkage per ampere is M. Reciprocity M12 = M21 is a theorem of the double integral of dl1 dot dl2 over r, symmetric in the two loops. Compute the easy direction: current in the long solenoid, flux through the small coil, never the reverse.

WHY / Reciprocity converts hard problems into easy ones; the small coil inside a long solenoid is the archetype.

WATCH / The EMF in the secondary opposes the CHANGE of primary current; a steady primary current gives no secondary EMF.

CHECK / M is bounded by root(L1 L2): perfect coupling on a shared core, less in air.

CHECK / Reciprocity lets you compute M the easy way round: current in the long solenoid, flux through the small coil.

Coaxial solenoids

$$M = \mu_0 n_1 N_2 A_1 = k\sqrt{L_1 L_2}$$

A second winding over a length of a long solenoid links B1 = mu0 n1 I1 through N2 turns of area A1: M = mu0 n1 N2 A1. With both windings sharing the same flux path M reaches root(L1 L2); the coupling coefficient k measures the fraction of flux shared.

WHY / One formula and one bound; every transformer question is somewhere between them.

WATCH / A transformer needs k near 1 on a shared iron core; air-coupled side-by-side coils run k = 0.1 to 0.5.

$$0 \le k \le 1, \qquad M = k\sqrt{L_1 L_2}$$

CHECK / Both windings sharing one flux path give M = root(L1 L2). Air-coupled side-by-side coils run k = 0.1 to 0.5; a shared iron core approaches 1.

When M vanishes

$$M = 0 \ \text{(perpendicular axes, aligned centres)}$$

Two coils whose axes are perpendicular with coincident centres share no flux however close: M = 0. Radio circuits use this to decouple coils. A rectangle beside a long wire has M = mu0 a ln((d+b)/d) / 2 pi, computed from the flux integral of the flux sheet.

WHY / The zero-coupling geometry is the standard true/false question; the answer follows from the flux, not the distance.

WATCH / M depends on orientation, not proximity; moving coils closer does not help if the flux threading is zero.

CHECK / Near a straight wire the mutual inductance is the flux per ampere: one integral answers both readings.

CHECK / Orthogonal pick-up coils on the same axis have M = 0 to first order: this is how receivers reject cross-talk.

Energy of coupled coils

$$U = \tfrac{1}{2}L_1 I_1^2 + \tfrac{1}{2}L_2 I_2^2 \pm M I_1 I_2$$

Two coils carrying I1 and I2 store their self energies plus a mutual term with the sign of the flux linkage. Holding currents constant the force on a movable coil is plus one half I squared dM/dx; holding flux constant the same force appears with the source disconnected.

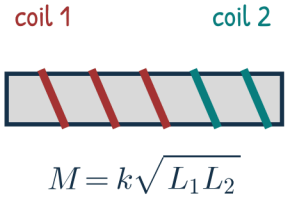
WHY / The sign of the mutual term is the sign of the shared flux: aiding currents add energy, opposing subtract.

WATCH / The force law needs the CURRENTS fixed by sources for the plus sign; isolated coils conserve flux and the bookkeeping flips.

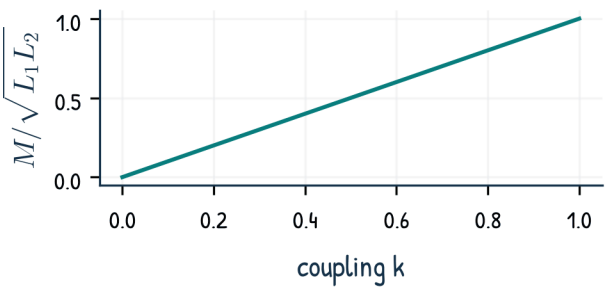
$$F = \tfrac{1}{2} I^2 \frac{dM}{dx} \ \text{(fixed currents)}$$

CHECK / The sign of the mutual term follows the shared flux. Aiding currents add energy, opposing subtract; measuring both series forms yields M without geometry.

Shared flux on a core links the coils; perpendicular ones decouple



Mutual inductance is k times the geometric mean: 1 on a shared core, 0 perpendicular



REMEMBER / Mutual inductance is reciprocity: M12 = M21, so compute it the easy way round.

The RL circuit and the inductive kick

Series R and L with battery V; switch at t = 0; no mutual coupling.

Rise: the exponential

$$i = \frac{V}{R}(1 - e^{-t/\tau}), \qquad \tau = \frac{L}{R}$$

Kirchhoff gives V = iR + L di/dt; the current rises from zero to V/R with time constant L/R. The inductor takes the full voltage at first and none at last; the resistor does the opposite. Check by substitution: L di/dt and iR sum to V at every instant.

WHY / Initial and final states first, then the exponential joins them: the inductor is an open circuit at t = 0 and a wire at infinity.

WATCH / The time constant uses the resistance the current ACTUALLY flows through after the switch; a different R before and after is the standard trap.

$$V_L = Ve^{-t/\tau}, \qquad iR = V(1 - e^{-t/\tau})$$

CHECK / 10 mH and 5 ohm on 10 V: tau = 2 ms, final 2 A, half-current after 1.4 ms.

Decay

$$i = i_0 e^{-t/\tau}, \qquad \int i^2 R\,dt = \tfrac{1}{2}Li_0^2$$

Short the battery out and the inductor drives the decaying current; its voltage reverses and the stored half L I squared appears as heat in R. The integral of i squared R equals the initial field energy exactly: the audit closes.

WHY / The inductor becomes the source during decay; the polarity of its voltage flips.

WATCH / Never leave the current path open when switching off a large inductance; give it a path or pay the voltage.

$$V_L = -L\frac{di}{dt} \text{ (driving during decay)}$$

CHECK / A freewheel diode gives the decaying current a safe path: the same L over R physics, with no arc at the switch.

The inductive kick

$$\mathcal{E} \sim \frac{Li_0}{\Delta t}$$

Interrupt 2 A in 10 mH within a microsecond and 20 kV appears across the opening contacts: the air breaks down, an arc carries the current until the energy is spent, and the contacts erode. The flyback diode gives the current a harmless path; the snubber capacitor turns the kick into an LC ring.

WHY / The energy always goes somewhere: arc, diode loop, snubber, or spark. Name the victim in every switching problem.

WATCH / The kick is a voltage spike from a small energy; it is the short time, not the joules, that breaks down the gap.

CHECK / An ignition coil exploits the same spike with a 1:100 turns ratio: 20 to 30 kV at the spark plug.

Slow charging of big magnets

$$\frac{dI}{dt} = \frac{V}{L}, \qquad t_{\text{charge}} = \frac{LI}{V}$$

Charging time scales as L times I over V: doubling the target current doubles the wait, whatever the resistance. Discharging faster than the supply allows needs somewhere to put megajoules; a quench dump resistor is that somewhere. The current rate is set by V/L, not by R.

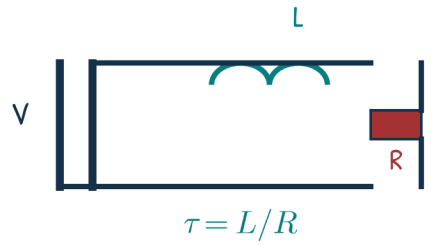
WHY / Large inductances make slow machines; the ramp rate is a design parameter, not an afterthought.

WATCH / With superconducting windings R is zero and the steady state is whatever the current reached; tau = L/R is meaningless there.

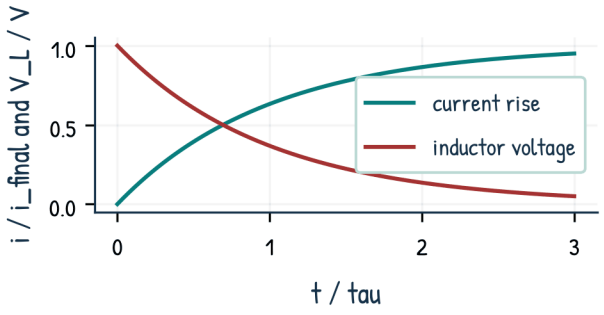
CHECK / The stored half L I squared at 500 A in 10 H is 1.25 MJ; the dump resistor must swallow it as heat.

CHECK / A superconducting magnet of 10 H charged from 10 V rises at 1 A/s: 500 A takes eight minutes.

The inductor fights change: current takes L/R seconds to make up its mind



RL transient: current rises as the inductor voltage falls; same exponential both ways



REMEMBER / Current cannot jump in an inductor; the exponential joins the states at rate tau = L/R.

Energy in the magnetic field

Linear media; long solenoid or toroid geometry; uniform field inside.

From circuit to field

$$U = \tfrac{1}{2}LI^2 = \frac{B^2}{2\mu_0} \times \text{volume}$$

While the current rises the source works against the back-EMF at rate L i di/dt; integrating gives half L I squared. Applied to the solenoid L = mu0 n squared A l and B = mu0 n I this becomes B squared over 2 mu0 times the interior volume: the energy lives in the field.

WHY / Two expressions for one energy: use the circuit form for bookkeeping and the field form for where it sits.

WATCH / The field form integrates over ALL space; for a long solenoid the interior dominates and the formula is nearly exact.

$$u_B = \frac{B^2}{2\mu_0} = 4 \times 10^5 \ \text{J/m}^3 \ \text{at} \ 1 \ \text{T}$$

CHECK / The circuit form handles bookkeeping; the field form says where the energy sits.

Electric versus magnetic storage

$$\frac{u_B}{u_E} = \frac{B^2/\mu_0}{\varepsilon_0 E^2} \sim 10^5$$

Air breaks down at 3 MV/m giving 40 J per cubic metre electric; 1 T is routine and gives 0.4 MJ per cubic metre magnetic. Capacitor or inductor, the same bookkeeping: half C V squared against half L I squared, with the mirror-image rule that capacitor voltage and inductor current cannot jump.

WHY / Field storage means magnetic fields; electric fields are limited by breakdown long before.

WATCH / Comparing storage at equal breakdown margins is the fair test; at equal volume magnetic wins overwhelmingly.

CHECK / The MRI magnet of 1.5 T over a cubic metre holds about 0.9 MJ per cubic metre: a car crash worth of energy in a bore.

What cannot jump

$$V_C \ \text{continuous}, \qquad i_L \ \text{continuous}$$

A capacitor holds its voltage through a switching instant; an inductor holds its current. This one rule fixes every initial condition: at t = 0 plus, i_L equals i_L before and V_C equals V_C before. The currents and voltages of the OTHER element can jump freely.

WHY / Continuity bookkeeping replaces solving at the instant; write the state table before the differential equations.

WATCH / Energy is NOT continuous at a switch; sparks and arcs are the visible proof.

$$U_C = \frac{q^2}{2C}, \qquad U_L = \tfrac{1}{2}LI^2$$

CHECK / Closing a switch on a charged capacitor wastes exactly half the stored energy in the leads every time.

The pressure reading

$$p = \frac{B^2}{2\mu_0} \ \text{(same number)}$$

Magnetic pressure equals energy density: 4 atmospheres at 1 T, 400 at 10 T. The winding of a solenoid carries this pressure as tension; the gapped core stores most of its energy in the gap where the field volume is.

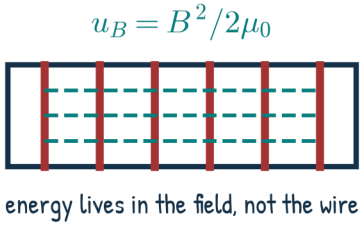
WHY / Pressure and energy density sharing one expression is why magnet design is pressure-vessel design.

WATCH / The pressure acts on the confining currents, not on free charges; it is the field pushing on its own container.

CHECK / 1 T over 10 cm squared end area pulls the halves together with 400 N: pressure times area.

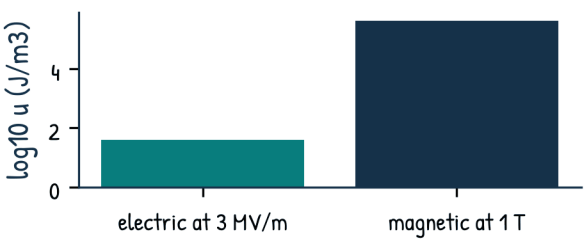
CHECK / The pressure acts on the confining currents, not on free charges. A 1 T solenoid of 10 cm2 bore pulls its halves together with 400 N.

The whole stored energy sits in the field volume between the windings



energy lives in the field, not the wire

Magnetic storage beats electric by five decades: breakdown limits air, not tesla



REMEMBER / Half L I squared is B squared over 2 mu0 per unit volume of field: magnetic storage beats electric by five decades.

The mechanical-electrical analogy and combinations

Linear elements; single loop or pair of coupled coils; ohmic R.

The table

$$m \leftrightarrow L, \quad k \leftrightarrow 1/C, \quad b \leftrightarrow R$$

Displacement is charge, velocity is current, force is EMF. Mass cannot change velocity abruptly exactly as inductance cannot change current; the spring force is the capacitor voltage; the dashpot is the resistor. Read tau = m/b as L/R and omega0 = root(k/m) as 1/root(LC).

WHY / The analogy reads unfamiliar circuits off known mechanics in one line: the capacitor rod is an added mass, the RL circuit a falling body.

WATCH / The analogy fails for mutual coupling: no mechanical element does what M does, and it says nothing about where the energy lives.

$$\omega_0 = \sqrt{\frac{k}{m}} \leftrightarrow \frac{1}{\sqrt{LC}}, \qquad \frac{m}{b} \leftrightarrow \frac{L}{R}$$

Series and parallel

$$L_{\text{series}} = L_1 + L_2 \pm 2M$$

Without coupling series adds and parallel combines by reciprocals, like resistors but because flux linkages add. With shared flux the series pair gains or loses 2M depending on whether the windings aid; two identical coils in opposition on one core give zero inductance.

WHY / The non-inductive winding is two coils subtracting their flux: precision resistors are wound that way.

WATCH / The sign of M is set by the winding sense relative to the shared axis; drawing both currents clockwise is not enough.

CHECK / Perfect coupling gives (root L1 plus or minus root L2) squared: check both limits against zero and infinite M.

CHECK / Two 1 mH coils on one core give 4 mH aiding and nearly zero opposed: the non-inductive winding again.

Energy of coupled coils

$$W = \tfrac{1}{2}L_1 I_1^2 + \tfrac{1}{2}L_2 I_2^2 \pm M I_1 I_2$$

The cross term is the work of establishing the mutual flux against the other coil. Its derivative at fixed current gives the force between the coils; at fixed flux the source is absent and the sign flips. Both bookkeepings give the same physical force.

WHY / One force law, two ledgers: quote the constraint before the derivative.

WATCH / Series aiding and opposing give the two observed inductances; measuring both yields M without any geometry.

$$M = \frac{L_{\text{aiding}} - L_{\text{opposing}}}{4}$$

CHECK / The cross term is the work of establishing mutual flux. At fixed current the force is one half I squared dM/dx; at fixed flux the source is absent and the ledger flips.

The inductor pair switched

$$I = \frac{L_1 I_0}{L_1 + L_2}, \qquad \Delta U = \frac{L_2}{L_1 + L_2} U_0$$

Switch a current-carrying L1 across an uncharged L2 with negligible resistance: flux linkage is conserved, giving a common current, while a fraction of the energy is lost to the switching arc however small the resistance. The inductive twin of the two-capacitor loss.

WHY / Flux conservation sets the final state; the loss is the price of the switch, and only the mechanism depends on R.

WATCH / Energy is not conserved in the joining; linkage is. State both before substituting.

CHECK / 10 mH at 3 A into 20 mH: final 1 A, energy falls 45 to 15 mJ, losing 30 mJ.

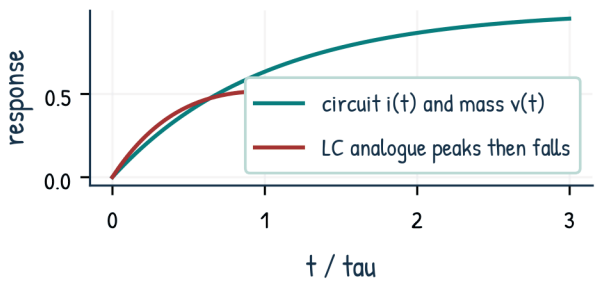
CHECK / Reopening the switch after the join forces the same flux through an arc: the spark energy is the 30 mJ deficit.

Mass-spring-dashpot and RLC share one differential equation



$$m \leftrightarrow L, k \leftrightarrow 1/C$$

Mechanical twins: mass is inductance, spring is capacitance, dashpot is resistance



REMEMBER / Every circuit equation has a mechanical twin: mass is inductance, spring is capacitance, dashpot is resistance.

LC oscillations and flux conservation

Ideal LC with no resistance, then small R; superconducting loops; switching transients.

The oscillator

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad q = q_0 \cos \omega_0 t$$

With no resistance the loop rule is L di/dt plus q/C = 0: SHM at omega0 = 1/root(LC). The current peaks a quarter-cycle after the charge; the energy sloshes between q squared / 2C and half L i squared, constant in sum, equal twice per cycle.

WHY / The quarter-cycle phase between charge and current is the AC section arriving early.

WATCH / The frequency uses the LOOP inductance and capacitance; stray wiring adds both and shifts the answer.

$$U_C = \frac{q_0^2}{2C}\cos^2 \omega_0 t, \qquad U_L = \frac{q_0^2}{2C}\sin^2 \omega_0 t$$

CHECK / 10 mH and 100 nF ring at 5.0 kHz with I0 = 0.32 A from 10 microcoulomb at 100 V.

Ring-down with resistance

$$q \propto e^{-Rt/2L}\cos \omega t, \qquad Q = \frac{\omega_0 L}{R}$$

Small R makes a damped oscillator: the amplitude decays with time constant 2L/R while the energy decays with L/R; the ringing lasts about Q cycles. Below critical resistance the circuit still oscillates; above, it creeps.

WHY / The quality factor counts the visible oscillations; a coil of Q = 157 rings a few hundred cycles.

WATCH / There is no tau = L/R exponential approach in LC circuits; two storage elements oscillate instead.

$$\omega = \sqrt{\omega_0^2 - (R/2L)^2}$$

CHECK / Below critical resistance the circuit still oscillates; above, it creeps. A coil of 10 mH with 2 ohm at 5 kHz has Q = 157 and rings hundreds of cycles.

Flux conservation

$$\Lambda = LI = \text{const} \ \ (\ R \to 0\ )$$

A zero-resistance loop keeps its flux linkage: any attempt to change it induces exactly the current that cancels the change. A switched inductor pair conserves L1 I0 across both; a squashed superconducting ring raises its current as L falls; a magnet dropped at a ring floats.

WHY / Flux linkage is the momentum of the superconducting loop; use it for the final state and an energy audit for the loss.

WATCH / Flux conservation holds for times short compared with L/R in ordinary circuits; afterwards Faraday with R applies.

CHECK / Squash a 1 microhenry ring at 100 A to half its inductance: the current doubles to 200 A and the energy rises 5 to 10 mJ.

The mechanical twins

$$q \leftrightarrow x, \quad i \leftrightarrow v, \quad \frac{1}{C} \leftrightarrow k$$

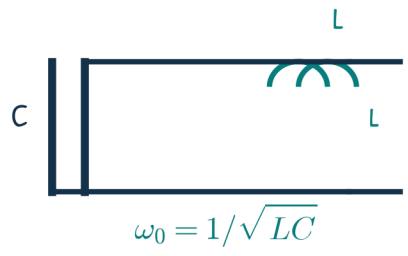
A mass on a spring is an LC circuit in disguise; the four snapshots of the oscillation map exactly: fully charged is maximum displacement, maximum current is maximum velocity. Damping maps to resistance and the driven case maps to forced oscillation with resonance.

WHY / Reading the mechanical picture solves circuit questions faster than the algebra when the geometry is unfamiliar.

WATCH / The analogy maps energy locations wrongly: the mass KE is the inductor field, not the wire energy.

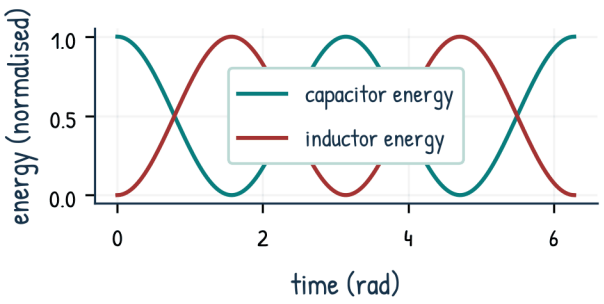
CHECK / The equal-energy instant is 45 degrees of phase, twice per cycle: same as the mass-spring equipartition crossing.

The LC tank: charge and current swap places omega0 times a second



$$\omega_0 = 1/\sqrt{LC}$$

Energy sloshes between C and L at twice the circuit frequency; the sum stays constant



REMEMBER / The LC circuit is simple harmonic motion: charge is displacement, inductance is mass, capacitance is the spring.

Why AC, and what RMS means

Sinusoidal sources; periodic waveforms; resistive heating as the definition of effective current.

Where AC comes from

$$v(t) = V_0 \sin(\omega t + \phi)$$

Every large generator is a rotating coil or magnet at fixed speed: 50 Hz in India and Europe, 60 Hz in the Americas. AC won because only a changing flux drives a transformer and only a transformer can step voltage up for transmission and down for use. Vocabulary: peak V0, peak-to-peak 2V0, period, frequency, phase.

WHY / The transformer is why AC exists; the sinusoid is what rotation gives for free.

WATCH / The rated 230 V is neither peak nor average; appliances must survive the 325 V peak.

$$V_{\text{peak}} = 325 \text{ V for } 230 \text{ V mains}$$

CHECK / Only a changing flux drives a transformer; only a transformer makes high-voltage transmission practical. The sinusoid comes free from rotation.

RMS from heating

$$I_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \frac{I_0}{\sqrt{2}}$$

The effective current is the steady current that heats a resistor at the same average rate; squaring removes the sign, averaging uses sin squared = 1/2, and the root gives I0 over root 2. The 230 V mains figure is an RMS value: its 325 V peak is what the insulation must survive.

WHY / RMS is defined by heating, not by calculus fashion; every waveform has its own factor.

WATCH / Mixing one peak and one RMS value in one formula is wrong by root 2; mixing two is wrong by 2.

CHECK / 230 V RMS is 325 V peak; a capacitor rated 250 V across the mains fails.

CHECK / RMS values multiply like DC: P = I rms squared R.

Average versus RMS

$$\langle i \rangle_{\text{half}} = \frac{2I_0}{\pi} = 0.637 I_0$$

The full-cycle average of a sinusoid is zero, which is why DC meters read nothing on AC. The half-cycle average is 2/pi of the peak: the reading of a DC meter fed through a full-wave rectifier. It is NOT the RMS: 0.637 against 0.707, and a meter calibrated for sinusoids misreads other waveforms.

WHY / Two different means of one sine: the arithmetic mean of the rectified wave and the root-mean-square of the raw wave.

WATCH / A moving-coil meter on a rectifier reads 0.90 of true RMS for sine input; the calibration is waveform-specific.

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = 0.707 I_0, \qquad I_{\text{avg}} = \frac{2I_0}{\pi} = 0.637 I_0$$

Other waveforms

$$\text{sq} : I_0; \qquad \text{tri} : \frac{I_0}{\sqrt{3}}$$

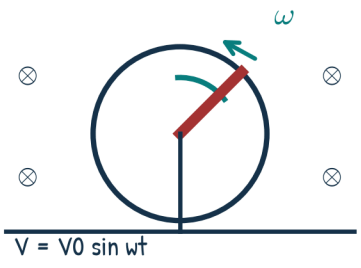
From the definition: a square wave of amplitude I0 has RMS I0 since i squared is constant; a triangular wave gives I0 over root 3; a half-wave rectified sinusoid gives I0/2 with average I0 over pi; a full-wave rectified sinusoid keeps the sine RMS I0 over root 2. Each waveform carries its own pair of factors.

WHY / Compute from the definition once and the factors become recognisable shortcuts.

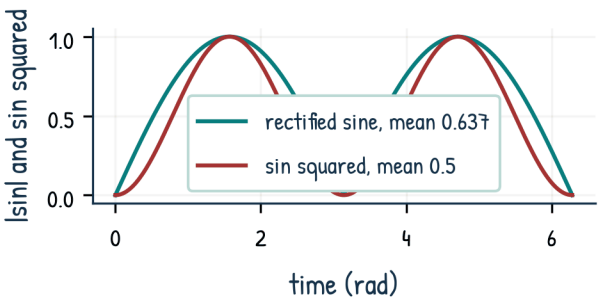
WATCH / Never assume 0.707 for non-sinusoidal AC; the defining integral is the only safe route.

CHECK / Half-wave rectified: RMS I0/2, average I0/pi; full-wave: RMS I0/root 2, average 2I0/pi.

The generator coil: EMF follows the sine of the rotor angle



RMS is the root of the mean of the square: 0.707 against the rectified mean 0.637



REMEMBER / The mains number is the RMS: 230 V is 325 V peak in disguise, and the effective current heats like DC.

R, L and C on AC: amplitude and phase

Each element alone across v = V0 sin omega t; linear; constant L and C.

The resistor

$$i = \frac{V_0}{R}\sin\omega t$$

Ohm point by point: the current is in phase with the voltage and its amplitude is V0/R. No phase puzzle and no energy storage; the resistor dissipates at the instantaneous rate i squared R whose average is I rms squared R.

WHY / The resistor is the reference phasor: everything else is measured against its phase.

WATCH / Instantaneous Ohm law still holds; using peak values in it is fine as long as both sides are peaks.

$$I_0 = \frac{V_0}{R}, \qquad \phi = 0$$

CHECK / Ohm point by point with peaks is fine as long as both sides are peaks. The resistor is the reference phasor; everything else is measured against its phase.

The inductor

$$i = \frac{V_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

From v = L di/dt the current is the integral of a sine, a negative cosine: amplitude V0 / omega L and a phase lag of 90 degrees. Physically the inductor is inertia: push and the velocity builds after. X_L = omega L grows with frequency: an inductor blocks high frequencies and passes DC.

WHY / The integral explains the lag; the inertia picture remembers it.

WATCH / X_L is not a resistance: the element stores and returns energy, dissipating nothing.

$$X_L = \omega L, \qquad \text{current lags } 90^\circ$$

CHECK / 50 mH at 50 Hz gives X_L = 15.7 ohm; at 5 kHz it is 1571 ohm: two decades of frequency swap the roles.

The capacitor

$$i = \omega C V_0 \sin\left(\omega t + \frac{\pi}{2}\right)$$

From q = Cv the current is the derivative: amplitude omega C V0 and a phase lead of 90 degrees. The current must flow first to build the charge that makes the voltage. X_C = 1/omega C falls with frequency: a capacitor blocks DC and passes high frequencies.

WHY / Derivative gives the lead; the spring picture: force is largest at maximum displacement, when the flow is zero.

WATCH / The mnemonic CIVIL holds: in C, I before V; in L, V before I.

$$X_C = \frac{1}{\omega C}, \qquad \text{current leads } 90^\circ$$

CHECK / 10 microfarad at 50 Hz gives X_C = 318 ohm; at 5 kHz only 3.2 ohm: L small and C large at mains frequency.

Reactance is not resistance

$$P_L = P_C = 0 \ \text{(cycle average)}$$

A pure L or C limits current amplitude but dissipates no average power: over a cycle the element takes energy from the source for a quarter period and returns it the next. Only R dissipates. The phase between v and i is the whole story of the next sheets.

WHY / Reactance sets amplitude; the phase decides power. Keep the two questions separate.

WATCH / Do not add X_L and R as scalars; the next sheets combine them as perpendicular phasors.

CHECK / At resonance the reactances cancel and the source sees pure resistance: the preview of the LCR sheet.

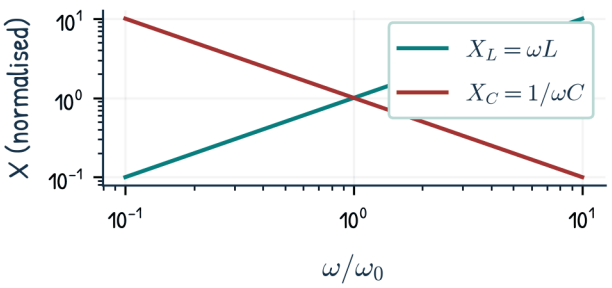
CHECK / A capacitor passes current while the voltage is already falling: that is why the current leads, never lags.

Three elements, three rules: R blocks nothing, L the fast, C the slow



$$\tilde{Z} = R + j\omega L + 1/(j\omega C)$$

Reactances cross at resonance: L blocks high frequencies, C blocks low



REMEMBER / Each element sets its own amplitude and phase: R in phase, L lagging 90 degrees, C leading 90 degrees.

Phasors: sinusoids as rotating vectors

Single frequency omega; linear circuits; amplitudes and phases constant in time.

The construction

$$a(t) = A\sin(\omega t + \alpha) \;\leftrightarrow\; A\angle\alpha$$

A sinusoid is the vertical projection of a vector of length A rotating anticlockwise at omega from initial angle alpha. Projection is linear, so the sum of two sinusoids of one frequency is the vector sum of their phasors; freezing the common rotation leaves a static diagram.

WHY / Linearity of projection is the legitimacy; at two frequencies the method fails and solutions superpose instead.

WATCH / Draw phasors anticlockwise; the projection is onto the vertical axis in the convention used here.

$$A\sin(\omega t+\alpha) + B\sin(\omega t+\beta) = C\sin(\omega t+\gamma) \text{ with phasors}$$

CHECK / Two equal phasors 120 degrees apart add to one of the same length: the three-phase sum is zero.

Series circuits: current is common

$$V_R = I_0 R \text{ along } I$$

In a series circuit the current is shared, so draw its phasor along the reference axis. Then V_R lies along I, V_L is rotated 90 degrees ahead, V_C 90 degrees behind. Their vector sum is the source phasor; its length is the voltage amplitude and its angle is the phase by which voltage leads current.

WHY / One current, three voltage phasors: the whole series analysis is geometry.

WATCH / Voltages across the elements are at different phases; the source is their vector sum, never their arithmetic sum.

$$V_L = I_0 X_L\ (+90^\circ), \qquad V_C = I_0 X_C\ (-90^\circ)$$

CHECK / The source phasor is the vector sum of element voltages.

The three triangles

$$RC:\ \phi = -\arctan\frac{X_C}{R}$$

$$RL:\ \phi = +\arctan\frac{X_L}{R}$$

Series RC: the source voltage lags the current by arctan(X_C/R); series RL leads by arctan(X_L/R); series LC has V_L and V_C antiparallel and the source is their difference, vanishing when X_L = X_C: resonance before it is named. Each is a right-angled triangle of phasors.

WHY / Three standard triangles; every series question is one of them or the full LCR triangle next.

WATCH / The sign of phi follows which reactance dominates; the arctan argument is always the net reactance over R.

CHECK / In series RC the current leads the source; in RL it lags. The capacitor leads, the inductor lags: CIVIL again.

Parallel circuits: voltage is common

$$I_R = \frac{V_0}{R} \text{ along } V$$

In a parallel circuit the voltage is shared: draw V as reference and the branch currents as phasors. I_R lies along V, I_C leads by 90 degrees, I_L lags by 90 degrees; the line current is their vector sum. Branch currents do not add as numbers.

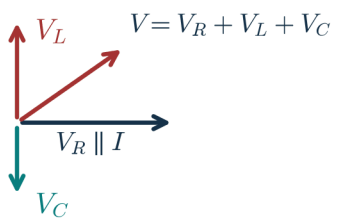
WHY / The mirror construction of the series case: swap the roles of current and voltage.

WATCH / Ammeters in the branches read magnitudes only; the line current is the phasor sum and can be smaller than either branch.

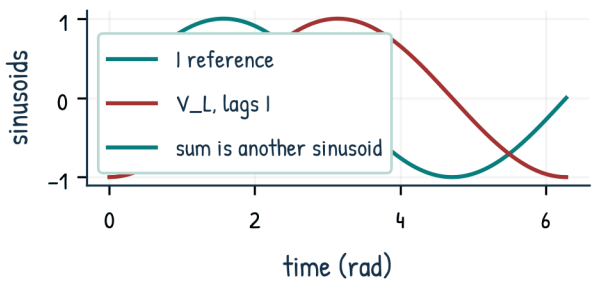
$$I_{\text{line}} = \sqrt{I_R^2 + (I_C - I_L)^2}$$

CHECK / With any two of the three voltages known, the phasor triangle closes the third: draw before you compute.

Phasor picture: horizontal resistive, vertical reactive, lengths at omega



Two sinusoids of one frequency add to a sinusoid: the phasor sum



REMEMBER / Two sinusoids of the same frequency add as vectors: that is why phasor diagrams work.

Series LCR: impedance and phase

R, L, C in series on v = V0 sin omega t; linear; steady state at one frequency.

The impedance triangle

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \qquad \tan\phi = \frac{X_L - X_C}{R}$$

The common current carries V_R along it and the net reactive voltage perpendicular; the source amplitude is their vector sum, defining impedance Z. The phase phi by which voltage leads current is positive when the circuit is inductive, negative when capacitive. The triangle has R on the base, net reactance vertical, Z as hypotenuse.

WHY / One triangle gives amplitude and phase at once; drawing it prevents the scalar-addition error.

WATCH / Z is not the sum of R, X_L and X_C; the reactances subtract first and enter as a leq.

$$V_0 = I_0 Z, \qquad \phi = \arg(Z)$$

CHECK / One triangle gives amplitude and phase at once. The phase is positive when inductive, negative when capacitive; Z is never the arithmetic sum.

Voltages do not add

$$V_R + V_L + V_C \neq V_{\text{source}}$$

A voltmeter across L may read more than the source; the readings of the three elements at 50 Hz in the worked circuit are 15, 12 and 242 V against a 230 V source. The phasor sum is 230 V exactly; the arithmetic sum 269 V is meaningless. Each of V_L and V_C can exceed the source: the magnification of the resonance sheet.

WHY / Read the meters as phasor magnitudes; only resistive runs add as numbers.

WATCH / The classic trap: adding inductive and capacitive voltages that are 180 degrees apart; read their difference.

CHECK / 20 ohm, 50 mH, 10 microfarad at 50 Hz: Z = 303 ohm, I = 0.76 A, phi = -86 degrees: almost a pure capacitor.

The worked circuit

$$X_L = 15.7\ \Omega, \qquad X_C = 318\ \Omega$$

At 50 Hz the net reactance is -302 ohm: the circuit is capacitive and the current leads by 86 degrees. At 225 Hz the reactances cancel and the same circuit resonates with Z = R = 20 ohm and eleven times the current. One circuit, two characters, set by frequency.

WHY / Frequency decides which element dominates; computing both extremes exposes the physics.

WATCH / The resonance frequency uses L and C only; R moves the peak width, not its location.

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 225\ \text{Hz}\ \text{(worked circuit)}$$

CHECK / One circuit, two characters: 225 Hz resonance with Z = 20 ohm against 50 Hz with Z = 303 ohm. Frequency decides which element dominates.

Reading the phase

$$\phi < 0\ \text{capacitive}, \qquad \phi > 0\ \text{inductive}$$

The sign of phi tells the character: below resonance X_C dominates and current leads; above, X_L dominates and current lags; at resonance phi = 0 and the source sees a pure resistance. The phase is measurable as the time shift between voltage and current zero-crossings.

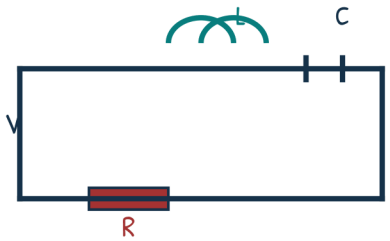
WHY / Three regimes, one sign test; exam questions often ask only which way the phase points.

WATCH / At exact resonance the voltages across L and C are equal and opposite, not zero each.

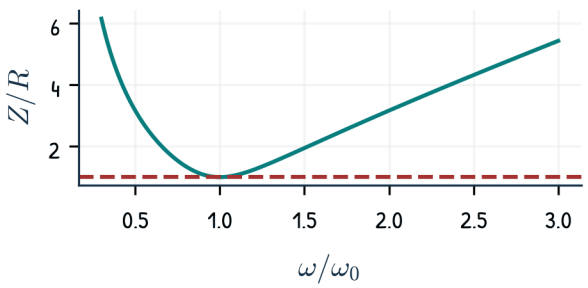
CHECK / Half-cycle shift of 86 degrees is about 2.4 ms at 50 Hz: measurable on an oscilloscope.

CHECK / Half of an 86 degree shift is about 2.4 ms at 50 Hz: measurable on a scope. At exact resonance the L and C voltages are equal and opposite.

Series LCR: one current, three voltages, one phase angle



Impedance dips to R at resonance where the reactances cancel exactly



REMEMBER / Impedance Z is the phasor sum of R and net reactance; voltages across the elements do not add as numbers.

Resonance, Q and bandwidth

Series LCR driven at variable frequency; linear; steady state.

The resonance condition

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad Z_{\min} = R$$

When X_L equals X_C the impedance is least, the current is greatest and the phase vanishes: the source sees a pure resistance. This is the LC oscillator natural frequency: a circuit driven at the frequency it would ring at by itself responds most. Below omega0 the circuit is capacitive, above it inductive.

WHY / Resonance is the driven oscillator finding its own frequency; the peak of I(omega) is the fingerprint.

WATCH / Parallel LC resonates at the same frequency with maximum impedance and minimum line current: the opposite response.

$$I_{\max} = \frac{V_0}{R} \text{ at } \omega_0$$

Voltage magnification

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{V_L}{V_0}$$

At resonance V_L equals V_C equals 10 X_L: with Q = 50 a 10 V source puts 500 V across the capacitor. The same Q is the ratio of reactance to resistance, and the danger and the radio front end in one number: microvolts at the aerial become millivolts at the capacitor.

WHY / Q multiplies voltages inside the circuit while the source sees only R; this is the tuned amplifier.

WATCH / The magnification applies at resonance only; off-peak V_L and V_C differ and their difference is the source voltage.

CHECK / A radio coil of 200 microhenry at 1 MHz with 5 ohm has Q = 251: a 10 microvolt aerial signal appears as 2.5 mV.

Bandwidth and sharpness

$$\Delta\omega = \frac{R}{L}, \qquad \frac{\omega_0}{\Delta\omega} = Q$$

The current falls to 1/root 2 of its peak where the net reactance equals R; solving near omega0 gives two half-power points omega0 plus or minus R/2L, a bandwidth R/L. The ratio of centre frequency to bandwidth is again Q: the same number measures magnification and narrowness.

WHY / The two definitions of Q are one identity; read the bandwidth straight off the resonance curve.

WATCH / The half-power points are symmetric only for Q much greater than 1; at low Q the peak shifts slightly.

$$I/I_{\max} = \frac{1}{\sqrt{1 + Q^2(\omega/\omega_0 - \omega_0/\omega)^2}}$$

CHECK / Bandwidth in hertz is R over 2 pi L: a 10 ohm, 10 mH coil is 160 Hz wide at any centre frequency.

Q as an energy ratio

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy lost per cycle}}$$

Stored half L I0 squared against dissipated half I0 squared R per cycle gives omega0 L/R: the third face of Q. A free oscillator rings about Q cycles before its amplitude falls appreciably; a coil of 10 mH with 2 ohm at 5 kHz has Q = 157 and rings a few hundred cycles.

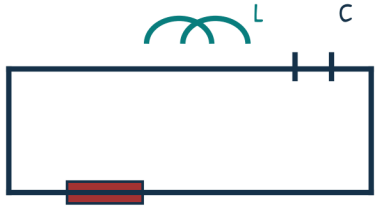
WHY / Magnification, sharpness and ring time are the same number; know all three readings.

WATCH / The energy ratio uses the PEAK stored energy convention; some texts use the mean and differ by a factor of 2.

CHECK / Q = 100 means a hundred-cycle ring and a one per cent bandwidth: the settle time of the driven circuit.

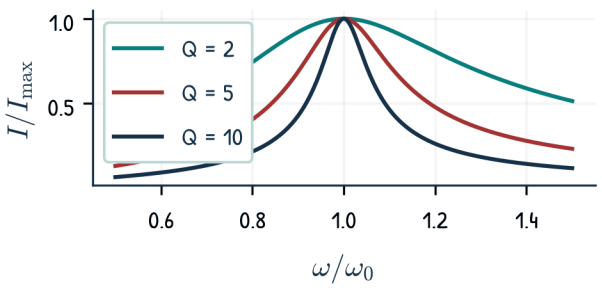
CHECK / Ring-down timing measures Q directly: count the cycles until the amplitude falls by a factor of e.

At resonance the source sees pure R; Q decides the sharpness



$\omega_0 = 1/\sqrt{LC}, Q = \omega_0 L/R$

Resonance curves sharpen with Q; half-power points sit omega0/Q apart



REMEMBER / A circuit driven at its natural frequency responds most; Q measures both the magnification and the sharpness.

Power in AC circuits

Sinusoidal steady state; linear loads with phase angle phi; RMS convention.

The average power

$$P = V_{\text{rms}} I_{\text{rms}} \cos\phi = \tfrac{1}{2} V_0 I_0 \cos\phi$$

Instantaneous p = vi expands into a constant term and a double-frequency term; averaging kills the oscillation and leaves one half V0 I0 cos phi. Only the in-phase part of the current does work; the quadrature part shuttles energy back and forth.

WHY / Derive once from the product; then remember P = V I cos phi with RMS values, or one half V0 I0 with peaks.

WATCH / Never mix one RMS and one peak value in the power formula; the factor of root 2 is the whole error.

$$P = \langle vi \rangle = \tfrac{1}{2} V_0 I_0 \cos\phi$$

CHECK / Only the in-phase part of the current does work. Remember P = V I cos phi with RMS values or one half V0 I0 with peaks: never mix.

The power triangle

$$S = VI, \qquad P = S\cos\phi, \qquad Q_r = S\sin\phi$$

Apparent power S in volt-amperes is the product of meter readings; real power P in watts is its projection on the current axis; reactive power Q_r in volt-amperes reactive is the perpendicular leg. Motors at 0.6 power factor draw 67 per cent more current than their watts suggest.

WHY / Three powers, one triangle; the billing meter reads watts but the wires carry volt-amperes.

WATCH / The Q of the power triangle is NOT the quality factor of resonance; same letter, different meaning.

CHECK / A 1 kW motor at 0.6 draws 1.67 kVA: 1.17 kVAR circulating through the supply.

CHECK / The billing meter reads watts; the wires carry volt-amperes. A motor at 0.6 power factor draws 67 per cent more current than its watts suggest.

Wattless current and correction

$$C = \frac{Q_r}{\omega V^2} \text{ (parallel correction)}$$

The 90-degree component of the current is wattless: it carries no average power but heats the wires and loads the generator. A parallel capacitor supplies the reactive volt-amperes locally, its current leading to cancel the motor current lagging; the supply then sees nearly unity power factor.

WHY / Correction is bookkeeping of reactive power done with a capacitor instead of the grid.

WATCH / The correction capacitor is sized at the actual supply frequency; the formula fails on harmonics.

CHECK / Correcting 0.6 to 0.95 cuts the line current by 37 per cent at equal real power.

Energy over a cycle

$$W = PT, \qquad U_L = U_C \text{ exchange each cycle}$$

In a full cycle the resistor absorbs PT; the inductor and capacitor each return exactly what they took, so the metered energy equals the real power times time. The instantaneous power oscillates at twice the source frequency for the reactive parts.

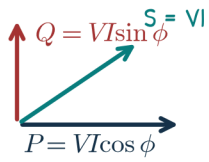
WHY / The energy ledger over one cycle separates the dissipating and shuttling components.

WATCH / Half-cycle energy accounting misleads; the reactive parts integrate to zero only over whole cycles.

CHECK / The capacitor stays cold; only the wires and generator carry the wattless current.

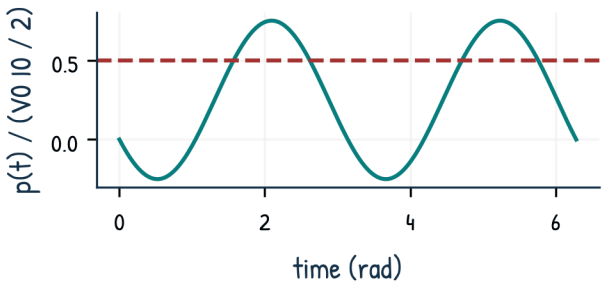
CHECK / The instantaneous power dips negative twice per cycle in a purely reactive branch: energy sloshes back to the source.

Real power does work; reactive power just sloshes between source and field



power triangle: watts out, vars sloshing

Instantaneous power pulses at 2 omega about the mean P = VI cos phi (dashed)



REMEMBER / Average power is V I cos phi; the power factor cos phi is what the customer pays for and the supply company hates.

Parallel AC circuits and the tank

Linear branches sharing a voltage; sinusoidal steady state; admittance language.

Admittance

$$Y = \frac{1}{Z} = \sqrt{G^2 + (B_C - B_L)^2}$$

Parallel branches share voltage so their admittances add as series impedances do: conductance G = 1/R in phase, susceptances B = 1/X perpendicular. The line current is V times Y; the branch currents are the legs of the current triangle.

WHY / Switch to admittance before writing parallel equations; impedance arithmetic in parallel is painful.

WATCH / Susceptance subtracts like reactance: capacitive and inductive branches cancel in the line.

$$I_{\text{line}} = VY, \qquad G = \frac{1}{R},\ B_C = \omega C,\ B_L = \frac{1}{\omega L}$$

CHECK / Convert each branch to G and B once; the network is then two numbers added component-wise.

The tank circuit

$$I_{\text{line}} = \frac{V}{R}, \qquad I_{\text{tank}} = Q I_{\text{line}}$$

At omega0 = 1/root(LC) the inductor and capacitor currents cancel in the supply line: the impedance is maximum and the line current minimum. Inside the LC loop a circulating current V omega0 C = Q times the line current flows back and forth, far larger than what the source supplies.

WHY / Anti-resonance: the opposite of the series case in every observable except the frequency.

WATCH / The trap: two branch currents of 2 A each give line current zero at resonance, not 4 A.

$$Z_{\text{parallel}}(\omega_0) = R \text{ (max) for the ideal tank}$$

CHECK / The trap: two branch currents of 2 A each give line current zero at resonance, not 4 A. Inside the tank a circulating Q times the line current flows.

Wave trap versus series

$$LC \text{ parallel rejects}, \qquad LC \text{ series passes}$$

A parallel LC across a line rejects signals at omega0: the radio wave trap; a series LC in the line passes only omega0: the band-pass filter. Same frequency, opposite placement, opposite function.

WHY / Placement decides function: shunt to reject, series to select.

WATCH / The rejection depth is set by Q; a low-Q tank is a shallow notch.

CHECK / A 200 microhenry, 127 pF tank at 1 MHz with Q = 250 attenuates the line by 250 at resonance.

CHECK / The IF transformer of a radio is this tank with taps: 455 kHz wide by design, not by accident. A detector diode after the tank needs only microvolts: selectivity first, gain second.

Solving parallel networks

$$Y_{\text{total}} = \sum Y_k, \qquad \phi = \arg Z$$

Add admittances of parallel branches, then convert back to Z for the phase. Mixed series-parallel networks go to complex impedance immediately; phasor diagrams alone get crowded fast. The worked R2 parallel C network gives Z = 25.9 ohm at -20 degrees for R1 = 10 in series.

WHY / One rule: series add Z, parallel add Y; mixed networks go complex.

WATCH / Do not add branch currents as magnitudes; the phase offsets decide the line current.

CHECK / 20 ohm parallel 31.8 ohm capacitive reactance at 50 Hz: branch impedance 16.9 ohm at -32 degrees.

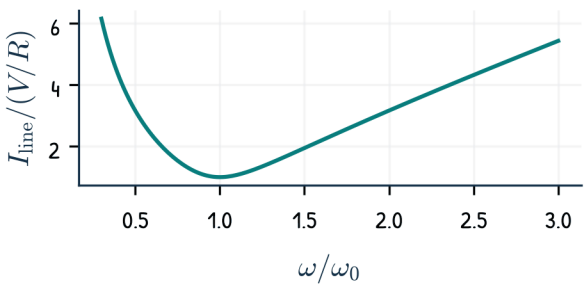
CHECK / Series add Z, parallel add Y; mixed networks go complex immediately. Do not add branch currents as magnitudes.

Parallel RLC: branch currents share V but not phase



three branch currents add as phasors

Parallel LC at resonance: line current minimum, impedance maximum: anti-resonance



REMEMBER / In parallel, admittances add; at resonance the line current is minimum and the tank circulates Q times it.

Complex impedance and networks

Linear circuits at one frequency; sinusoidal steady state; j-squared = -1 convention.

The transform

$$\tilde{Z}_R = R, \qquad \tilde{Z}_L = j\omega L, \qquad \tilde{Z}_C = \frac{1}{j\omega C}$$

Represent v = V0 sin(omega t + alpha) by V0 exp(j alpha); then d/dt becomes j omega and each element multiplies current by a complex impedance. Series impedances add and parallel combine by reciprocals: exactly like resistors, with phases carried in the argument.

WHY / The method converts phase bookkeeping into algebra; networks that defeat drawing fall to one line.

WATCH / The physical quantity is the real part of the complex amplitude times exp(j omega t); a j surviving into a physical answer is an error.

$$\tilde{V} = \tilde{Z}\tilde{I}, \qquad Z = |\tilde{Z}|, \;\; \phi = \arg \tilde{Z}$$

CHECK / Multiplying by j is a quarter-period phase lead: two j factors give the minus sign of the LC oscillator.

Series LCR revisited

$$\tilde{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

The complex impedance of the series circuit has magnitude Z and argument phi: the impedance triangle read as a complex number. Resonance is where the imaginary part vanishes; the phase is the arctan of imaginary over real.

WHY / Complex arithmetic reproduces the whole LCR sheet in one expression.

WATCH / Keep the j convention consistent: j omega L for inductors, 1/(j omega C) = -j/(omega C) for capacitors.

CHECK / The worked network R1 = 10 in series with 20 ohm parallel 100 microfarad at 50 Hz gives 24.3 - j 9.0 ohm.

CHECK / The imaginary part vanishes at resonance. Keep the j convention consistent: j omega L for inductors and minus j over omega C for capacitors.

Networks both ways

$$\tilde{Z} = R_1 + \frac{R_2}{1 + j\omega R_2 C}$$

For the mixed network, rationalise the parallel part into real and imaginary parts; the real part dissipates and the imaginary part does not. Phasor drawing does the same in three lines of geometry: one line of algebra wins.

WHY / Read off Z and phi directly from the rationalised form; the physical split is real versus imaginary.

WATCH / Power from complex amplitudes needs the conjugate: P = one half real part of V times conjugate I.

$$P = \tfrac{1}{2}\mathrm{Re}(\tilde{V}\tilde{I}^{\,*})$$

CHECK / Power from complex amplitudes needs the conjugate: P = one half the real part of V times conjugate I. The rationalised form separates dissipation from storage.

Limits of the method

$$\omega \text{ single: complex;} \quad \text{two } \omega\text{: superpose}$$

Complex impedance is valid at one frequency; transients and harmonics need superposition of solutions at their own frequencies. The transient of the driven LCR is a decaying mode at the natural frequency plus the steady state at the drive.

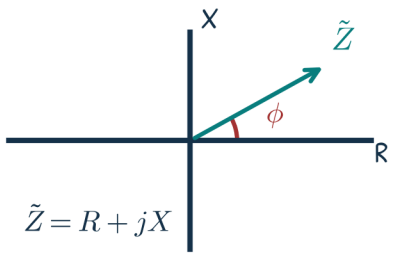
WHY / A j that fails to cancel is the signal that two frequencies have been mixed.

WATCH / Rectifier circuits draw harmonics; their correction needs each harmonic treated separately.

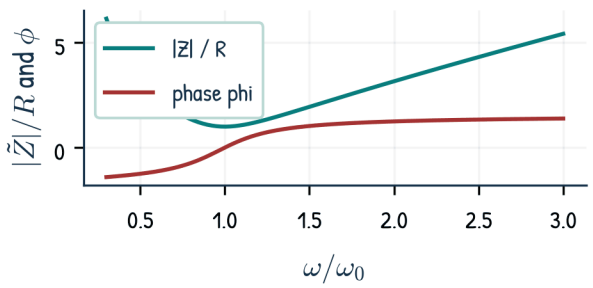
CHECK / Switching transients die in about 2L/R seconds; after that the phasor solution is the circuit.

CHECK / A surviving j in a physical answer flags a dropped conjugate or a mixed frequency.

Impedance is a complex number: real part dissipates, imaginary stores



Complex impedance: magnitude and argument in one picture; phi flips sign at resonance



REMEMBER / Phasors are complex numbers: impedance arithmetic then does what phasor drawing cannot.

Transformers and transmission

Two coupled coils on a closed iron core; k near 1; sinusoidal steady state.

Flux linkage and voltage

$$\frac{v_s}{v_p} = \frac{N_s}{N_p}$$

The same flux links every turn of both windings: v = N dPhi/dt gives the voltage ratio equal to the turns ratio. The flux is set by the PRIMARY voltage: Phi0 = Vp0 / Np omega whatever the load draws. A transformer on DC makes no secondary EMF and the primary, seeing only its winding resistance, burns.

WHY / The load does not set the flux; the supply voltage does. The current follows the load.

WATCH / Never connect a transformer to DC; steady flux means no induction and a shorted primary.

$$\Phi_0 = \frac{V_{p0}}{N_p \omega} \text{ (set by the primary)}$$

Currents and power

$$\frac{i_s}{i_p} = \frac{N_p}{N_s}, \qquad v_p i_p = v_s i_s$$

With the secondary open only a small magnetising current flows, 90 degrees behind the voltage and wattless. Connect a load: the secondary current would change the flux, the primary voltage forbids it, and the primary draws the extra current that cancels the secondary ampere-turns. Ideal power in equals power out.

WHY / The primary current is set by the LOAD; an unplugged charger draws almost nothing.

WATCH / The magnetising current is missing from the ideal current ratio; at no load it is all there is.

CHECK / 230 V to 12 V gives 19.2:1; a 6 ohm load draws 2 A at 12 V and the primary takes 0.104 A.

Impedance reflection

$$Z_p = \left(\frac{N_p}{N_s}\right)^2 Z_s$$

A load Zs on the secondary looks from the primary like the turns-ratio squared times Zs: the principle of impedance matching. An 8 ohm loudspeaker on a 1:20 output transformer looks like 3.2 kohm to the amplifier; a matched cable termination works the same way.

WHY / The square of the turns ratio is the impedance transformer; matching networks are the same physics.

WATCH / Reflection is exact only for ideal coupling; leakage inductance adds a series reactance that matters at band edges.

$$n = \frac{N_p}{N_s}, \qquad Z_p = n^2 Z_s$$

CHECK / An 8 ohm speaker on a 1:20 transformer looks like 3.2 kohm.

Losses and transmission

$$P_{\text{loss}} = I^2 R_{\text{line}} = \frac{P^2 R_{\text{line}}}{V^2}$$

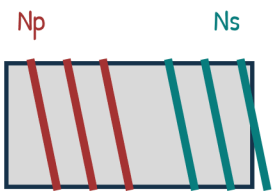
Real transformers lose a few per cent to copper loss, eddy currents in laminated cores, hysteresis and flux leakage. Transmission lines lose I squared R: delivering 10 MW over 5 ohm wastes 41 per cent at 11 kV and 0.1 per cent at 220 kV. Step up at the station, down at the substation.

WHY / The whole grid architecture is the V-squared law; the transformer is the enabling device.

WATCH / High-voltage DC now does the same with power electronics and wins on very long lines by avoiding reactive current.

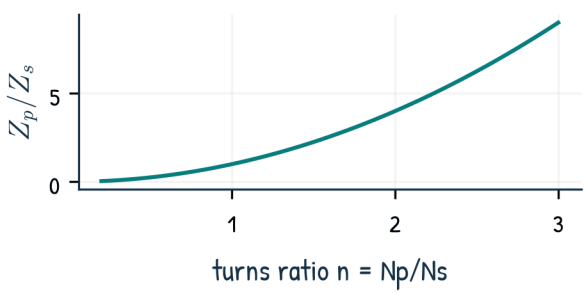
CHECK / Twenty times the voltage cuts the line loss by 400: the factor behind every transmission line.

Shared flux forces the volts-per-turn to match on both windings



$$V_s/V_p = N_s/N_p = n$$

Impedance reflects as the square of the turns ratio: the matching principle



REMEMBER / The turns ratio sets voltage and current; the reflected impedance is the square of that ratio; the grid runs high voltage because losses scale as one over V squared.

Rectification and smoothing

Diodes with 0.7 V silicon drop; capacitor smoothing; roughly constant load current.

Half and full wave

$$V_{\text{avg}} = \frac{V_0}{\pi}\ \text{(half)}, \qquad \frac{2V_0}{\pi}\ \text{(full)}$$

A single diode passes only positive half-cycles: average V0 over pi, RMS V0/2, and gaps between humps. A bridge of four diodes, or a centre-tapped secondary with two, delivers both humps: average 2V0 over pi, ripple at twice the frequency. Subtract 0.7 V per conducting diode.

WHY / The DC output is the AVERAGE; the RMS describes heating. Do not interchange them.

WATCH / A meter on the output reads average; quoting the RMS as the DC level overestimates by 11 per cent for full-wave.

$$V_{\text{rms}} = \frac{V_0}{2}\ \text{(half)}, \qquad \frac{V_0}{\sqrt{2}}\ \text{(full)}$$

CHECK / A meter on the output reads average; quoting RMS as DC overestimates by 11 per cent for full-wave.

Smoothing and ripple

$$\Delta V \approx \frac{I}{2fC}\ \text{(full-wave)}$$

A capacitor across the load charges to the peak on each hump and discharges through the load between humps; for a roughly constant load current the sag is I times the interval over C: I/(2fC) full-wave, I/(fC) half-wave. Sizing: 0.5 A at 50 Hz with 1 V ripple allowed needs 5 millifarad.

WHY / The ripple formula is charge bookkeeping between peaks; the sawtooth approximation is good for small ripple.

WATCH / Half-wave ripple is twice full-wave at the same f; the interval between humps is what enters.

CHECK / 100 Hz ripple into 5 millifarad at 0.5 A gives 1 V of sawtooth: the big can in every supply.

CHECK / Half-wave ripple is twice full-wave at the same f: the interval between humps is what enters. 0.5 A at 50 Hz with 1 V ripple needs 5 millifarad.

The impedance view

$$X_C(2f) = \frac{1}{2\pi(2f)C} \ll R_{\text{load}}$$

The smoothing capacitor shunts ripple around the load: at 100 Hz a 5 millifarad capacitor has 0.32 ohm reactance against a load of several ohms, while passing no DC. Filters are the same idea with inductors added: the supply chain transformer, rectifier, capacitor, regulator sits inside every charger.

WHY / Each stage of a charger is one section of this chapter in order.

WATCH / An inductor-input filter reverses the ripple advantage at light load; know which topology is drawn.

$$\tilde{V} = \tilde{I}\,\tilde{Z}$$

CHECK / A charger block in order: transformer for level, bridge for direction, capacitor for ripple, regulator for stiffness.

Diode limits and checks

$$V_{\text{drop}} \approx 0.7\ \text{V per Si diode}$$

Silicon drops about 0.7 V when conducting and blocks to its breakdown voltage in reverse; the peak inverse voltage on a bridge is the peak secondary voltage, on a centre-tap twice it. Small currents raise the drop only logarithmically; the ideal-diode approximation is fine above an ampere.

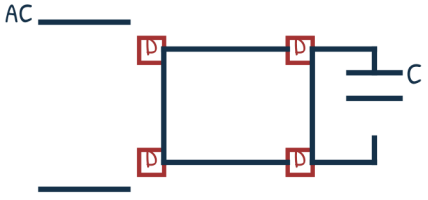
WHY / The drop matters in low-voltage supplies: a 5 V rail from a 6 V transformer loses two diode drops in a bridge.

WATCH / Reverse breakdown is a hard limit; silicon rectifiers fail shorted, not open.

CHECK / A 12 V RMS secondary gives 16.3 V peak, 15.6 V after a bridge at load: subtract 1.4 V before smoothing.

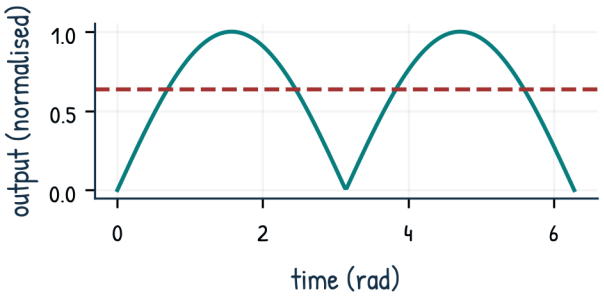
CHECK / Peak inverse voltage on a bridge is the peak secondary; on a centre-tap it is twice. Silicon rectifiers fail shorted, not open.

Bridge rectifier plus capacitor: chop, steer, smooth



four diodes steer both half-cycles one way

Full-wave output with its DC average 2/pi; a smoothing capacitor rides the humps



REMEMBER / Rectifiers make one-sign current; capacitors smooth it; ripple sets the capacitor size.